

WSR-88D SYSTEM SPECIFICATION



Prepared by the WSR-88D Radar Operations Center

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WSR-88D SYSTEM SPECIFICATION

2810000J

REVISION RECORD

REVISION	-	A	INT B	INT B (2nd Draft)	INT B (3rd Draft)	B	C	D	E	F	G	H	I	J
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SECTION 1	-	A	INT B	INT B	INT B	B	C	C						
SECTION 2	-	A	INT B	INT B	INT B	B	C	D		F	G		I	J
SECTION 3	-	A	INT B	INT B	INT B	B	C	D	E	F	G	H	I	J
SECTION 4	-	A	INT B	INT B	INT B	B	B	D				H		
Appendix A	-	A	Deleted	Deleted	Deleted	Deleted	Deleted	Deleted						
Appendix B	-	A	INT B	INT B	INT B	B	C	D	E	F	G	H		J
Appendix C	-	A	INT B	INT B	INT B	Deleted	Deleted	Deleted						
Appendix D	-	A	INT B	INT B	INT B	B	B	B		F	G			J
Appendix E	-	A	INT B	INT B	INT B	B	B	B		F	G	H	I	J
Appendix F	-	A	Deleted	Deleted	Deleted	Deleted	Deleted	Deleted						
Appendix G	-	A	INT B	INT B	INT B	B	C	D		F				J
Appendix H	-	A	INT B	INT B	INT B	B	B	B		F				
Appendix I	-	A	INT B	INT B	INT B	B	B	B	E	F	G			J
Appendix J	-	A	INT B	INT B	INT B	B	B	B					I	
Appendix K	-	A	INT B	INT B	INT B	B	B	B		F	G			
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Appendix M	-	A	Deleted	Deleted	Deleted	B	Deleted	Deleted						
Annex A	NONE	1.0	2.0	2.0	2.0	2.1	Deleted	Deleted						

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1 SCOPE

This document establishes the performance, design, development and test constraints for the Weather Surveillance Radar-1988 Doppler (WSR-88D) System. This document constitutes the System Specification, as defined by MIL-STD-490. The WSR-88D System is evolving from a proprietary hardware platform and operating system to a standards-based architecture such that improvements in data quality, operational utilization of advances in scientific algorithms and interoperability of the WSR-88D system with other agency modernized systems (e.g. Advanced Weather Interactive Processing System (AWIPS), Meteorological Operational Capability (MOC), Weather and Radar Processor (WARP), Integrated Terminal Weather System (ITWS)) are more easily realized. While requirements for unique equipment procured for specific systems often included a design for 20 year useful life, an acquisition philosophy that accommodates the infusion of new technology that is regularly available in the open market depends upon the use of standards-based commercial off-the-shelf (COTS) hardware that is not expected to be used for 20 years. The Tri-Agency Requirements (TAR) for Operational Use of Weather Radar Data contains guidance for the evolution of the WSR-88D system. The remaining sections are as follows:

Section 2 - Applicable Documents

Section 3 - Requirements

Section 4 - Quality Assurance

2 APPLICABLE DOCUMENTS

2.1 Government Documents

The following military and Federal specifications and standards are available from the sources shown.

2.1.1 Military Standards and Handbooks

MIL-STD-130	Identification and Marking of U.S. Military Property
MIL-HDBK-454	General Guidelines for Electronic Equipment
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-469	Radar Engineering Design Requirements, Electromagnetic Compatibility
MIL-HDBK-471	Maintainability Verification/Demonstration/Valuation
MIL-STD-490	Specification Practices
MIL-HDBK-781	Reliability Testing for Engineering Development, Qualification, and Production, (All- Equipment Production Reliability Test Plan)
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-961	Department of Defense Standard Practice, Defense Specifications
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities
SAE AMS 2695	Connections, Electrical, Solderless, Wire-Wrapped
MIL-HDBK-2036	General Requirements for Electronic Equipment Specifications
MIL-STD-188-124B	Grounding, Bonding, and Shielding
MIL-HDBK-217	Reliability Predictions of Electronic Equipment
MIL-HDBK-419/1	Grounding, Bonding, and Shielding for Electronic Equipment and Facilities - Basic Theory
MIL-HDBK-419/2	Grounding, Bonding, and Shielding for Electronic Equipment and Facilities – Applications
AFI 21-116	Maintenance Management of Communications Electronics
DoD 3235.1-H	Test and Evaluation of System Reliability, Availability, and Maintainability

Source: Defense Printing Service Office
700 Robbins Avenue
Philadelphia, PA. 19111-5094

Memorandum Of Agreement among the DoC, DoD, and DoT for Interagency Security Requirements and Certification & Accreditation of the WSR-88D

Source: WSR-88D ITSO
Radar Operations Center (ROC)
1313 Halley Circle
Norman, OK 73069

2.1.2 Federal Documents

FED-STD-595	Colors Used in Government Procurement
NIST Pub 500-230	NIST Pub 500-230 Application Portability Profile (APP) The U.S. Government's Open System Environmental Profile Version 3.0
NSTISSI No. 1000 April 2000	National Information Assurance Certification and Accreditation Process (NIACAP)
OSHA-29 CFR 1910	OSHA Safety and Health Standards
FMH-11	Federal Meteorological Handbook 11 NWS Transient Susceptibility Standard (May 1978) National Weather Service Operations Manual
Source:	National Weather Service 1325 East-West Highway Silver Spring, MD 20910
AC 70/7460-1K, Change 1	FAA Advisory Circular, Obstruction Marking and Lighting
DOT/FAA Order 6000.15D	General Maintenance Handbook for National Airspace System (NAS) Facilities
DOT/FAA Order 6000.30C	National Airspace System Maintenance Policy
DOT/FAA Order 6000.40	Monitoring Policy for the Maintenance Control Center (MCC) of the NAS
DOT/FAA Order 6000.50B	Airway Facilities NAS Operations Procedures
DOT/FAA Order 6090.1B	Development and Implementation of Remote Monitoring Subsystem (RMS) within the NAS
FAA-STD-019b	Lightning Protection, Grounding, Bonding and Shielding Requirements for Facilities

Source: Federal Aviation Administration
Washington, DC 20591

Manual of Regulations and Procedures for Federal Radio Frequency
Management, National Telecommunications and Information
Administration

R400-AR401 NEXRAD Algorithm Report

R400-IS301C NEXRAD Integrated Logistics Support Plan (ILSP), 15 March 2002

Interface Control Documents (ICDs):

2620000	RPG/RMS ICD
2620001	RPG/CLASS 1 USERS ICD
2620002	RDA/RPG ICD
2620004	RDA/RMS ICD
2620006	RPG/CLASS 2 USERS ICD
2620040	RPG X.25 PROTOCOL ICD
2620041	RPG TCP/IP Protocol ICD

Engineering Drawings

1213851 Plate, Identification

Source: WSR-88D Radar Operations Center (ROC)
1313 Halley Circle
Norman, OK 73069

Tri-Agency Requirements (TAR) for Operational Use of Weather Radar Data (May 1997)

Source: NEXRAD Product Improvement Office
1325 East-West Highway
Silver Spring, MD 20910

Radar Product Generation Algorithm Survey and Analysis Report, June 1996

FAA National Airspace Systems (NAS) Documents:

Functional Requirements for NAS System Level Specification for NAS Infrastructure

NAS-IR-51070000, Interface Requirements Document, NAS Infrastructure Management System Manager/Managed Subsystem

NAS-IC-51070000-1, Interface Control Document, NAS Infrastructure Management System Manager/Managed Subsystem using the Simple Network Management Protocol Version 1 (SNMPv1)

Source: Federal Aviation Administration
Washington, DC 20591

2.2 Non-Government Documents

The following non-Government documents are available from the sources shown.

2.2.1 American National Standards Institute (ANSI) Standards

ANSI INCITS 4 Information Systems - Coded Character Sets 7-Bit
American National Standard Code for Information
Interchange

Source: ANSI, Inc.
1430 Broadway
New York, NY 10018
<http://www.ansi.org>

2.2.2 Telecommunications Industry Association/Electronics Industries (TIA/EIA) Standards

TIA/EIA-222-G Structural Standards for Steel Antenna Towers and Supporting
Structures

Source: <http://global.ihs.com>

2.2.3 Other Non-Government Documents

NFPA 70 National Electrical Code

IEEE C95.1- Safety Levels with Respect to Human Exposure to Radio Frequency
2005 Electromagnetic Fields, 3 kHz to 300 Ghz

NFPA 780	Standard for the Installation of Lightning Protection 1995 Edition
Source:	National Fire Protection Association Batterymarch Park Quincy, MA 02269 http://www.nfpa.org
IEEE Std 1413-1998	IEEE Standard Methodology for Reliability Prediction and Assessment for Electronic Systems Equipment, 8 Dec 1998
Source:	http://www.standards.ieee.org/
IEEE C62.41.1	IEEE Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits
IEEE C62.41.2	Recommended Practices on Characterization of Surges in Low-voltage (1000 V and less) AC Power Circuits
UL 60950	Safety of Information Technology Equipment
FCC, Part 15	Radio Frequency Devices

2.3 Precedence of Documents

An attempt has been made to prevent conflicts between this Specification and the referenced documents. However, in the event of a conflict between requirements, the following order of precedence (highest=a) will apply:

- a. WSR-88D System Specification
- b. Military Standards
- c. Federal Standards
- d. ANSI Standards
- e. EIA Standards
- f. Other

3 REQUIREMENTS

This section delineates the performance and design requirements for the WSR-88D System. Throughout this document, many performance requirements are stated as integer numbers. A minimum precision of 2 decimal places is assumed for all performance requirements given as an integer.

3.1 System Definition

The WSR-88D System comprises Doppler radars, telecommunications, computer data communications, data processing hardware and software, display and data entry equipment, documentation, and certain facilities and support capabilities required to detect, process, distribute, and display weather information in a manner which allows the Department of Commerce (DOC), the Department of Defense (DoD) and the Department of Transportation (DOT) to fulfill their mission needs.

3.1.1 General Description

A general description of the WSR-88D System and its functions are provided in the following paragraphs.

3.1.1.1 System Users

The system users include both principal users and other users.

3.1.1.1.1 Principal Users

The principal users of the WSR-88D System are operationally oriented agencies within the participating departments (DOC, DoD, and DOT) which use weather radar information to perform or support their activities. These activities include warning of hazardous weather and flash flood prediction, predicting weather conditions, ensuring safety of flight, protecting base resources, and planning military missions.

3.1.1.1.1.1 Department of Commerce

Within DOC, the National Weather Service (NWS), a component of the National Oceanic and Atmospheric Administration (NOAA), is the Federal Government's principal civilian meteorological agency. As such, it requires weather radar information to fulfill its assigned responsibilities. This information is required at National Weather Service Forecast Offices (NWSFOs), National Weather Service Offices (NWSOs), River Forecast Centers (RFCs), the Center for Environmental Prediction (NCEP), the Tropical Prediction Center (TPC), and the Storm Prediction Center (SPC).

3.1.1.1.1.2 Department of Defense

In the case of DoD, Air Force Weather (AFW) is the principal user agency which provides meteorological services to the Air Force, the Army, and certain other DoD elements. The Naval Meteorological Oceanography Command (NMOC) is the principal meteorological agency which supports the Navy, Marine Corps, and other elements of DoD. Air Force Weather requires weather radar information at its Base Weather Stations (BWSs), regional hubs, and at the Air Force Global Weather Central (AFGWC). NMOC requires weather radar information at its Naval Oceanography Command Detachments (NOCs).

3.1.1.1.1.3 Department of Transportation

In the case of DOT, the Federal Aviation Administration (FAA), as the principal civilian agency for ensuring air traffic safety, requires weather radar information. This information is required for Air

Traffic at Air Route Traffic Control Centers (ARTCCs), Terminal Radar Control facilities (TRACONs), airport towers, Automated Flight Service Stations (AFSSs), and the FAA's Air Traffic Control System Command Center (ATCSSCC). Airway Facilities requires this information at the National Operations Control Center (NOCC), Operational Control Centers (OCCs), and at Service Operations Center Work Centers (SOC/WCs).

3.1.1.1.2 Other Users

Other users of the WSR-88D System include Federal Government agencies other than the principal users; state and local government agencies; and private sector users such as airline companies, consulting meteorologists, the media, and university laboratories. Selected weather radar information will be made available to these users according to policy established by NEXRAD Program Management Committee (PMC).

3.1.1.2 Product Types

To support the common needs of the principal and other users for weather radar information, the WSR-88D System will generate products which can be used for distribution and presentation to operational personnel. These include base data, base products, derived products, alphanumeric products, and derived data array products (Defined in Appendix B).

3.1.1.2.1 Base Data

Base data consist of the Doppler moments at the configured resolution available from the signal processor (at least reflectivity, mean radial velocity and spectrum width).

3.1.1.2.2 Radar Products

The base products consist of digital fields of reflectivity, mean radial velocity, and radial velocity spectrum width (standard deviation) estimates, differential reflectivity, differential phase, and correlation coefficient data presented in a format suitable for display on a color graphic display. The derived products provide processed radar data, concentrated to provide large volumes and varieties of data in a form suitable for display on color graphic displays and, to the extent possible, requiring minimum interpretive effort. The alphanumeric products provide processed radar data for users without a graphic display capability. The derived data array products provide data in a non-displayable format for transmission to systems external to the WSR-88D System for further processing.

Product generation will be automated to the extent feasible. However, at selected sites, an interactive display and data entry capability will be available for human analysis of radar data and assistance in the generation of certain radar products.

3.1.1.3 Product Distribution

The WSR-88D System will distribute products to the operational facilities (Defined in Appendix B) of the principal users and have an access capability so that other users can obtain selected products.

3.1.1.3.1 Principal User Product Distribution

3.1.1.3.1.1 NWS Local Sites

In the case of NWS, operational personnel at NWSFOs and NWSOs will automatically be provided products from a preselected WSR-88D. Certain NWSFOs may also regularly be provided products from one or more additional preselected WSR-88D Systems since their geographic area of responsibility is usually larger than for NWSOs. NWSFOs and NWSOs may choose products from

the preselected WSR-88D(s) for regular receipt or on-demand call-up, subject only to their communications bandwidth limitations. In addition, NWSFOs and NWSOs will be able to obtain a limited number of products from any other WSR-88D in the United States on a dial request basis.

3.1.1.3.1.2 NWS National and Regional Sites

In the case of NWS, regional facilities and national facilities (Defined in Appendix B) will be provided selected products based on their needs. For each RFC, precipitation related products will be made available from each WSR-88D within its area of responsibility. Each RFC may choose products from any WSR-88D within its area of responsibility via dial-up requests. For the NWS national centers (NCEP, the TPC, and SPC), relatively low resolution products will be routinely made available from each relevant WSR-88D. In addition, the national centers will be able to request higher resolution products from any WSR-88D within the conterminous United States when needed.

3.1.1.3.1.3 DoD Local Sites

In the case of DoD, operational personnel at BWSs and NOCDs will automatically be provided products from a preselected WSR-88D. The BWSs and NOCDs may choose any product from the preselected WSR-88D for regular receipt or on-demand call-up, subject only to their communications bandwidth limitations. In addition, BWSs and NOCDs will be able to obtain a limited number of products from any other WSR-88D on a request basis.

3.1.1.3.1.4 DoD National Facility and Regional Centers

The only DoD national facility which requires WSR-88D products is AFGWC. It, like the NWS national centers, will routinely have available low resolution products and will be able to request more detailed products from any WSR-88D. The Air Force Weather regional hubs will receive products from pre-selected WSR-88D Systems.

3.1.1.3.1.5 DOT Facilities

In the case of the FAA, the NEXRAD WSR-88D System will provide products to all ARTCCs. The Center Weather Service Unit (CWSU) in each ARTCC will be provided with selected products from every WSR-88D within, or with extensive coverage within, its area boundary. (However, the further processing of the data received at the CWSU and its distribution to controllers within the ARTCC and to personnel at other FAA operational facilities is the responsibility of the FAA.) If one or more WSR-88Ds provide coverage for each of a selected set of major airport terminals, certain base data products and other RPG-derived products will be sent from those RPGs to the FAA facility at that airport/terminal.

3.1.1.3.2 Other User Products

Other users will have access to selected products, according to policy established by the NEXRAD Program Management Committee (PMC).

3.1.1.4 Principal User Processing and Display

Each of the principal users of the WSR-88D System will provide product ingest, processing, and display capability according to their operational needs. This capability will be considered an external user system from the WSR-88D to be used by operational personnel for the selection, analysis, and further annotation and distribution of products.

3.1.1.5 Radar Siting

The WSR-88D radar (antenna) sites must meet the area and point coverage of the principal users. These sites are within the conterminous States; in Alaska, Hawaii, and Puerto Rico; and at selected overseas locations. A number of these sites are in the vicinity of selected airports to provide airport and terminal coverage. Some radar sites are collocated with a principal user site.

3.1.1.6 Support Facilities

The WSR-88D system support environment includes support facilities (Defined in Appendix B) to assure satisfactory equipment operation and depot maintenance; provide training for operational and maintenance personnel; and support development, test, and evaluation activities.

3.1.1.6.1 Maintenance and Supply Depots

A centralized maintenance facility, the National Reconditioning Center (NRC), and supply depot, the National Logistics Support Center (NLSC), supports all WSR-88D equipment. These depots contain WSR-88D hardware, including radar, data processing, and display and data entry equipment configured as at an operational site. In addition, other support hardware and software required for the maintenance function and related documentation are located at the NRC. Spare parts and consumable supplies are located at the NLSC.

3.1.1.6.2 Maintenance Training Facilities

WSR-88D maintenance training facilities are located at existing principal user's training sites. Each facility contains WSR-88D equipment, including radar, data processing, and display and data entry hardware, and appropriate test software, dedicated to the hardware maintenance training function.

3.1.1.6.3 Operational Training Facilities

WSR-88D system operational training facilities are located at existing principal users' training sites. Each training facility possesses the equipment necessary for operational training including a data playback capability.

3.1.1.6.4 Radar Operations Center

The WSR-88D Radar Operations Center (ROC) provides support to the WSR-88D System. This facility provides operational support, test and evaluation activities and life cycle support. The ROC also manages the development of hardware enhancements and retrofits, technical documentation, and software changes. To support test and evaluation activities, the ROC has data recording, data playback, and data analysis capabilities. Radar data sets, recorded in climatologically different geographical areas and under different meteorological conditions, are provided to the ROC to support test and evaluation of the data processing, distribution, and display portions of the WSR-88D System. To support maintenance, enhancement, and distribution of the WSR-88D System operational software, support software (such as assemblers, compilers, operating systems, test aids, and adaptation data generation software) is available. Support hardware (such as simulation equipment, data storage devices, and printers) is available to support the software maintenance and enhancement functions.

3.1.1.7 Functional Areas

An overview of the WSR-88D System functional areas, together with an identification of the functions of each area, is provided in Figure 3-1. Functional area requirements are presented in more detail in Section 3.7 of this document. The WSR-88D System Specification has been defined in terms of the functional areas listed below. Accordingly, a system design may be selected that allocates functions to equipment items and computer programs in a manner different than implied by the functional area breakdown, provided all of the system requirements are met. Specifically, conclusions should not be drawn, unless explicitly stated, regarding the relationship between a physical location (site) and the requirements of an individual functional area.

- a. System Requirements
- b. Radar Data Acquisition
- c. Radar Product Generation
- d. Facilities
- e. Support

FUNCTIONAL AREAS

SYSTEM	RADAR DATA ACQUISITION	RADAR PRODUCT GENERATION	FACILITIES	SUPPORT
Security	Antenna Control	Product Generation	Buildings	Off-Line Diagnostics
	Transmit	Product Storage	Towers	Support Software
	Receive	Product Distribution	Real Property Installed Equipment (RPIE)	Adaptation Data Maintenance
	Signal Processing	Display & Data Entry		Map Generation
	Ground Clutter Suppression	Control		Computer Performance Monitoring
	Control	Status Monitoring & Error Detection		Test Data Generation
	Monitoring & Error Detection			Test Software
	Display	Hydrometeorological Processing		Configuration Management
	Calibration	Base Data Ingest		
	Base Data Output	Base Data Distribution		

FIGURE 3-1 FUNCTIONAL AREA OVERVIEW

3.1.1.7.1 Radar Data Acquisition

The Radar Data Acquisition (RDA) functional area includes the Doppler radars which are located at the radar sites. This functional area also includes hardware and software required to perform the signal processing, clutter suppression, base data distribution, equipment and software control, status monitoring and error detection (both local and through the FAA Remote Monitoring System (RMS) (Defined in Appendix B.), display, and calibration functions related to radar operation. It also includes any hardware and software, including communications equipment, needed for wideband links that may be required for transmission of data between non-collocated RDA and RPG.

3.1.1.7.2 Radar Product Generation

The Radar Product Generation (RPG) functional area includes all hardware and software required for real time generation, storage, and distribution of products (to dedicated and dial up interfaces) for operational use. It also includes hardware and software required for base data ingest from the RDA, control, status monitoring and error detection (both local and through the FAA RMS) and hydrometeorological processing. To the extent required, this functional area includes display and data entry hardware and software provided for human participation in the generation of the products. It also includes any hardware and software, including communications equipment, needed for wideband links that may be required for transmission of data between non-collocated RDA and RPG.

3.1.1.7.3 Principal User Processing

This section is no longer applicable to this specification.

3.1.1.7.4 Facilities

The Facilities functional area includes buildings, towers, and Real Property Installed Equipment (RPIE) such as power generation and air conditioning equipment.

3.1.1.7.5 Support

The Support functional area includes an off-line diagnostics capability, support software, adaptation data maintenance, map generation, computer performance monitoring, test data generation, test software, and Configuration Management (CM). The support software includes capabilities required for the development, test, maintenance, and enhancement of the operational software.

The NEXRAD Integrated Logistics Support Plan (ILSP) outlines the logistics, maintenance depot equipment, spares, maintenance support equipment, and infrastructure to support the WSR-88D systems. The ILSP also describes and assigns responsibility for maintenance and operational training.

3.1.2 Missions

DOC, DOD, and DOT have major weather related missions and responsibilities that require weather radar information.

3.1.2.1 Department of Commerce Missions

DOC, through the National Weather Service (NWS), is the principal civilian meteorological agency of the Federal Government. Specifically, DOC is responsible for detection of hazardous weather such as severe thunderstorms, tornadoes, excessive rainfalls, and winter weather events; and warning the public. It is also responsible for providing essential weather information to other activities such as civilian aviation, marine, agricultural and forestry operations, and the entire river and flood prediction program. DOC operates many weather radars and uses information from some radars operated by DOD and DOT to meet these responsibilities.

3.1.2.2 Department of Defense Missions

Within DOD, the Air Force Weather Agency (AFWA) provides worldwide meteorological and aerospace environmental services to the Air Force, Army, and certain other DOD elements; the Naval Meteorological and Oceanography Command (NMOC) supports the Navy, Marine Corps, and some other elements of DOD. These organizations are responsible for providing and relaying severe weather warnings for the protection of DOD resources and personnel, providing weather information to aid the decision making process at specific locations, and supporting military aviation. To meet these responsibilities, DOD operates weather radars in the United States and overseas and uses information from these and from DOC and DOT WSR-88D systems in the United States.

3.1.2.3 Department of Transportation Missions

DOT, through its Federal Aviation Administration (FAA), is responsible for the safe and efficient utilization of United States airspace. In meeting these responsibilities, the FAA provides information on the location and intensity of potentially hazardous weather conditions to air traffic personnel, flight service specialists, airline dispatchers, pilots, and others concerned with aviation. There has been an increased emphasis on providing real time hazardous weather information. DOT has a number of programs planned to exploit the WSR-88D data. DOT presently obtains its information from other equipment, to include its own air traffic control radars (not designed for weather detection), FAA terminal and en route weather radars, DOD and NWS radars via remote displays, and from other FAA and NWS products, and NWS personnel located at FAA facilities.

3.1.2.4 System Mission

The mission of the WSR-88D System is the acquisition, processing, and distribution of weather radar information to aid in reducing loss of life, injuries, and damage to property. The radar system addresses the common needs among three Federal departments (DOC, DoD, and DOT). The WSR-88D System is expected to provide real-time information on the location, intensity, and movement of both routine and hazardous weather phenomena. These phenomena, although not mutually exclusive, are: precipitation, wind and wind shear, tornadoes, fronts and fine lines, tropical cyclones, mesocyclones, thunderstorms, turbulence, icing conditions, hail, and freezing/melting level. The WSR-88D System provides routine information useful for monitoring and assessing general weather conditions such as rainfall amounts and distribution. To meet the minimum objectives of the principal users, the WSR-88D System must acquire, process, and display data in a readily interpretable fashion. Although the WSR-88D System is intended to be a standalone system, it must also interface with certain existing principal user display and standard communications systems. This will be accomplished by providing a standard set of products over a standard communications interface.

3.1.3 System Diagrams

In addition to the functional area overview (Figure 3-1) already presented, the following diagrams are also presented:

a. Figure 3-2 shows four possible WSR-88D example configurations.

Example 1 consists of the DoD Collocated RDA and RPG and external interfaces for Distant MSCF (serial PPP or WAN) and product users (Classes, 1, 2, 98 & 99).

Example 2 consists of the NWS Remote RDA, the NWS RPG, and its external interfaces.

Example 3 consists of the NWS Remote Redundant RDA, the NWS RPG and its external interfaces.

The external interfaces to an NWS RPG in example 2 and 3 include a TCP/IP LAN connection to AWIPS; class 1, 2, 98, & 99 serial interfaces; and class 2 & 98 TCP/IP WAN interfaces to other AWIPS locations.

Examples 2 and 3 indicate that the RDA and RPG are physically separated by some distance that requires commercial/private telco service.

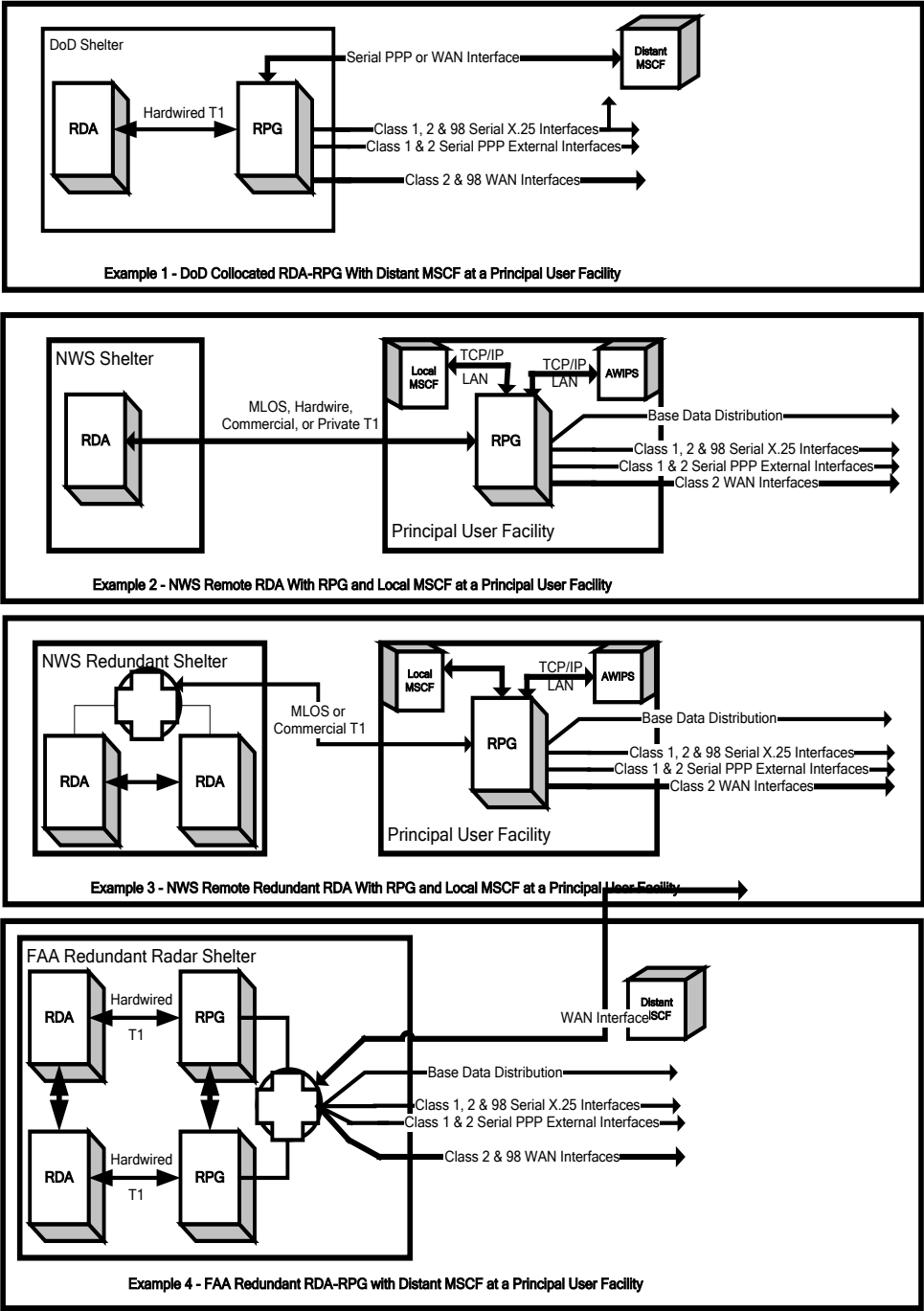
Example 4 consists of the FAA Remote Redundant RDA and RPG, and external interfaces for Base Data Distribution (option), Distant MSCF via WAN, and product users (Classes, 1, 2, 98 & 99).

In examples 1 and 4, the RDA(s) and respective RPG(s) are in physical proximity under 50 ft.

The tri-agency external user systems connected to the RPG in all 4 examples may be located in the same building facility as the RPG or at respective agencies remote locations.

b. Table 3-1 presents a NEXRAD principal user organization overview.

3.1.4 User Interfaces



Proposed New Fig 3-2, WSR-88D Example Configurations
Figure 3-2 WSR-88D EXAMPLE CONFIGURATIONS

TABLE 3-1 NEXRAD PRINCIPAL USER ORGANIZATION OVERVIEW

FACILITY	RELEVANT RESPONSIBILITY	NEED FOR/USE OF NEXRAD INFORMATION
• NCEP	Issue weather analysis and forecast guidance	Overview information for generation of radar summary products; detailed rainfall information; and high resolution data for mesoscale models.
• SPC	Issue severe convective storm and tornado watches	Low resolution information in storm areas.
• TPC	Issue hurricane watches and warnings	Low resolution information in hurricane areas.
• RFCs	Derive flood forecasts.	High resolution area/precipitation estimates covering river basins.
• NWSFOs/NWSOs	Issue severe weather, flood, and flash flood warnings for local area.	High resolution information on hazardous weather in or approaching local area.
DoD • BWSs	Issue weather warnings and advisories for AF base area.	High resolution information on hazardous weather in or approaching base area.
• AFGWC	Issue weather warnings and advisories where no BWS operating.	High resolution information on hazardous weather where no BWS operating; Low resolution overview information elsewhere.
• NOCDs	Relay weather warnings and advisories in Navy base/station area.	High resolution information on hazardous weather in or approaching base area
FAA • ARTCCs/CWSUs	Disseminate hazardous weather information internally and to other FAA users (e.g., ARTCC, TRACON, and Tower) in ARTCC area of responsibility.	High resolution on hazardous weather in or approaching ARTCC area of responsibility
• ATCSCC	Interface with each ARTCC to establish air traffic flow patterns, enhance flight safety, and minimize delays.	Low resolution overview information on a national basis.
• AFSSs	Provide hazardous weather information within air-to-ground coverage area; provide En route Flight Advisory Service for area of responsibility (if applicable); provide preflight briefings on a national basis.	High resolution information on hazardous weather in or approaching FSS area of responsibility; low resolution information on a national basis.
• TRACONS	Provide hazardous weather information to controllers and pilots.	High resolution current information on hazardous weather in terminal airspace (e.g., gust fronts, down drafts, wind shear, and turbulence), especially on final approach and take-off.

3.2 System Characteristics

This section contains the performance, physical, reliability, maintainability, availability, environmental, and transportability requirements for the WSR-88D System.

3.2.1 Performance Characteristics

This section contains the performance requirements. The WSR-88D can be operated manually or is capable of operation in a fully automatic mode. Any performance limitations associated with a fully automated operation are subject to NEXRAD Configuration Control Board (CCB) approval.

3.2.1.1 WSR-88D Full Load

The full load requirements of the Radar Data Acquisition, Radar Product Generation, and Principal User Processing functional areas are defined in this section. The full load defines the total number of functions and activities to be performed concurrently within each functional area while meeting all required response time and throughput requirements (Sections 3.2.1.3 and 3.2.1.4) without product generation load-shedding, loss of any radar data, or loss of control/status messages. This does not preclude load-shedding of products to be distributed if the capacity of the distribution interfaces is exceeded. The column labeled "Full Load" in Table 3-4 represents the system configuration that must meet the full load requirements in this section. The full load requirements are the following:

- a. The WSR-88D shall (1) perform the control and monitoring functions of the RDA functional area, as defined in Section 3.7.2
- b. The WSR-88D shall (2) monitor and transmit RDA performance data.
- c. In the volume scan time specified in paragraph 3.7.2.2.1.2, the WSR-88D shall (3) provide base data from the RDA to the RPG in accordance with the unique elevation angles described for each scan strategy.
- d. The WSR-88D shall (4) process RDA control commands from the RPG.
- e. The WSR-88D shall (5) monitor and control all activities performed by the RPG.
- f. The WSR-88D shall (6) generate the full load product mix defined in Table E-1a.
- g. The WSR-88D RPG shall (7) output products selected from the full products column of Table E-1a to drive four Class 1 serial interfaces, four Class 99 serial dedicated interfaces, three Class 99 WAN dedicated interfaces, and either one Class 98 WAN dedicated interface or one Class 98 LAN dedicated interface to their full capacity. Requirements for interface capacity are defined in Section 3.7.1.8.
- h. The WSR-88D RPG shall (8) output products selected from the products column of Table E-1a to four Class 2 serial and four Class 2 WAN interfaces. Requirements for interface capacity are defined in Section 3.7.1.8.
- i. The WSR-88D RPG shall (9) process a mix of peak requests (see Section 3.2.1.2) from four serial Class 1, four serial Class 99, three WAN Class 99, four WAN Class 2, and one LAN class 98 or WAN Class 98 users.

j. The WSR-88D shall (10) satisfy all processing requirements defined in Section 3.7.3 for the RPG functional area.

k. Each user shall (11) receive all requested products from the RPG interface(s), to which the user connects, in accordance with the corresponding WSR-88D Interface Control Documents (ICDs). Requirements for interface capacity are defined in Section 3.7.1.8.

l. Each Class 2 dial-in and WAN user shall (12) receive requested products selected from the products column of Table E-1a from an RPG in accordance with the applicable WSR-88D Interface Control Documents (ICDs).

m. The RPG shall (13) output the unaltered base data stream to four Base Data Distribution Interfaces.

n. The RPG shall (14) output the following RPG statistics for monitoring and troubleshooting:

- System uptime in seconds
- Percentage of time processors are idle
- System load average for previous 1 minute
- System load average for previous 5 minutes
- System load average for previous 15 minutes
- Total size of memory (RAM) in KBs
- Total size of unused (free) memory (RAM) in KBs
- Total size of supplemental memory (SWAP) in KBs
- Total size of unused (free) supplemental memory (SWAP) in KBs
- Total size of disk space in KBs
- Total size of unused (free) disk space in KBs

3.2.1.2 Peak Request Rates

The RPG shall (1) handle up to 4 product requests (3 existing products and 1 product requiring generation) per second.

3.2.1.3 Response Time Requirements

Table 3-3 summarizes the response time requirements for the WSR-88D system. All response times (except for Startup and Restart) shall (1) be met while the system is processing a full load as defined in Section 3.2.1.1.

System response time requirements shall (2) be met if 90% or greater of all measurements for a statistically significant sample of the response time parameter are equal to or less than the defined requirement. Requirements referencing Table 3-3 are in the following sections. Unless specifically stated, display functionality requirements do not apply to the RPG.

TABLE 3-3 SYSTEM RESPONSE TIME REQUIREMENTS

FUNCTION	RDA	RPG	Associated User/ RMS Interface
Status Message Output / Display	-	10 sec.	10 sec.
Alarm Output	2.5 sec.	2.5 sec.	-
Local Display of Alarm	5 sec.	5 sec.	-
Alert Message Output (Note 2)	-	2.5 sec.	-
Power Fail Recovery	-	1.5 min	-
Hard Copy Generation	-	2 min.	-
Cold Startup	15 min.	6 min.	-
Warm Startup	10 sec.	-	-
Restart	Note 1	1.5 min	-
Engine Generator On-line	25 Sec	-	-
FAA Redundant Channel Switchover	15 Sec	15 Sec	-

Note 1: The RDA Restart time is a function of outage time and the state of the RDA, but will in no case exceed the RDA Cold Startup time.

Note 2: Alert message output response time is from detection to transmission of the alert message to the RPG product user distribution queue. Delay in product distribution queue not included.

3.2.1.3.1 Deleted

This section is no longer applicable to this specification.

3.2.1.3.2 Off-Center/Preset Center

This section is no longer applicable to this specification.

3.2.1.3.3 Resolution Change

This section is no longer applicable to this specification.

3.2.1.3.4 Display Annotation

This section is no longer applicable to this specification.

3.2.1.3.5 Display of Cursor Coordinates

This section is no longer applicable to this specification.

3.2.1.3.6 Cursor Home

This section is no longer applicable to this specification.

3.2.1.3.7 Test Pattern Selection/Deselection

This section is no longer applicable to this specification.

3.2.1.3.8 Status Message Display

Display of Status Messages shall (1) be completed within the time indicated in Table 3-3 to include the time from generation of status message (within the functional area) or reception of status message from another functional area to completion of display. Transmission of status messages

from the RPG to dedicated users and the FAA RMS interface shall (2) be completed within the response time indicated in Table 3-3 to include the time from generation of status message (within the RPG) or reception of status message from the RDA, but not including delay in the communication queue.

3.2.1.3.9 Alarm/Alert Output/Display

The following alarm response times shall (1) be within the times indicated in Table 3-3:

- a. Time from the detection of an alarm condition (within the functional area) or reception of alarm from another functional area to the activation of a local alarm (display at the RDA position, the RPG position, and MSCF position).
- b. Time from the detection of an alarm condition or reception of alarm from another functional area to the transmission of the alarm to the appropriate functional area (e.g. RPG to Class 1 User or RDA/RPG to the FAA RMS interface). Delay in communication queue is not included.

The alert message response times shall (2) be within the times indicated in Table 3-3. All products automatically generated when alert criteria are met shall (3) meet the appropriate response times for one-time product requests as shown in Table E-1b.

3.2.1.3.10 Power Fail Recovery

When normal power levels have returned following a power interruption, the WSR-88D System shall (1) automatically resume normal operation (with all functions available to users and all previously displayed data displayed) within the response time required (see Table 3-3). The functional requirements for the power failure recovery capability, including limitations, are stated in Section 3.7.1.4 for the system requirements.

3.2.1.3.11 Magnification

This section is no longer applicable to this specification.

3.2.1.3.12 Time Lapse Display

This section is no longer applicable to this specification.

3.2.1.3.13 Master Cursor Function

This section is no longer applicable to this specification.

3.2.1.3.14 Hard Copy Generation

The time from the request for hard copy output until it is available shall (1) be within the response time required in Table 3-3.

3.2.1.3.15 Startup/Restart

The entire cold start transition shall (1) be capable of being completed within the time specified in Table 3-3.

The warm startup transition shall (2) finish within the response time specified in Table 3-3.

The restart transition shall (3) finish within the time specified in Table 3-3.

3.2.1.3.16 Engine Generator Online

In the event of primary power failure or manual selection of backup power, the backup engine generator shall (1) become operational and on-line within the response time required in Table 3-3.

3.2.1.3.17 FAA Redundant Channel Switchover

The redundant channel switchover shall (1) be capable of being performed by one person within the time specified in Table 3-3 with the response time measured from the initiation of the command (from the MSCF, RDA HCI or through the FAA RMS interface) until data transfer from the RDA to the RPG has begun.

3.2.1.4 Unit Throughput

Products generated on a routine basis by the RPG shall (1) be available to operational positions within the throughput time requirements defined in Table E-1b.

All throughput time measurements shall (2) be taken while the WSR-88D System is performing under a full load condition. The WSR-88D throughput time requirements will be met if 90% or greater of all measurements for a statistically significant sample of the throughput parameter are equal to or less than the defined requirement.

3.2.1.5 Configuration

The WSR-88D shall (1) provide the minimum operational configuration as indicated in Tables 3-4a and 3-4b for each functional area. Performance requirements for the Full Load configuration listed in the table are detailed in Section 3.2.1.1.

Functionally, the following configurations shall (2) be possible without modification (facilities excluded) of the WSR-88D equipment:

- a. Collocated RDA and RPG in radar shelter, MSCF at one of the Principal User Facilities, Principal Users' systems located with MSCF and at other locations
- b. RDA in radar shelter, RPG and MSCF at one of the Principal User Facilities, Principal Users' systems located with RPG/MSCF and at other locations
- c. Collocated redundant RDAs in radar shelter (Section 3.2.1.5.2); RPG and MSCF at one of the Principal User Facilities; Principal Users located with RPG/MSCF and at other remote locations
- d. Collocated redundant RDAs and RPGs in radar shelter (Section 3.2.1.5.1); MSCF at one of the Principal User Facilities; Principal Users' systems located in shelter and at other locations; RMS located in shelter

3.2.1.5.1 FAA Redundant Configuration

The driving principle behind the FAA redundant configuration is increased availability. A capability for a collocated RDA/RPG in a fully redundant configuration shall (1) be provided. All RDA and RPG functional area equipment, but excluding the antenna, pedestal, tower, telco service, and backup power generator, shall (2) be redundant.

The point for recombining the redundant channels shall (3) be before telco service access. A single environmentally controlled area shall (4) be provided to house both redundant channels. The goal of the equipment arrangement is to facilitate convenient and efficient access for maintenance activities. One Master System Control Function (MSCF) display/control hardware position shall (6) be provided in the FAA redundant configuration.

One RDA HCI display/control position shall (6a) be provided at each RDA in the redundant configuration. One RPG HCI display/control position shall (6b) be provided at each RPG in the redundant configuration. The back-up power generator and the air conditioning equipment shall (7)

be sized and configured to support one channel at full operation and the second channel at full power without high voltage being applied to the klystron.

3.2.1.5.2 NWS Redundant Configuration

A capability for a configuration comprised of a redundant RDA (see Section 3.2.5 for availability requirements) and a single-channel remote RPG which is collocated with the MSCF hardware position shall (1) be provided. All RDA functional area equipment excluding telco service (as applicable), wideband interface to telco, antenna, pedestal, tower, and backup power generator, shall (2) be redundant. The point for recombining the redundant channels shall (3) be the output for the wideband (telco service) interface.

A single, environmentally controlled area shall (4) be provided to house both redundant RDA channels. The goal of the equipment arrangement is to facilitate convenient and efficient access for maintenance activities. One RDA HCI and its associated display/control position shall (6b) be provided at each RDA in the redundant configuration.

A selection capability shall (7) be provided at the Remote RDA maintenance terminal to bring either channel to Standby or Operational, regardless of the state of the other channel.

The back-up power generator and the air conditioning equipment shall (8) be sized and configured to support this configuration.

TABLE 3-4a WSR-88D RDA CONFIGURATIONS

FUNCTIONAL AREA FEATURE	Operational Configurations	
	Minimum	Full Load Test
Base Data Interface to the RPG	1	1
Archive Level I Interface	1	0
Remote RDA Control Interface	1	1
FAA RMS Interface	0	1

Table 3-4b WSR-88D RPG CONFIGURATIONS

FUNCTIONAL AREA FEATURE	Operational Configurations	
	Minimum	Full Load Test
Base Data, Status, and Control Interface to the RDA	1	1
Dedicated Serial Port Class 1 or Class 99 Interfaces	0	8
Dial-in Serial Port Class 2 Interfaces	2	4
LAN Principal User (AWIPS) Product Interface or WAN Principal User (AWIPS) Product Interface	0	1
Supplemental WAN Principal User (AWIPS) Product Interfaces	0	3
Base Data Distribution LAN Interfaces	0	4
MSCF Interface Serial Dedicated, LAN, or WAN (as applicable)	1	1
Remote Monitoring System (RMS) Interface	0	1
Remote RPG Control Interface	0	4
RPG Statistics Requirements	0	1

TABLE 3-4c WSR-88D MSCF HARDWARE CONFIGURATIONS

FUNCTIONAL AREA FEATURE	Operational Configurations	
	Minimum	Full Load Test
RPG Interface Serial Dedicated, LAN, or WAN (as applicable)	1	1
Graphic/Alphanumeric Hardcopy Device	0	1
Total Class 1, 2 Ports that must be supported without expansion is a minimum of 8.		

3.2.1.6 Data Processing Reserve

The equipment performing the RPG functions when the hardware is refreshed shall (1) have a processing reserve such that when processing the full load requirements (defined in Section 3.2.1.1) the utilization of each of the processing, addressable memory (available to software not including the operating system), and direct access mass storage functions, and removable storage functions averaged over a volume scan time period, does not exceed the following:

RESOURCE	UTILIZATION
Processing	30%
Addressable memory	30%
Mass Storage	30%
Removable Storage	N/A (Note 1)

Note 1: The utilization averaged over a volume scan time period is not reasonable when referring to removable storage since the functions described in 3.2.1.8 typically do not occur in that time frame.

This reserve is based upon the anticipated increase, over the first several years of operation after the hardware refresh, of resource requirements as indicated in the Radar Product Generation Algorithm Survey and Analysis Report, June 1996.

The equipment performing the RDA functions when the hardware is refreshed shall (2) have a processing reserve such that when processing the full load requirements (defined in Section 3.2.1.1) the utilization of each of the processing, addressable memory (available to software not including the operating system), and direct access mass storage functions, averaged over a volume scan time period, does not exceed the following:

RESOURCE	UTILIZATION
Processing	30%
Addressable memory	30%
Mass Storage	30%

This reserve is based upon the anticipated increase, over the first several years of operation after the hardware refresh.

3.2.1.7 Expandability

WSR-88D Systems shall (1) be incrementally expandable by field modification. The expansion capability is in reference to the WSR-88D equipment to meet the full load, as defined in Section 3.2.1.1, including the reserve as defined in Section 3.2.1.6.

3.2.1.7.1 RPG Expandability

The RPG functional area shall (1) be expandable to at least the following:

- a. Processing Capacity: 3 times
- b. Addressable Memory: 3 times
- c. Direct Access Mass Storage Devices: 3 times
- d. Product User/Logical Port Interfaces: 44 Maximum
- e. Base Data Distribution Ports: maximum of 2 physical connections

3.2.1.7.2 PUP (Collocated and Non-Collocated) Expandability

This section is no longer applicable to this specification.

3.2.1.7.3 RDA Expandability

The RDA functional area using Legacy as a basis shall (1) be expandable to at least the following:

- a. Processing capacity: 10 times
- b. Addressable memory: 3 times
- c. Direct Access Storage devices: 3 times
- d. Communication Interfaces: 2 times
- e. Base data distribution interfaces: 2 times

3.2.1.8 Removable Storage

WSR-88D systems shall (1) provide multiple read /multiple write storage device(s) with removable media. The device(s) shall (2) be capable of archiving any one or all of the system administrative functions including backup of operating system, backup of application software, backup of all adaptation data, backup of security audits/logs, and system status/error/performance data. This same device may also be used to support other similar archive, logging, backup, or recording functions. The expansion capability is in reference to the WSR-88D equipment to meet the full load, as defined in Section 3.2.1.1, including the reserve as defined in Section 3.2.1.6.

3.2.1.8.1 RPG Expandability

The RPG functional area shall (1) be expandable to at least the following:

- a. Capacity of removable storage media: 5 times

3.2.1.8.2 RDA Expandability

The RDA functional area using Legacy as a basis shall (1) be expandable to at least the following:

- a. Capacity of removable storage media: 5 times

3.2.2 Physical Characteristics

Section 3.7.5 further defines physical characteristic limitations consistent with existing access capabilities. The WSR-88D shall (1) operate within the limits of the existing facilities.

3.2.3 Reliability

This section defines the system reliability requirements for the WSR-88D System using definitions prescribed by DOD 3235.1-H.

Reliability is defined as a characteristic of design and installation. This characteristic is expressed as the probability that a system will perform its required mission essential functions for a period of time when used under specified operating conditions. The WSR-88D system reliability for functions listed in Table 3-5 is expressed in terms of Mean Time Between Failure (MTBF) and is based on a duty cycle of continuous operation, i.e., 8,760 hours per system per year. The MTBF goals for the WSR-88D System which are derived from the tri-agency Operational Availability requirements are stated in Paragraph 3.2.5.1.

By definition, MTBF is calculated using the following formula:

$$\text{MTBF} = \frac{\text{Total Operating Time}}{\text{Number of Failures}}$$

3.2.4 Maintainability

This section defines the system maintainability requirements for the WSR-88D System.

Maintainability is defined as a characteristic of design and installation. This characteristic is expressed as the probability that an item will be retained in, or restored to, a specified condition within a given period of time if prescribed procedures and resources are used. From the definition of maintainability it follows that the ability and need to perform maintenance actions is the underlying consideration in assessing maintainability. Maintenance is defined as all actions required to retain a system in, or restore it to, a specified condition. The factors which affect the frequency with which maintenance is required are reliability (frequency of corrective maintenance) and the preventive maintenance schedule.

3.2.4.1 Corrective Maintenance

Corrective Maintenance is defined as the maintenance performed on a non-scheduled basis to restore the system to a fully operational condition by correcting a malfunction.

The WSR-88D System shall (1) have a Mean Time To Repair (MTTR) of 2.5 hours or less. The MTTR is defined as the total elapsed time (clock hours) for corrective maintenance divided by the total number of corrective maintenance actions during a given period of time. MTTR is expressed in terms of the statistical mean of the distribution of repair times for each replaceable component in the WSR-88D System or hardware Configuration Item (CI).

The times that are included in the calculation of MTTR are as follows:

- a. Localization - The time required to locate a failure without using support equipment.
- b. Preparation - The time to obtain, set up, interconnect, and warm up support equipment and/or tools.
- c. Isolation - The time required to locate a failure using support equipment.
- d. Disassembly - The time required to disassemble equipment to gain access to the Line Replaceable Unit (LRU) that is to be replaced.
- e. Interchange - The time required to remove the defective LRU and install the replacement.
- f. Reassembly - The time required to close and reassemble the equipment after the LRU replacement has been made.

g. Alignment - The time required to perform alignment and/or adjustment made necessary by the repair action.

h. Checkout - The time required to perform those minimum checks necessary to verify correction and equipment serviceability.

The Maximum Time To Repair (MAXTTR) is the maximum corrective maintenance downtime within which 90% of all corrective maintenance actions can be accomplished. *This requires that no more than 10% of all corrective maintenance actions take longer than MAXTTR.*

For the WSR-88D System the following MAXTTR limits shall (2) be imposed on the functional areas identified:

MSCF	4.0 hours
RDA	10.0 hours
RPG	5.5 hours
TPS	6.0 hours

3.2.4.1.1 Functional Area Mean-Time-To-Repair (MTTR)

Based on the System MTTR of 2.5 hours, the following MTTR values shall (1) apply to the WSR-88D Functional areas:

MSCF	1.75 hours
RDA	2.81 hours
RPG	1.8 hours
TPS	3.0 hours

These values shall (2) be used in calculating reliability, maintainability and availability at the Functional Area level.

3.2.4.2 Preventive Maintenance

Preventive Maintenance is defined as systematic inspection, detection and correction of incipient failures either before they occur or before they develop into major defects. Adjustment, lubrication, and scheduled checks are included in the definition of preventive maintenance. The WSR-88D System Total Preventive Maintenance (TPM) time which requires the system to be placed in such a state as to cause the radar to be unable to perform its mission essential functions shall (1) not exceed 23 hours each year.

3.2.5 Availability

Availability is:

System Up Time

System Total Time (Up Time + Down Time)

Actual assessment of availability is accomplished by substituting time-based elements defined below into variations of this basic equation. Different combinations of these elements combine to formulate different definitions of availability which are used to assess support effectiveness.

The elements of Availability applicable to the WSR-88D System are:

TT = Total Time = 8,760 hours per system per calendar year

OT = Operating Time per system per calendar year = 8,760 hours per calendar year

ST = Standby Time = 0 hours

TPM = Total Preventive Maintenance down time per system per calendar year:

FAA Redundancy:	67.61 hours
NWS Single Channel:	57.02 hours
DOD Single Channel:	57.02 hours
NWS Redundancy:	73.21 hours

TCM = Total Corrective Maintenance down time per system per calendar year =

$$\frac{TT \times MTTR}{MTBF}$$

ALDT = Administrative Down Time (ADT) and Logistics Down Time (LDT) spent waiting for parts, maintenance personnel, or transportation per system per calendar year where:

FAA LDT =	72 Hours
NWS LDT =	24 Hours
DOD LDT =	24 Hours

ADT = 1 hour average response time per Corrective Maintenance action times the number of Corrective Maintenance actions=

$$\frac{1 \text{ hr} \times TT}{MTBF}$$

LDT = Agency average response time for obtaining the required parts not already on site assuming a sparing level that ensures having a replacement part already on site and available for 95% of Corrective Maintenance actions time the number of Corrective Maintenance actions =

$$\frac{\text{Agency LDT hrs} \times .05 \times TT}{MTBF}$$

There are four types of availability stated for the WSR-88D System. All are derived from Operational Availability as it covers all segments of time that the system is intended to be operational. These four types of availability are Operational Availability, Service Availability, Inherent Availability, and Achieved Availability. Operational Availability or the tri-agency defined Service Availability is the most desirable form of availability to be used in assessing a system's mission readiness. Achieved and to a lesser extent Inherent Availability are primarily the concern of the system's design activity in its interface with the contractor or other co-developing agencies.

3.2.5.1 Operational Availability (A_o)

A_o is an important measure of system effectiveness because it relates system hardware, support, and support characteristics into one meaningful parameter - a figure of merit depicting the system's state at the start of its mission. Because it is an effectiveness-related index, A_o or some tailored variation of it is used as a starting point for nearly all effectiveness analyses. The WSR-88D tri-agencies by agreement utilize a variation which excludes TPM called Service Availability to measure WSR-88D operational performance. Service Availability goals are calculated for the WSR-88D in paragraph 3.2.5.2.

A_o covers all segments of time that the system is intended to be operational. A_o is used to support design goals, operational testing assessment, and life cycle costing. As a stated requirement of the tri-agencies, the WSR-88D System shall (1) have an A_o of at least 0.960 for system functions listed in Table 3-5, and for redundant DOT WSR-88D Systems the A_o shall (2) be at least 0.989.

The formula for calculation of A_o is:

$$A_o = \frac{OT + ST}{OT + ST + TPM + TCM + ALDT}$$

Given the elements of Availability applicable to the WSR-88D System as stated in Paragraph 3.2.5, the A_o formula can be used to derive the MTBF requirements which must be achieved by the system's design as follows:

A. DOT System MTBF is calculated as follows:

$$System\ MTBF = A_o + [8760 - 67.61\ hrs - (TT * 2.5\ hrs) / MTBF - (TT * 1\ hr) / MTBF - (TT * .05\ hrs * 72\ hrs) / MTBF] \div 8760$$

$$System\ MTBF = 7.1 / (.9923 - A_o)$$

Substituting the stated A_o requirements of .989 into the equation, just derived, the WSR-88D System MTBF for DOT Systems shall (1) be at least 2151.51 hours

B. DOD System MTBF is calculated as follows:

$$System\ MTBF = A_o + [8760 - 57.02\ hrs - (TT * 2.5\ hrs) / MTBF - (TT * 1\ hr) / MTBF (8760 * .05 * 24\ hrs) / MTBF] \div 8760$$

$$System\ MTBF = 4.7 / (.9935 - A_o)$$

Substituting the stated A_o requirements of .960 into the equation, just derived, the WSR-88D DOD System MTBF shall (2) be at least 140.29 Hours.

C. NWS System MTBF for single thread System is calculated as follows:

$$\text{System MTBF} = A_o + [8760 - 57.02 \text{ hrs} - (TT * 2.5 \text{ hrs}) / \text{MTBF} - (TT * 1 \text{ hr}) / \text{MTBF} (8760 * .05 * 24 \text{ hrs}) / \text{MTBF}] \div 8760$$

$$\text{System MTBF} = 4.7 / (.9935 - A_o)$$

Substituting the stated A_o requirements of .960 into the equation, just derived, the WSR-88D NWS Single Thread System MTBF shall (2) be at least 140.29 Hours.

D. NWS System MTBF for Redundant Systems is calculated as follows:

$$\text{System MTBF} = A_o + [8760 - 57.02 \text{ hrs} - (TT * 2.5 \text{ hrs}) / \text{MTBF} - (TT * 1 \text{ hr}) / \text{MTBF} (8760 * .05 * 24 \text{ hrs}) / \text{MTBF}] \div 8760$$

$$\text{System MTBF} = 4.7 / (.99318 - A_o)$$

Substituting the stated A_o requirements of .960 into the equation, just derived, the WSR-88D NWS Redundant Systems shall (4) be at least 135.84 hrs.

3.2.5.2 Service Availability (A_s)

The WSR-88D tri-agencies by agreement utilize a variation of Operational Availability which excludes TPM. It is called Service Availability (A_s) and is used to measure WSR-88D operational performance, i.e. mission effectiveness.

Given the elements of Availability applicable to the WSR-88D System as stated in Paragraph 3.2.5, the A_s formula is:

$$A_s = \frac{8760 - \text{TCM} - \text{ALDT}}{8760}$$

The A_s for each agency is as follows:

DOT Service Availability: The Service Availability for the DOT Systems Shall be (1) at least .9967

NWS Redundant Service Availability: The NWS Redundant Service Availability Shall be (2) at least .9964

NWS and DOD Single Thread Service Availability: The NWS and DOD Single thread Systems Service Availability shall be (3) at least .9964

Substituting the WSR-88D System's MTBF requirements into the equation, the WSR-88D A_s for an MTBF of **125.7** hours is **0.9626**, and the A_s for an MTBF of **559.5** hours is **0.992**.

3.2.5.3 Inherent Availability (A_i)

A_i is defined as the probability that, when used under stated conditions in an ideal support environment without consideration for preventive maintenance actions, a system will provide the functional area capabilities stated in Table 3-5. Each functional area must provide all of the capabilities listed in the corresponding section of the Table in order to be classified as available for mission performance. An ideal support environment exists when the stipulated tools, parts, skilled

manpower, manuals, support equipment and other support items required are available. Inherent Availability excludes whatever preventive maintenance down time, supply down time, and administrative down time may be required. A_i is useful in determining basic system operational characteristics under conditions which might include testing in a contractor facility or other controlled test environment. Since this definition of availability is easily measured, it is frequently used as a contract-specified requirement.

The (A_i) formula is:

$$A_i = (TT - TCM) \div OT$$

Where TT = 8760 hours; TCM=(TT x MTTR) /MTBF; MTTR= 2.5 Hours; and OT = 8760 hours.
MTBF

$$A_i = [8760 - (8760 * 2.5 \text{ hrs} \div MTBF)] \div 8760$$

The A_i for each agency is as follows:

DOT Inherent Availability: The inherent Availability for DOT Systems shall be (1) at least .9995.

NWS Redundancy Thread Inherent Availability: The Inherent Availability for the NWS Redundant Systems shall be (2) at least .9815

NWS and DOD Single Thread Inherent Availability: The Inherent Availability for NWS and DOD Single Thread Systems shall be (3) at least .9821

WSR-88D System shall (1) have an A_i of at least 0.9801 for system functions listed in Table 3-5, and for redundant DOT WSR-88D Systems the A_i shall (2) be at least 0.9955.

The formula for calculating A_i for a CI is expressed as:

$A(i) = f(\text{all } A(j))$, where: $A(j)$ is the availability of the “jth” CI and is given by:

$$A(j) = MTBF(j) \div [MTBF(j) + MTTR(j)]$$

Where:

MTBF (j) = the statistical mean of the summation of MTBF values for each LRU contained in the CI.
MTTR (j) is the statistical mean of the summation of MTTR values for each LRU in the CI.

Simplified:

$$MTTR = [(1-A_i)/A_i] * MTBF$$

3.2.5.4 Achieved Availability (A_a)

A_a is defined as the probability that, when used under stated conditions in an ideal support environment without consideration of administrative and logistics down times, a system will provide the functional area capabilities stated in Table 3-5. A_a differs from A_i only in its inclusion of consideration for preventive maintenance actions. A_a excludes logistics down time and administrative down time.

$$A_a = (TT - TPM - TCM) \div OT$$

Where TT= 8760 Hrs; TPM = 23 hours;

$$TT = (TT \times MTTR) \div MTBF \quad \text{and OT= 8760 hours.}$$

$$A_a = [8760 - 23 - (8760 \times 2.5 \div MTBF)] \div 8760$$

The A_a for each agency is as follows:

DOT Redundancy Achieved Availability: The Achieved Availability for DOT Systems shall be (1) at least .9911.

NWS Redundancy Achieved Availability: The Achieved Availability for the NWS Redundant Systems shall be (2) at least .9732.

NWS and DOD Single Thread Achieved Availability: The Achieved Availability for the NWS and DOD Single Thread Systems shall be (3) at least .9757.

The WSR-88D System shall (1) have an A_a of at least 0.9774 for system functions listed in Table 3-5, and for redundant DOT WSR-88D Systems the A_a shall (2) be 0.9929 or higher. Any corrective and preventive maintenance that can be accomplished while redundant components are functional and the system is operationally capable is not assessed against A_a .

TABLE 3-5 FUNCTIONALITY FOR SYSTEM AVAILABILITY REQUIREMENTS

RDA Functional Area	
Transmit/Receive	a. Control Antenna b. Generate Horizontal (H) and Vertical (V) polarized RF pulses c. Receive H and V echo returns d. Separate H and V RF and IF data
Signal Processing (Reflectivity)	a. Provide IF analog-to-digital conversion for Base data generation. b. Generate digitized reflectivity estimates c. Clutter filtering
Signal Processing (velocity and spectrum width)	a. Generate digitized mean radial velocity and spectrum width estimates b. Clutter filtering
Signal Processing (polarimetric variables)	a. Generate digitized differential reflectivity, correlation coefficient and differential phase estimates from digitized IF data
RPG Functional Area	
Processing I	a. Base product generation b. Base product distribution
Processing II & III	a. Signature recognition b. Tracking c. Product generation (other than base products) d. On-line product storage e. Product distribution (other than base products) f. Control/status monitoring g. Accept request/reply commands

3.2.6 Environmental Conditions

The WSR-88D System must be capable of operating in and withstanding the environmental conditions specified in the following sections. The term operating criteria defines the conditions under which the system must operate and meet all functional and performance specifications. Non-operating criteria (including transportation and storage) define conditions which the system will withstand without damage.

3.2.6.1 Temperature and Humidity

All equipment within the shelter or user agency facility shall (1) meet the indoor environment temperature and humidity requirements for operating and non-operating conditions (Table 3-6).

TABLE 3-6 TEMPERATURE AND HUMIDITY
OUTDOOR ENVIRONMENT

CONDITION	TEMPERATURE		HUMIDITY
	Minimum	Maximum	
Operating	-40°C	+49°C	15 - 95%
Non-operating	-62°C	+60°C	15 - 95%

INDOOR ENVIRONMENT

CONDITION	TEMPERATURE		HUMIDITY
	Minimum	Maximum	
Operating	+10°C	+35°C	20 - 80%
Non-operating	-35°C	+60°C	15 - 95%

All equipment within the radome shall (2) meet the outdoor environment temperature and humidity requirements for operating and Non-operating conditions (Table 3-6). For internal radome air temperatures above +25 degrees C, forced air ventilation shall (3) be provided to keep the radome interior within 5 degrees C of the exterior ambient air temperature.

3.2.6.2 Pressure and Altitude

Equipment located at a radar site shall (1) be capable of operating at altitudes up to 3300 meters above mean sea level. Equipment located at a user site shall (2) be capable of operating at altitudes up to 2100 meters above mean sea level (MSL).

3.2.6.3 Solar Radiation

Exposed (Exposed equipment is that equipment not normally located in a controlled environment, including radomes, towers, shelters, etc.) WSR-88D equipment shall (1) be capable of withstanding without failure the following intensities of ultraviolet, visible, and infrared radiation:

TABLE 3-7 RADIATION INTENSITIES

Portion	Wavelength± (Microns)	Intensities (Wm ⁻²)[(Wft ⁻²)]	Percent of Total
Ultraviolet	< 0.38	75.35 [7] or less	7 or less
Visible	0.38 - 0.78	538.20 [50] or less	48 or less
Infrared	> 0.78	775.0 [72] or less	69 or less
Total Intensity = 1,119 ± 10.76 Wm ⁻² [104 ± 1 Wft ⁻²]			

(Wm⁻²) = Watts per square meter
Wft⁻²= Watts per square foot

3.2.6.4 Rainfall

Exposed WSR-88D equipment shall (1) be operable and sustain no physical damage or significant degradation in performance when subjected to wind and rain under the following conditions:

a. Operational An instantaneous rain rate equivalent to 300 mm per hour with a maximum wind speed of 18 ms

b. Non-operational

(1)A 1-hour average rain rate of 130 mm per hour (with an instantaneous rate of 400 mm per hour) with a wind speed of 33 ms,

(2)A 12-hour average rain rate of 30 mm per hour with a maximum wind speed of 26 ms, and

(3)A 24-hour average rain rate of 18 mm per hour with a maximum wind speed of 21 ms

3.2.6.5 Fungus

Exposed WSR-88D equipment shall (1) be compliant with Guideline 4 (Fungus-Inert Materials) of MIL-HDBK-454.

3.2.6.6 Salt Fog

Exposed WSR-88D equipment shall (1) be capable of continuous operation in a salt laden atmosphere without degradation of material surfaces or performance.

3.2.6.7 Sand and Dust

All system capabilities shall (1) be met when exposed WSR-88D equipment has been subjected to the effects of blowing fine sand and dust particles with wind speeds up to 18 ms, and particle concentration of 0.177 grams per cubic meter with particle sizes up to 150 micrometers in diameter.

3.2.6.8 Wind

The RDA shall (1) provide a pointing accuracy in both azimuth and elevation of $\pm 1/3$ degrees rms in a steady wind of 25 ms, and ± 1 degree rms in a steady wind of 50 ms. Exposed WSR-88D equipment shall (2) be capable of withstanding, without physical damage for both operational and Non-operational conditions, a steady wind or gust up to 60 ms. All wind speeds of Section 3.2.6 are measured at the maximum height of the tower.

3.2.6.9 Snow and Ice Load

Exposed WSR-88D Systems shall (1) sustain no physical damage when subjected to 235 kilograms per square meter of snow and ice loading.

3.2.6.10 Lightning (Transient Susceptibility)

All WSR-88D equipment shall (1) be compliant with IEEE Recommended Practice on surge Voltages in Low-Voltage AC Power Circuits.

3.2.6.11 Lightning Protection

The lightning protection devices at FAA WSR-88D sites shall (1) be installed in accordance with NFPA 780-2011 (Standard for the Installation of Lightning Protection Systems) established "Rolling Sphere Concept" and FAA-STD-019b (Lightning Protection, Grounding, Bonding and Shielding Requirements for Facilities).

3.2.6.12 Electromagnetic Compatibility (EMC), Electromagnetic Interference (EMI) and Radio Frequency Interference (RFI)

For Developed Items, the WSR-88D System shall (1) comply with all recommended general requirements (Part 1) and all recommended specific requirements as Class A3 (USAF requirements), as covered in MIL-STD-461, as detailed in sections 3.2.6.11.1 through 3.2.6.11.4. For COTS equipment, the WSR-88D System shall (2) comply with FCC, Part 15, Class A emission standard.

3.2.6.12.1 Emissions (Conducted)

The emissions (conducted) requirements for developed items are as follows:

TABLE 3-8 EMISSION (CONDUCTED)

	Applicable Specification	Range of Requirements
A.C Power (50/60HZ)	CE03	15 KHz - 50 MHZ
	CE07	±50% of nom. RMS volt
D.C Power & Control Lines	CE03	15 KHz - 50 MHZ
	CE07	+50%, -150% of nom. volt
Antenna Terminals		
Receivers	CE06	(NB) 34 dBmV (BB) 40 dBmV/MHZ
Transmitters (Key up)	CE06	(NB) 34 dBmV (BB) 40 dBmV/MHZ
Transmitters (Key down)	CE06	80 dB down from f ₀

3.2.6.12.2 Emissions (Radiated)

The emissions (radiated) requirements for developed items are as follows:

TABLE 3-9 EMISSION (RADIATED)

	Applicable Specification	Range of Requirements
All WSR-88D Radar and RDA Equipment	RE02	(NB) 14 KHz - 10 GHz (BB) 14 KHz - 1 GHz

3.2.6.12.3 Susceptibility (Conducted)

The susceptibility (conducted) requirements for developed items are as follows:

TABLE 3-10 SUSCEPTIBILITY (CONDUCTED)

	Applicable Specification	Range of Requirements
All Power lines	CS101	30 Hz - 50 KHZ
	CS102	50 KHz - 400 MHZ
	CS106	Spikes & Transient
Antenna Terminals	CS103	Intermodulation, 2-signal
	CS104	Residual of undesired signals
	CS105	Cross Modulation
	CS107	Squelch (if any) Test 1 90 db/mv/MHZ Signal-Signal method Test 2 2-Signal Method

3.2.6.12.4 Susceptibility (Radiated)

The susceptibility (radiated) requirements for developed items are as follows:

TABLE 3-11 SUSCEPTIBILITY (RADIATED)

	Applicable Specification	Range of Requirements
Spikes	RS102	Spike Transient
Power Frequency	RS102	Inductive Transient
Electrical Field	RS103	14 KHz - 10 GHz

3.2.6.12.5 Reference Test Data

For developed items, Magnetic field radiated emissions, RE 101, and magnetic field radiated susceptibility, RS 101, tests shall (1) be performed in accordance with MIL-STD-461.

3.2.7 Transportability

All items of equipment comprising the WSR-88D System shall (1) be capable of being transported via common surface, air, or sea carrier in its packed-for-shipment configuration, excluding the WSR-88D radar antenna. Size, weight, and physical characteristics shall (2) not exceed the limits imposed for commercial equipment by any particular common carrier available. Transportation methods for the WSR-88D radar antenna shall (3) be subject to government approval on a case by case basis.

3.3 Design and Construction Requirements

The design and construction of the WSR-88D System will be accomplished in accordance with the requirements identified below:

- a. For new equipment designs, the general equipment requirements contained or referenced in MIL-HDBK-454 for the design (except for human engineering, maintainability, reliability, climatic service conditions, electronic countermeasure vulnerability, and encapsulation and embedment); parts; materials; processes; nomenclature, identification marking, and labels; and workmanship shall (1) apply.
- b. For Commercial-Off-the-Shelf (COTS) equipment, applicable commercial standards such as ASTM, ANSI, shall (2) apply. MIL-HDBK-2036 may also be used for guidance.

c. In addition, the following Standards shall (2a) apply to FAA facilities:

1. 29CFR 1910, Occupational safety and Health Standards (General Industry)
2. 29CFR 1926, Safety and Health Regulations for Construction
3. National Fire Protection Association (NFPA) 70, National Electric Code
4. NFPA 70E Standard for Electrical Safety in the Workplace.
5. FAA Order 3900.19B, FAA Occupational Safety and Health Program
6. FAA Standard HF-STD-001, Human Factors Design Standard

The requirements in this section should not preclude consideration of Non-Developmental Items (NDI), including COTS. Deviations from these requirements to permit an economical use of NDI shall (3) be subject to approval on a case by case basis, evaluating the risk of non-compliance.

3.3.1 Materials, Processes, and Parts

The selection of materials, processes, and parts to be used in WSR-88D System equipment will conform to the requirements specified in the following sections. Consistent with the limitations imposed by COTS equipment usage, at least two sources shall (1) exist for all parts used in the WSR-88D System.

3.3.1.1 Trade Practices and Non-specified Processes and Materials

Where the design, quality, or function of the system or any part thereof is not controlled by the requirements of this document, such design, quality, or function shall (1) comply with applicable ASTM and National Electrical Code standards and requirements.

3.3.1.2 Selection of Electrical and Mechanical Parts

The selection of electrical and mechanical parts will satisfy the following requirements:

- a. The specified performance of the system shall (1) be obtained without the special selection of any part(s).
- b. The system shall (2) comply with this document when any part meeting the part manufacturer's specification is used.
- c. Parts whose characteristics are more restrictive than those specified by the part manufacturer shall (3) not be used.
- d. The use of parts shall (4) not be dependent upon characteristics not specified by the part(s) manufacturer.
- e. Specified system performance shall (5) be obtained by installing acceptable parts at random as received from the part manufacturer without subsequent replacement or interchanging thereof.

3.3.1.3 Part Limits and Ratings (Electrical and Mechanical)

The part manufacturer's specifications and limits for any part shall (1) not be exceeded when the system is subjected to and operated under the conditions specified in Section 3.2.6 and in accordance with all other requirements of this document. Derating of parts shall (2) be accomplished as necessary to assure the required reliability within the specified operating conditions.

3.3.1.4 Mechanical Design

Unless basic component size and/or maintenance accessibility prohibit, all major equipment used in the WSR-88D System (e.g., computers and storage devices) shall (1) be designed for a standard 19 inch rack mounting. All parts shall (2) be securely mounted and attached and shall (3) withstand the rigors of transportation, operation, and maintenance without damage.

The WSR-88D System design will facilitate ease of maintenance. Specifically, the system design:

- a. shall (4) allow localization of malfunctions to a defective LRU without disassembly or removal (extender boards allowed).
- b. shall (5) allow routine operation or maintenance without the removal of structural members.
- c. shall (6) allow direct access to LRUs, test points, and maintenance adjustments without disassembly or removal (extender boards allowed).

3.3.1.5 Toxic, Hazardous Substances, and Ozone Depleting Chemicals

For FAA Facilities, the Equipment installed in the WSR-88D System Shall (1) comply with occupational safety and health requirements in 29 CFR 1910, Subpart H (Hazardous Materials), Subpart I (Personal Protective Equipment) and Subpart J (General Environmental Controls) and the environmental requirements in 40 CFR, Chapter I, Subpart I (Solid Waste).

The use of Hazardous materials in the design, manufacture, installation, and operation of equipment at FAA Facilities shall (2) be minimized to the extent possible, in accordance with Executive Order 13423, Strengthening Federal Environment, Energy, and Transportation Management, and Executive Order 13513, Federal Leadership in Environmental, Energy, and Economic Performance.

A Reutilization and Disposition Plan shall (3) identify handling, storage, and disposal requirements for any hazardous materials in the equipment in accordance with the Resource Conservation and Recovery Act (RCRA) and FAA Order 4600.27, Personal Property Management.

In addition, any hazardous material used in program activities will include a Material Safety Data Sheet (MSDS) which will be furnished with delivery of the equipment and will be made available to personnel in the work area in accordance with 29CFR 1910.1200, Hazardous Communications.

Equipment will be delivered free of Class I Ozone-depleting substances (ODSs) HCFC-22, HCFC-141b, and HCFH-142b, in accordance with 40 CFR82, Protection of Stratospheric Ozone.

See Section 3.7.5.3, Environment, for requirements.

3.3.1.6 Recycled, Recovered, or Environmentally Preferable Materials

Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. See Section 3.7.5.3, Environment, for requirements.

3.3.1.7 Use of Corrosion Resistant Parts

Parts and materials used throughout the WSR-88D equipment should be resistant to corrosion to prevent deterioration through the equipment life cycle with minimum maintenance.

3.3.2 Electromagnetic Radiation

System requirements for electromagnetic radiation in Section 3.2.6.12 and Section 3.7.5 and its subsections will be satisfied.

3.3.3 Nameplates and Product Markings

All WSR-88D equipment repairable items shall (1) have a unique serial number. Additionally, nameplates shall (2) be required for major WSR-88D configuration items as specified in Drawing #1213851, Plate, Identification, updated for any changes in manufacturers identified. LRUs shall (3) contain reference designators in accordance with ANSI/ IEEE Standard Y32.16.

3.3.4 Workmanship

The minimum standard of workmanship to be applied during fabrication of non-COTS equipment shall (1) be compliant with Guideline 9 of MIL-HDBK-454. MIL-HDBK-2036 may be used as a guideline for NDI including COTS.

3.3.5 Interchangeability

Each type of LRU shall (1) be physically, mechanically, structurally, and electrically interchangeable in and between all WSR-88D equipment procured under this document without modification and without causing any unit to deviate from the requirements of this document.

3.3.6 Safety

The WSR-88D System shall (1) be compliant in all respects with OSHA Safety and Health Standards (29 CFR 1910), Occupational Safety and Health Standards (General Industry) and 29CFR 1926, Safety and Health Regulations for Construction, and with American National Standards Institute Standard ANSI C95.1 (2005).

FAA Facilities shall also (1a) be compliant with FAA Order 3900.19B, FAA Occupational and Health Program, FAA Standard HF-STD-001, Human Factors Design Standard, National Fire Protection Association (NFPA) 70, National Electric Code, NFPA 70E Electrical Safety in the Workplace and relevant fire codes and building codes.

In addition, the following requirements of MIL-HDBK-454 will be satisfied. Specifically, the WSR-88D system:

- a. shall (2) comply with Guideline 1 for Safety (Personnel Hazard).
- b. shall (3) comply with Guideline 3 for Flammability.
- c. shall (4) comply with Guideline 45 for Corona and Electrical Breakdown Prevention.

3.3.7 Human Performance/Human Engineering

The WSR-88D System equipment will be designed to ensure human factors principles are applied for those aspects of the equipment requiring both operator and maintenance personnel interface. The objectives of the operator and maintainer require different aspects of human factors principles to be applied. For the WSR-88D maintainer, the requirements of Section 5.9 of MIL-STD-1472D shall (1) apply to the WSR-88D developed item equipment design. For the WSR-88D operator, the requirements of Section 5.15 of MIL-STD-1472D shall (2) apply for the User-computer interface characteristics associated with WSR-88D developed item equipment design.

3.3.7.1 Acoustic Noise Requirements

Acoustic noise requirements for equipment accessed by maintenance personnel only and for equipment in areas occupied by operational personnel are defined in the following sections. FAA maintenance and operations personnel will minimize their noise exposure in compliance with the guidance listed in FAA Order 3900.19B, FAA Occupational Safety and Health Program.

3.3.7.1.1 Equipment Accessed by Maintenance Personnel Only

The acoustic noise level of WSR-88D equipment which is normally accessed by maintenance personnel only shall (1) not exceed condition A of the tabulation of noise limits defined in Section 3.3.7.1.3. Acoustic noise level requirements apply to operating free-standing equipment, with all internal fans necessary for proper air flow and cooling operating normally. Acoustic noise level measurements shall (2) be performed on operating equipment. The measurements shall (3) be taken a distance of 1.0 meters from the cabinet sides and 1.5 meters up from the floor.

Measurements shall (4) be taken on the air inlet and air exhaust sides of cabinets wherever air is drawn in or exhausted via the sides of cabinets. Measurements shall (5) be performed in an environment where ambient noise levels do not constitute more than 5 percent of the noise generated by the equipment at any frequency.

3.3.7.1.2 Equipment in Areas Occupied by Operational Personnel

The acoustic noise level of WSR-88D equipment installed in Principal User Facility areas occupied by operational personnel shall (1) not exceed condition B of the tabulation of noise limits defined below. This shall (2) include all RPG functional area equipment and all MSCF functional area equipment. Optional noise suppression devices (acoustic panels, baffles, etc.) may be used to satisfy this requirement. Individual measurement sets may be used for the RPG and MSCF. Acoustic noise level requirements apply to operating equipment with all internal fans necessary for proper air flow and cooling operating normally.

Acoustic noise level measurements shall (3) be performed on operating equipment in an environment where ambient noise levels do not constitute more than 5 percent of the noise generated by the equipment at any frequency. Measurements shall (4) be performed 0.8 meters from the front, rear, and each side of the equipment (not the noise suppression device) at a height of 1.2 meters.

3.3.7.1.3 Acoustic Noise Limits

The following tabulation provides the noise limits for WSR-88D equipment expressed in decibels (dB) with a reference of 0.0002 dynes per square centimeter.

TABLE 3-12 ACOUSTIC NOISE LIMITS

CONDITION	FREQUENCY BANDS (HZ)								
	20-75	75-150	150-300	300-600	600-1200	1200-2400	2400-4800	4800-9600	9600-20000
A	100	89	82	76	73	70	68	67	66
B	79	68	60.2	52	48	45	43	42	41

3.3.7.2 Heating Requirements

At selected sites, a modular heating capability in the radome shall (1) be provided. One of three levels of heat shall (2) be provided to correspond to three geographic temperature zones, the coldest being -40 degrees Celsius.

Resistive and/or infrared electric heating shall (3) be provided to produce:

- a. An acceptable working environment for maintenance personnel to work on the azimuth and elevation bearing and drive areas of the pedestal
- b. A minimum temperature of 5 degrees Celsius on the internal pedestal components in the vicinity of the azimuth and elevation drives for safety concerns of skin-to-metal contact.
- c. Automatic activation of radome heating when the ambient air temperature within the radome drops below 0 degrees Fahrenheit.

The heating mode shall (4) be automatically activated via thermostat located in each radome heating unit. Total power consumption for this heating capability shall (5) not exceed 30 kW. Backup power support for the radome heat capability is not required.

3.3.8 Computer Software

The software architecture should use recognized standards for services when available and appropriate to facilitate system evolution and technology infusion. Additional guidance is contained in the Application Portability Profile (APP), NIST Special Publication 500-230. Requirements for use of the software standards are listed in the following sections. Requirements for communications standards are in Section 3.7.1.8, Interface Requirements. Requirements for the graphical user interface are in Section 3.3.7, Human Performance/Human Engineering.

3.3.8.1 Software Portability

To support portability and maintainability of the software, the selection of operating systems and programming languages will be made on the following requirements (assuming availability on chosen hardware platforms).

- a. Kernel services are part of the operating system and perform essential operating system tasks, such as handling disk input and output and managing internal memory. They also provide functions that may be called by a user process to interact with the hardware in some fashion. The Systems Application Program Interfaces are standardized operating system services provided to a high-order language, in this case the C programming language. Kernel Services shall (1) comply with current Portable Operating System Interface (POSIX) standards.
- b. The shell is a program that provides the user interface between the operating system and the user. The utilities or Unix and shell commands provide the mechanism for operations at the operator level such as comparing, printing, and displaying file contents, editing files, pattern searching, evaluating expressions and logging messages. The shell and utilities shall (2) comply with current POSIX standards for shell and utilities.
- c. Real-time extensions provide programs with a collection of services designed to support predictable responses to asynchronous events. Real-time embedded systems usually have different requirements than systems which are not real-time or embedded. When using real-time extensions they shall (3) comply with current POSIX standards for real-time extensions, as far as practical.
- d. Software code written in a particular computer language shall (4) maintain compatibility with the current software and shall (5) follow the current industry standard for that computer language as long as compatibility with the current software can be maintained.

e. Firmware is software code stored in ROM, PROM, or EPROM. Firmware is hardware specific and, in general, not portable. The choice of hardware will determine the firmware language and standards compliance.

3.3.8.2 Database Requirements

New development for the WSR-88D requiring the use of a relational database shall (1) conform to FIPS PUB 127-2 for the database management system. The database management services include:

- a. Data Definition - The capability to create, alter and delete database elements
- b. Data Manipulation - The capability to insert, select, update and delete database elements
- c. Data Query - The capability to specify search conditions consisting of a combination of select lists, predicates, and comparison operators.
- d. Data Integrity - The capability to perform data locking, transaction control, referential constraints for data integrity, and synchronous writing of data.

3.3.8.3 Extensibility

The functionality and capability of the WSR-88D will evolve as the needs of the NEXRAD agencies change. Extensibility (or flexibility) is the ease in which the WSR-88D System can be modified to provide new capabilities to meet these needs.

3.3.8.3.1 Application Programming Interface (API)

Applications related to WSR-88D Open Systems software (RPG) product generation shall (1) be implemented using an application programming interface (API) that facilitates ease of implementation and modification.

3.3.8.3.2 Software Modularity

A goal of the WSR-88D software is that the architecture of the software is modular. Modular software should allow changes in system capabilities and/or requirements in one area with minimal impact on system capabilities in another area. Another goal of the software architecture is that the software is implemented in a manner which maximizes reusability with the WSR-88D system.

3.3.8.4 Expandability

Another goal of the software architecture is to provide the capability to add or delete hardware components to support varying or expandability requirements for computer resources.

3.3.9 Firmware

Firmware is defined to consist of those programs or microprograms that reside in the class of memory that cannot be dynamically modified by the computer during its processing. The class of memory includes read-only memory (ROM), programmable read-only memory (PROM), or writeable control store (WCS). Documentation of the firmware memory should be included in the hardware development specification, where applicable. Documentation of the firmware program should be included in the software development specification and product specification, (as appropriate).

3.3.10 Flexibility and Scalability Requirements

The system architecture shall (1) support the addition of hardware components to accommodate the expandability requirements in Section 3.2.1.7.

3.4 Logistics

The WSR-88D System shall (1) be designed so that it can remain fully operational, when supported in accordance with the NEXRAD Integrated Logistics Support Plan (ILSP), throughout its specified useful life period. The logistics concept is summarized in this section.

The logistics system for NEXRAD coordinates the hardware configuration, to include the identification of those hardware components that are line replaceable units (LRU), with the availability and maintainability requirements. The NEXRAD logistics system is based on existing NEXRAD agency logistic systems and augments those existing systems to the maximum extent possible. The logistics system includes both periodic and corrective maintenance actions.

3.4.1 Maintenance/Logistics Concept

The general WSR-88D maintenance approach is to monitor system performance locally at the RDA and RPG, or by using the MSCF, to localize the failures through the use of software and hardware maintenance features, and to replace the failed LRU from onsite spares, whenever available. The actual repair of a replaced item will be managed by the NRC for depot repairables, or at a designated on-site repair area for those items that are coded for local repair or disposal. The WSR-88D maintenance practices, which will augment the maintenance practices of the principal users, require that the individual who performs on-equipment, onsite maintenance (removal and replacement of the LRU) will also perform any off-equipment, onsite maintenance (repair of the faulty LRU). If a failed LRU is not repaired onsite, it will be coded for offsite repair and sent to a repair depot. The concept of removal and replacement will be implemented, and maximum use will be made of depot level repair, consistent with the Life-Cycle Cost (LCC) analysis.

3.4.1.1 Calibration

Calibration will be conducted to maintain operational equipment performance parameters and characteristics to specification tolerances. To the degree warranted by good engineering practices, development of performance diagnostic techniques using external support equipment will be oriented toward providing measurements more precisely than the tolerances of the operating equipment/performance parameters to be measured.

3.4.1.2 Maintenance Environment

Equipment will be located at fixed worldwide locations. For DOD, onsite maintenance will be accomplished in accordance with AFI 21-116. For FAA, on-site maintenance will be accomplished in accordance with DOT/FAA Order 6000.15D, 6000.30C, 6000.40, 6000.50B, and 6090.1B. For NWS, on-site maintenance will conform to Weather Service Operations Manual (WSOM) Chapter H-50, Equipment Maintenance Program, and WSOM Chapter H-98, Engineering Directives Subsystem. Off-site maintenance will be performed by a designated NEXRAD agency organization. That organization will arrange for service engineering modifications.

3.4.1.3 Depot Maintenance

The NRC is the single central depot to support all LRUs not repaired on-site.

3.4.1.4 Supportability

WSR-88D system support will be consistent with: a single Operational Support Facility which provides engineering, operations support, and software systems maintenance; a single depot which provides central repair and procurement services; and a separate supply depot for each principal user.

3.4.2 Personnel Skill Level

There are primarily two types of personnel which will have hands-on access to the WSR-88D equipment; maintenance personnel and operations personnel.

3.4.2.1 Maintenance Personnel

The WSR-88D shall (1) be designed to allow the system to be maintained by the personnel of the NEXRAD agencies having the skills and training stated below. The skill of these personnel varies. For those sites staffed by DOD, maintenance personnel will consist of skill level 5 (journeyman) airmen or equivalent. Individual qualifications of DoD maintenance personnel vary from graduates of basic electronic and systems technical schools with several years experience to new graduates of these schools. NWS and FAA will, in general, use skilled maintenance personnel that have been trained in the proper maintenance procedures for the WSR-88D. See Section 3.5.1 for maintenance training.

3.4.2.2 Operations Personnel

The operational users of the WSR-88D equipment are varied across the three agencies but are primarily targeted towards meteorologically oriented personnel. Skill levels range from meteorologists to hydrologists, meteorological technicians, weather forecasters and weather observers. See Section 3.5.2 for operations training.

3.4.3 Supply

Site spares and initial depot spare parts are provided. The site spares consists of line (lowest) replaceable units (LRUs). Provisioning documentation consisting of provisioning parts list, numerical parts list, long lead-time item list, and item identification/item logistics data records are provided. A single principal user (the NWS) will be responsible for ordering all non-consumable WSR-88D supplies. This agency will redistribute supplies to the other NEXRAD agencies utilizing normal supply systems where possible. Spare parts should be made available in sufficient time to replenish inventories without impacting system operation.

3.5 **Training**

The WSR-88D System is supported by operations and maintenance training, enabling Government personnel to operate and maintain the WSR-88D equipment. FAA maintenance and operation personnel will receive Occupational Safety and Health Training in compliance with 3900.1B, FAA Occupational Safety and Health Program.

3.5.1 Maintenance Training

All maintenance personnel will complete the training directed by their agency before being responsible for maintenance of the WSR-88D equipment.

3.5.2 Operations Training

All operators of the WSR-88D equipment will complete the training directed by their agency before being responsible for operation of the WSR-88D equipment.

3.6 **Intentionally Left Blank**

FUNCTIONAL AREAS

3.7 Functional Area Characteristics

Characteristics are specified herein for the WSR-88D functional areas, consisting of the following:

- a. System Requirements
- b. Radar Data Acquisition
- c. Radar Product Generation
- d. Facilities
- e. Support

SYSTEM REQUIREMENTS SECTION

3.7.1 System Requirements

System requirements include those requirements not specific to an individual functional area or common to more than one functional area. These requirements are necessary to meet system performance and operational characteristics. An attempt is made to identify as many of these requirements as possible, however system evolution and equipment design may identify additional requirements unforeseen at this time. The identification of a requirement as a system requirement precludes that required function from being forced into a functional area that may not prove optimal as the system design takes shape.

3.7.1.1 Data Quality

The following sections define the range unfolding and velocity dealiasing requirements.

3.7.1.1.1 Range Unfolding

The WSR-88D shall (1) provide mean radial velocity and spectrum width data processing with the accuracies specified in Section 3.7.2.2.3.1 over the unambiguous range interval of at least 115 km as defined by the following equation:

$$R(\text{Unambiguous Range}) = \frac{cT_s}{2}$$

where: T_s = Doppler Pulse Repetition Time (PRT)
 c = Speed of Light

A technique shall (2) be provided to present the mean radial velocity and spectrum width data at the correct range (range unfolding). This range shall (2a) be configurable to a minimum range of 230 km or a minimum range of 300 km. See Section 3.7.2.2.1.1 for unambiguous range coverage. Data significantly contaminated due to overlaid echoes shall (3) be identified.

3.7.1.1.2 Velocity Dealiasing

Velocity dealiasing to extend the measurement of the mean radial velocity interval to at least twice the Nyquist velocity shall (1) be provided in the WSR-88D System.

The Nyquist velocity (V_a) shall (2) be at least 11 m/s for long pulse, and 25 m/s for short pulse, as given by the following equation:

$$V_a = \pm \frac{\lambda}{4T_s}$$

where: λ = wavelength
 T_s = Doppler Pulse Repetition Time (PRT)

3.7.1.2 Pulse Repetition Frequency Selection

A technique shall (1) be provided for automated selection of a Pulse Repetition Frequency (PRF) on a volume scan basis from a minimum of five different PRFs meeting the operational constraints on unambiguous range interval (Section 3.7.1.1.1) and on Nyquist velocity (Section 3.7.1.1.2).

The PRF selected shall (2) be the one calculated to result in minimum echo area obscuration (relative power is within 5 dB) at the lowest elevation scan.

This PRF shall (3) be used during the next volume scan. There shall (4) be a manual override for this capability. The set of PRFs available for selection shall (5) be adjustable, on a site specific basis,

over a range of at least $\pm 1\%$. This will permit the selection of PRFs to allow certain types of receiver interference suppression circuitry to be effective.

3.7.1.3 Level II

The Level II data shall (1) be the digital base data output of the signal processor, status information, and metadata (which includes: RDA performance/maintenance data, volscan data, clutter filter bypass map data, clutter filter map data, and RDA adaptation data).

3.7.1.3.1 Level II Interface Connection

The WSR-88D System shall (1) provide interface connections to allow Level II data to be collected. Synchronization, calibration, and other data required to properly interpret the archive data, including date, time, antenna position, status, etc., shall (2) be available at the Level II interface connections.

3.7.1.3.2 Level II Equipment Data Distribution

The WSR-88D System shall (1) also have the capability to ingest previously recorded data from the Level II system. Starting and stopping of the Level II Data Distribution shall (2) be controllable from the RPG HCI.

3.7.1.4 Power Failure Recovery

The RDA and RPG shall (1) retain all required data necessary for full load operation for any number of outages per 24 hour period with no single outage exceeding 15 seconds and the total duration of outages for a 24 hour period not exceeding five minutes with the interval between outages not less than 0.1 second. All unit adaptation and control information shall (2) be stored in such a manner that required response times can be met for a restart up to at least one hour after power interruption, and that all PUPs can be returned to the state existing immediately prior to the outage.

3.7.1.5 Overload Control

The recovery from an overload condition (known as load shedding) shall (1) be performed automatically for each resource. Load shedding shall (2) be performed in a manner so as not to allow a rapid cycling between overload and non-overload conditions.

3.7.1.5.1 Processing Overload Control

When a processing overload condition occurs, allocation of resources shall (1) be performed dynamically, if possible, to alleviate the overload condition. The goal of dynamic resource allocation should be to, without operational impact, balance processor loads caused by excessive demand or processor failure.

3.7.1.6 Master System Control Function (MSCF)

3.7.1.6.1 General Requirements

The WSR-88D system shall (1) include the Master System Control Function (MSCF). The RDA HCI and RPG HCI shall (1a) run on the MSCF display/control position. The MSCF, RPG HCI and RDA HCI provide operator/maintenance access to all RDA and RPG control, status monitoring, and error detection functions defined in the subsections below. The MSCF, RPG HCI and RDA HCI shall (2) include a human/machine interface (control position) in the form of a graphical user interface for data entry and display. Requirements for the graphical user interface are in Section 3.3.7. The RPG

HCI shall (3) be accessible from any display/control position within the RPG, MSCF and FAA RMS Interface.

The RDA HCI shall (3a) be accessible for each RDA from a single MSCF display/control position and a single display/control position for each RDA within the RDA and the FAA RMS interfaces. The RDA control capabilities shall (4) be only available at one and only one of the display/control positions or the FAA RDA/RMS interface at any given time. This means that the subset of RDA HCI control capabilities that are provided to the FAA RDA/RMS interface, are either being controlled through the FAA RMS interface or the RDA HCI or RPG HCI on the MSCF or one of the RDA HCI control positions at any one time.

The capability shall (5) be provided for a user with the proper privileges at an MSCF through the RPG HCI or RDA HCI to take control from the FAA RMS interface (or the RPG or an RDA HCI control position). Status and monitoring information shall (6) be simultaneously available at all display/control positions and the FAA RMS interface. The MSCF display/control position and the RDA display position shall (7) be capable of being located remotely from the remaining RPG and RDA equipment via a dedicated communications line. The dedicated communications line to a remotely located Master System Control Function display/control positions shall (8) have a data rate of no less than 9600 bits per second.

3.7.1.6.1.1 Access Controls

Provisions will be made to limit the access to selected classes of control functions through level of change authority, access control, or other suitable means as described in this section. To prevent unauthorized access to the MSCF, RPG HCI and RDA HCI or selected classes of its control functions, a level of change authority hierarchy shall (1) be provided. The selected classes of control functions which require additional security will include but are not limited to the following: System monitoring, Vulnerability Scanning, System Administration, Adaptation Data Control, and Maintenance.

3.7.1.6.1.2 Operator Commands

All operator commands shall (1) be validated. Whenever an operator provides an invalid command, an appropriate English language error message shall (2) be provided to the operator. Errors in operator input or invalid commands shall (3) not require a restart for recovery.

3.7.1.6.2 Overload Status and Control

Notification shall (1) be provided of a condition of resource utilization overload of any part of the WSR-88D System. The capability shall (2) also be provided to the RPG HCI and the FAA RMS Interface to adjust, by means of adaptation data, the thresholds for these notifications. Through the RPG HCI, the capability shall (3) be provided, by means of adaptation data, to define the priority for load-shedding processing tasks (within each processor) when a processing overload situation occurs.

Through the RPG HCI, the capability shall (4) be provided, by means of adaptation data, to define the priority for load-shedding products. Recovery from product load-shedding shall (5) be in the reverse of the shedding order. The capability shall (6) exist for the operator to specify an on-line storage retention time for each product type.

3.7.1.6.3 Status Monitoring

The capability shall (1) be provided to detect, record, and display all status and error messages. Such messages can be displayed through graphical symbols, text messages or other means. Any text

messages displayed shall (2) be in descriptive English language. However, for the recording of each status and error message, a descriptive English language message shall (3) be provided.

A record of each message shall (4) be automatically stored in non-volatile memory for at least 24 hours.

3.7.1.6.3.1 Radar Status Monitoring

The capability shall (1) be provided for radar equipment monitoring, reporting, and displaying of status of radar functions to the RPG HCI and RDA HCI. The capability shall (2) be provided to request and output selected or all of the RDA monitored parameters. The capability shall (3) be included to monitor the performance of selected functions on a non-interfering basis with the processing of on-line data and perform sufficient tests to ascertain that no degradation in operation has occurred.

The following parameters, at a minimum, shall (4) be monitored:

- a. Peak Transmitter Power (at transmitter output and the antenna pedestal)
- b. Receiver Calibration, both Horizontal and Vertical channels
- c. Waveguide Arcing
- d. Pulse Repetition Frequency (PRF)
- e. Memory Checks
- f. Signal Processor Performance
- g. IF A/D Converter Performance, both Horizontal and Vertical channels
- h. Timing Controls
- i. Over/Under Voltages
- j. Over/Under Currents
- k. Over Temperature
- l. Air Flow Failures
- m. Pedestal Oil Level
- n. Radar Data Output, both Horizontal and Vertical channels
- o. Backup Power
- p. Protective Circuitry Parameters (as defined in Section 3.7.2.4.5.1)

Alarms shall (5) be provided for RDA conditions which might result in further equipment problems or failures if the indicated condition is not corrected. This shall (6) include the case when the automatic radar calibration function can no longer maintain performance parameters within specification limits. All alarms shall (7) result in alarm messages being sent to the RDA HCI and RPG HCI, with further dissemination as defined in Section 3.7.3.5.

3.7.1.6.3.2 System Status Monitoring

Through the MSCF, the capability shall (1) be provided to display the status of the RDA, the RPG (including components of the local area network), and each communications interface. Through the RDA HCI, the capability shall (1a) be provided to display the status of the RDA (including components of the local area network), and each communications interface. This status shall (2) include operational modes, conditions, and parameters, as well as error conditions.

Alarm messages received shall (3) be considered status messages for the purposes of further dissemination. Status shall (4) be provided to the MSCF, RPG HCI and the FAA RMS interface: (a) when requested; (b) periodically, as specified; and (c) when there is a change in the status of the

RDA, the RPG, or any of the WSR-88D communications interfaces. The capability shall (5) be provided to turn on or turn off display of selected status and error messages.

This capability shall (6) be provided individually for the RPG itself, each of its major subcomponents (e.g., a communications server, product server, control workstation), the RDA, and each of the communications interfaces. The capability shall (7) be provided to selectively display the record of status and error messages by functional areas or other subsets based on operator-selected keywords.

3.7.1.6.4 System Control

MSCF, RPG HCI and RDA HCI capabilities shall (1) provide for operation of the WSR-88D System. These capabilities are accessible through the FAA RMS interface. These capabilities include, but are not limited to, the following:

- a. RPG Startup
- b. RPG Restart
- c. RDA Restart
- d. RDA Warm Startup
- e. User Interface Connection Control
- f. Controlled Shutdown
- g. Mode Control
- h. RPG Reconfiguration
- i. Level II Control
- j. Product Generation Control
- k. Product Storage Control
- l. Adaptation Control
- m. RDA Backup Power Control
- n. User Interface Configuration
- o. RPG and RDA Redundant Channel Control
- p. Product Distribution and Data Ingest Control
- q. Volume Coverage Pattern Control
- r. Clutter Suppression Control
- s. System Administration
- t. Stop/Start Base Data Transmission
- u. Spot Blanking Control
- v. RDA Maintenance

See Sections 3.7.2.8, 3.7.3.4 and 3.7.3.9 for more detailed requirements.

3.7.1.6.4.1 RPG Startup

The RPG HCI shall (1) provide the capability to manually initiate the RPG startup function.

3.7.1.6.4.2 RPG Restart

The RPG HCI shall (1) provide the capability to manually initiate the RPG restart function. The RPG HCI shall (2) provide the capability to manually initiate restart of any RPG subcomponent.

3.7.1.6.4.3 RDA Restart

The MSCF, RPG HCI and RDA HCI shall (1) provide the capability to manually initiate the RDA restart transition and warm startup functions. The restart capability shall (2) transition an RDA from Ready to Halt and back to Ready.

3.7.1.6.4.4 RDA Warm Startup

The RPG HCI shall (1) provide the capability to manually initiate the RDA warm startup transition. The warm startup capability shall (2) transition the RDA from the Offline-Operate or Standby to the Operate state.

3.7.1.6.4.5 User Interface Connection Control

The RPG HCI shall (1) provide the capability to manually connect, terminate, or prevent connection with any and all user interfaces (listed in Section 3.7.1.8) within its WSR-88D System.

3.7.1.6.4.6 Controlled Shutdown

The MSCF shall (1) provide the capability to shutdown in a controlled manner the following parts of the system:

- a. The communications links.
- b. The RPG, or its subcomponents.

The capability shall (2) exist to shut down the entire RPG functional area with a single command, or to shut down individual subcomponents with a single command. Shutdown of the RPG, or individual subcomponents of the RPG, shall (3) have no effect on the operation of the RDA if the MSCF or RDA HCI is active. Shutdown of the RDA, or individual subcomponents of the RDA, shall (4) have no effect on the RPG's capabilities related to sending existing products to users.

3.7.1.6.4.7 Mode Control

3.7.1.6.4.7.1 RDA Mode Control

The RDA HCI shall (1) provide the capability to manually change RDA Modes by selecting various RDA parameters.

The parameters which shall (2) be selected are the following:

- a. Mode Control: Operational/Test/Off-line Maintenance

The RPG HCI and RDA HCI shall (2a) provide the capability to manually change RDA Operational Modes by selecting various RDA parameters.

- b. Operational Modes: A, B, and others specified in Appendix I

Selection of Mode B, for clear air observations, shall (3) require a manual action through the RPG HCI.

3.7.1.6.4.7.2 RPG Mode Control

3.7.1.6.4.8 RPG Reconfiguration

The RPG shall (1) provide the capability to manually control the shifting of loads between RPG processors, if the RPG has more than one processor. At redundant sites, the RPG shall (2) provide the capability to manually reallocate RPG resources. For Open Build 1 this means reallocation of major subcomponents (e.g., a communications server, product server, control workstation).

3.7.1.6.4.9 Level II

The RPG HCI shall (1) provide the capability to activate, deactivate, and monitor the status of the data archive function. The RPG HCI shall (2) also provide the capability to select the product types to be routinely archived and their frequency of archiving. Additionally, the RPG HCI shall (3) provide the operator the capability to routinely archive status and error messages and operational parameters.

The RPG HCI shall (4) also provide the capability to pause and resume archive recording. The RPG shall (5) be capable of displaying operator selected status and error messages previously recorded on archive media. The RPG HCI shall (6) provide the capability to activate and deactivate the Level II function.

3.7.1.6.4.10 Product Generation Control

The RPG HCI shall (1) provide the capability, by means of adaptation data, to select the set of products to be routinely generated. In addition, the capability shall (2) be provided for the RPG HCI to specify, on a per product basis, specific product parameters (e.g., center az/ran, layer heights, elevation angle/slice) and the frequency at which the product is generated. The RPG HCI shall (3) provide the capability to select the product for generation for each alert category (see Appendix J).

3.7.1.6.4.11 Product Storage Control

The RPG HCI shall (1) provide the capability, by means of adaptation data, for defining and changing the product mix, frequency, and retention time (up to a maximum of six hours for all NEXRAD products) for products to be retained in the on-line RPG storage.

3.7.1.6.4.12 Adaptation Control

The RPG HCI shall (1) provide the capability to display and modify all RPG adaptation parameter values. The RDA HCI shall (1a) provide the capability to display and modify all RDA adaptation parameter values. The RPG HCI shall (2) provide the capability to select the use of the baseline RPG adaptation data. The RDA HCI shall (2a) provide the capability to select the use of the baseline RDA adaptation data.

3.7.1.6.4.13 RDA Backup Power Control

The RPG HCI and the RDA HCI shall (1) provide the capability to initiate with a single command a systematic, controlled power transfer between commercial and backup generator power for all WSR-88D equipment at the radar site.

3.7.1.6.4.14 Interface Configuration

The RPG shall (1) provide the capability to configure or reconfigure, by means of adaptation data, any product distribution and ingest interface to connect to any of the WSR-88D communication classes available.

3.7.1.6.4.15 RDA and RPG Redundant Channel Control

The RPG HCI and its associated display/control positions shall (1) be capable of full status reporting, alarms, and control of both RDA channels. One RDA HCI and its associated display/control position shall (2) be provided at each RDA in the redundant configuration. The RDA HCI and RPG HCI shall (3) provide the capability to control the state of the FAA redundant channels with a single command.

The MSCF and the RDA HCI shall (3a) provide the capability to change active channels with a single command. When active channels are switched, either automatically or by command, there

shall (4) be no change in operating characteristics (i.e., no manual input of status or adaptation data to the new channel).

3.7.1.6.4.16 Product Distribution and Data Ingest Control

The RPG HCI shall (1) provide the capability to specify and change, through adaptation data, which products will be available and their frequency of availability to Legacy Class 2 users. The available distribution methods shall (2) include as a minimum: a predefined single set of products, a predefined set of products routinely, and a limited selection of products on a one-time request basis.

3.7.1.6.4.17 Volume Coverage Pattern Control

The RPG HCI shall (1) provide the capability for selecting a single scanning strategy to be employed at any given time from the set of defined (local and remote) scan strategies (Note: "Local" scan strategies are stored at the RDA. "Remote" scan strategies are stored at the RPG). The RPG HCI shall (2) provide a manual override of the defined strategies for maintenance actions. A manual override for automated Pulse Repetition Frequency (PRF) selection capability shall (3) be provided through the RPG HCI.

The capability shall (4) be provided to select one of the predetermined Doppler PRFs through the RPG HCI when the manual override capability has been selected. The RPG HCI shall (5) provide the capability to modify VCP parameters by elevation scan where appropriate. The parameters shall (6) include, but are not limited to Doppler PRF selection, SNR threshold values (as defined in Section 3.7.2.6), sector definition, and velocity measurement increment.

3.7.1.6.4.18 Clutter Suppression Control

Through the RPG HCI and RDA HCI, the capability shall (1) be provided, at a minimum, to:

- a. Define clutter suppression regions;
- b. Initiate generation of a new bypass clutter filter map.

The definition of clutter suppression regions shall (2) include the capability to select the elevation angle(s), minimum and maximum ranges and azimuths, and type of suppression.

The capability shall (4) be provided to modify, through adaptation data, the selections in a above. The bypass clutter filter map shall (5) consist of a site-specific elevation angle independent clutter suppression map which specifies the ranges and azimuths to be used for clutter suppression.

3.7.1.6.4.19 System Administration

The RPG HCI AND RDA HCI shall (1) provide status and control for system fault management, performance accounting, system security, and configuration management. The RDA HCI shall (2) provide status and control for system fault management, performance accounting, system security and configuration management. A multiple read /multiple write storage device with removable media shall (3) be provided on the MSCF for archiving any one or all of the system administrative functions including backup of operating system, backup of application software, backup of all adaptation data, backup of security audits/logs, and system status/error/performance data. This same device may also be used to support other similar archive, logging, backup, or recording functions.

3.7.1.6.4.20 Stop/Start Base Data Transmission

The RDA HCI shall (1) provide the capability to start or stop the transmission of base data from the RDA to the RPG. Parameters of this command shall (2) allow selection of any combination of base data, reflectivity, mean radial velocity, and/or spectrum width data.

3.7.1.6.4.21 Spot Blanking Control

There shall (1) be the capability of invoking or prohibiting spot blanking through the RPG HCI.

3.7.1.6.4.22 RDA Maintenance

The RDA HCI shall (2) provide the capability to manually override of the automatic calibration function defined in Section 3.7.2.11.

3.7.1.7 Remote System Access

The capability shall (1) be provided (at selected sites) for dial-up access to the WSR-88D RDA control capabilities. The characteristics of these interfaces are in Section 3.7.1.8.2, External Interfaces.

3.7.1.8 Interface Requirements

The meteorological and hydrological products of the WSR-88D are accessible by external user display systems. Requirements for the distribution of WSR-88D products are in Section, 3.7.3.3, Product Distribution. Access to products includes, but is not limited to base data, base products, and derived products. This section states the required characteristics supported for external user interfaces; capabilities provided at the interfaces are covered elsewhere. Characteristics not specified or referenced in this section are left for system design. Requirements for the number and type of interfaces to be supported by functional area are in Section 3.2.1.5, Configuration and Section 3.2.1.1, WSR-88D Full Load. Expandability requirements are in Section 3.2.1.7.

An invalid password entry shall (1) be processed in accordance with the WSR-88D System Security Plan (SSP).

3.7.1.8.1 Internal Interfaces

The RPG shall (1) support dedicated and dial modems for inter-operability with WSR-88D product users. All RPG modems shall (2) have a test capability to include local analog and remote digital loop-backs.

3.7.1.8.1.1 Dedicated PUP Connections

This section is no longer applicable to this specification.

3.7.1.8.1.2 Dial-up PUP Connections

This section is no longer applicable to this specification.

3.7.1.8.2 External Interfaces Requirements

3.7.1.8.2.1 Legacy Class 1

Legacy Class 1 users shall (1) have full time dedicated access to WSR-88D System products. Example Class 1 interfaces include the AWIPS, WARP, ITWS, and WFO Advanced. These interfaces shall (2) transfer data at a minimum rate of 14.4 Kbps, unless otherwise specified.

The defining ICD for this interface is number 2620001. These users shall (3) be capable of processing all message types in Table II of this ICD, except for the Sign On Message.

3.7.1.8.2.2 Legacy Class 2

Class 2 users shall (1) have dial access to WSR-88D System products. Example Class 2 interfaces include the AWIPS, WFO Advanced, and SSM (See Appendix G). The defining ICD for this interface is number 2620006. In addition, these users shall (2) be capable of processing all message types described in Section 3.5 in this ICD.

This interface shall (3) transfer data at a minimum rate of 14.4 Kbps. This interface shall (4) conform to the security and privacy requirements specified in paragraph 3.7.1.8 for system access.

3.7.1.8.2.3 Deleted

3.7.1.8.2.4 Base Data Distribution Interface

Base Data Distribution users shall (1) have dedicated access to the WSR-88D base data stream. This interface shall (2) transfer data at a rate sufficient to capture the real-time data stream. This interface shall (3) conform to the Archive II to User ICD, 2620010. The RPG shall (4) maintain an LDM queue size of 9 megabytes. The intent of this requirement is to hold approximately 15 minutes of data. As the VCP's change, the queue size may need to be adjusted in order to reach the 15 minute buffering goal.

3.7.1.8.2.5 FAA Remote Monitoring Subsystem (RMS)

The FAA Remote Monitoring Subsystem interface provides dedicated access to all system status and control capabilities as stated in this specification. The system shall (1) accept transfer of data that effectively controls the system at a rate in accordance with RPG/RMS ICD 2620000 and RDA/RMS ICD 2620004. The format of status and control messages and other characteristics shall (2) be in accordance with RPG/RMS ICD 2620000 and RDA/RMS ICD 2620004. A subset of the RDA HCI and RPG HCI interface specified in 2620004, the RDA to RMS ICD and 2620000, RPG to RMS ICD, shall (3) be made available at the FAA Remote Monitoring System (RMS).

3.7.1.8.2.6 Remote Access Interfaces

The Remote RDA Control Interface shall (1) provide dial up access to the RDA control capabilities. These interfaces shall (2) transfer data at a rate to effectively control the system but not less than 9.6 Kbps. These interfaces shall (3) conform to the security and privacy requirements specified in Section 3.7.1.8 for system access.

3.7.1.9 Diagnostic Techniques

Fault isolation to the LRU without component removal shall (1) be incorporated into the system design. This shall (2) be achieved through the use of Built-In Test Equipment (BITE), automatic alarms, indicator lights, and computer diagnostic and system diagnostic software. The equipment shall (3) be designed to minimize the use of special support equipment.

Easily accessible test points shall (4) be provided to facilitate the use of external test equipment. The BITE system shall (5) provide remote performance monitoring, parameter measurements, and equipment adjustments sending performance information to the appropriate maintenance control section of the unit responsible for maintaining the system.

3.7.1.10 Security

As per the system definition, the WSR-88D system is not an isolated system, where simple security measures, such as limiting physical access would suffice to adequately protect the system. The WSR-88D now encompasses features of many similar federal computer systems, including networked

telecommunications for data dissemination, open operating systems, user displays and data entry computer equipment at multiple locations, and both manned and unmanned facilities. The requirement for security controls is mandated for the WSR-88D by both public law and by each of the individual tri-agency bodies. In the event of difference or conflict between the agency's policies, the WRS-88D system shall (1) adhere to the most stringent of the three. Specifically, the security policy, requirements, and procedures applicable to the WSR-88D are defined at the highest level in a security specific agreement, MOA for Interagency Security Requirements and Certification & Accreditation (C&A) of the Weather Surveillance Radar -1988, Doppler (WSR-88D).

The WSR-88D Security Memorandum of Agreement (MOA), prescribes the tri-agency security policies and procedures for mutual certification and accreditation (C&A) among WSR-88D Principal Users. The responsibilities of the departments required to implement the policies and procedures are also defined in this agreement. The MOA identified as reference (a) prescribes the procedures and standards of operation for the WSR-88D system for the Principal Users. The authority for this agreement is based on the security directives contained in Office of Management and Budget (OMB) Circular A-130, Appendix III, Security of Federal Automated Information Resources, the requirements of the Computer Security Act of 1987 (P.L. 100-235), national standards for certifying and accrediting systems identified in the NIST SP 800-37, 800-53 and 800-53.

RADAR DATA ACQUISITION (RDA) SEGMENT

3.7.2 Radar Data Acquisition Functional Area

The Radar Data Acquisition (RDA) functional area provides the capability to acquire, perform signal processing on, and transfer weather radar data to the Radar Product Generation (RPG) functional area.

3.7.2.1 General Requirements

3.7.2.1.1 Monitor/Control

The RDA shall (1) include an antenna, pedestal, radome, transmitter, receiver, signal processing capability, control capability, monitoring and error detection capability, and an interfacing capability/maintenance position. These capabilities shall (2) provide for the detection and measurement of weather phenomena to the extent defined in Section 3.7.2.2, while meeting the reliability, maintainability, and availability requirements defined in Sections 3.2.3, 3.2.4, and 3.2.5, and while operating within the environmental conditions defined in Section 3.2.6. RDA monitoring and control functional capabilities (defined in Sections 3.7.1.6 and 3.7.2.9) shall (3) be included in the RDA to support both operations and maintenance activities. The RDA shall (4) be designed for unattended operation.

3.7.2.1.2 Operational Modes

The RDA shall (1) be capable of operation in at least two modes, designated Mode A and Mode B. Mode A is for precipitation detection, tracking, and measurement. Mode B is for clear air observations. Except where noted, all requirements of Section 3.7.2 shall (2) apply to both Modes A and B. Switching from Mode B shall (3) be accomplished automatically through the Mode Selection Function described in Appendix D. Criteria for automatic mode switching are provided in Appendix I.

3.7.2.1.3 Long Pulse

A long pulse capability shall (1) be provided for Mode B. Radar system parameter changes allowed for this long pulse capability shall (2) be:

- (2a) a 3 dB pulse volume depth less than 750 meters
- (2b) a Nyquist velocity greater than 11 ms
- (2c) H and V returns out to a minimum range of 460 km
- (2d) a longer system dwell time to allow attainment of the requirements of Section 3.7.2.2.3.1.

3.7.2.1.4 Short Pulse

Radar system performance characteristics for short pulse shall (1) be:

- (1a) a 3 dB pulse volume depth less than 250 meters
- (1b) a Nyquist velocity greater than 25 ms⁻¹
- (1c) H and V returns out to a minimum range of 460 km

3.7.2.1.5 RDA Power Failure Recovery Requirements

The RDA shall (1) operate using commercial power as specified in Section 3.7.5.2.1. In the event that primary power feeds to the RDA fails or degrades below acceptable limits, the RDA shall (2) switch to auxiliary power automatically. Acquisition of radar data during the switch between either primary or backup power feed is required when the WSR-88D system is equipped with a Transitional Power Source.

Following a power outage of less than or equal to 15 seconds while acquiring base data, the RDA shall (3) resume acquisition at the interrupted elevation scan as soon as the antenna and transmitter have returned to stable operational conditions. The elevation scan which was in progress at the point of power outage shall (4) be restarted with a full collection of base data, and, thereafter, the remaining portions of the volume coverage shall (5) be completed. For power outages in excess of 15 seconds, the volume coverage shall (6) be restarted.

3.7.2.2 Performance Characteristics

The RDA shall (1) consist of a coherent Doppler radar capable of operating within a Radar Spectrum Engineering Criteria (RSEC) allocation bandwidth within the 2.7-3.0 GHz frequency band. The RDA shall (2) be capable of continuous, preprogrammed surveillance to provide, at a minimum, reflectivity, mean radial velocity, and spectrum width, differential reflectivity, correlation coefficient and differential phase base data (Defined in Appendix B). Base data shall (3) meet the requirements defined in this section.

3.7.2.2.1 Coverage

The following sections define the range, volume coverage and spot blanking requirements.

3.7.2.2.1.1 Base Data Coverage

The WSR-88D collects base data for each elevation angle of the operational volume coverage pattern. The horizontal resolution and unambiguous range coverage are defined by the volume coverage pattern definition subject to the following requirements.

3.7.2.2.1.1.1 Base Reflectivity

The RDA shall (1) provide quantitative reflectivity estimates over an unambiguous range of 1 km to a minimum of 460 km (or up to a height of 70,000 ft MSL if this height occurs prior to reaching 460 km in horizontal range). The quantitative reflectivity estimates from 2 km to a minimum of 230 km shall (2) meet the accuracies specified in Section 3.7.2.2.3.2. The same signal processing scheme shall (3) be used to provide all reflectivity estimates.

3.7.2.2.1.1.2 Base Velocity and Spectrum Width

The RDA shall (1) provide quantitative mean radial velocity and spectrum width estimates from 1 km to the unambiguous range of the employed pulse repetition frequency. The quantitative velocity estimates from 1 km to the unambiguous range of the employed pulse repetition frequency shall (2) meet the accuracies specified in Section 3.7.2.2.3.1. The range unfolding and velocity dealiasing requirements are specified in Sections 3.7.1.1.1 and 3.7.1.1.2 respectively.

3.7.2.2.1.1.3 Differential Reflectivity

The RDA shall (1) provide the ratio of horizontal to vertical reflectivity from 1 km to 300 km. The quantitative differential reflectivity estimates from 2 km to the unambiguous range of the employed pulse repetition frequency shall (2) meet the accuracies specified in Section 3.7.2.2.3.2.

3.7.2.2.1.1.4 Correlation Coefficient

The RDA shall (1) provide the correlation between horizontal and vertical signals from 1 km to 300 km. The quantitative correlation coefficient estimates from 2 km to the unambiguous range of the employed pulse repetition frequency shall (2) meet the accuracies specified in Section 3.7.2.2.3.2. The same signal processing scheme shall (3) be used to provide all correlation coefficient estimates.

3.7.2.2.1.1.5 Differential Phase

The RDA shall (1) provide the phase difference between the horizontal and vertical signals from 1 km to 300 km. The quantitative differential phase estimates from 1 km to the unambiguous range of the employed pulse repetition frequency shall (2) meet the accuracies specified in Section 3.7.2.2.3.1. The range unfolding and velocity dealiasing requirements are specified in Sections 3.7.1.1.1 and 3.7.1.1.2 respectively.

3.7.2.2.1.2 Volume Coverage

The RDA shall (1) provide the capability for a volume coverage defined by a rotated triangular section extending in range from the antenna to the specified range, 360 degrees in azimuth relative to the antenna's rotational axis, and from -1 degree to 45 degrees in elevation relative to the antenna's rotational axis and the horizontal plane at the antenna. However, there is no requirement for coverage above 70,000 feet Mean Sea Level (MSL). The capability of at least 20 unique automatic antenna scanning strategies shall (2) be provided.

3.7.2.2.1.2.1 Dynamic Scanning

Certain meteorological conditions may limit the effectiveness of standard VCP scanning. Therefore, flexible, dynamic scanning options are designed for integration into the baseline VCP scanning schemes.

Early Volume Coverage Pattern Termination. The RDA shall (1) support the ability to terminate a volume scan, based on detected return, prior to the completion of the last elevation in the VCP definition. This early termination shall (2) cause the system to enter its normal end-of-volume scan transition (RDA antenna retrace, RPG conclude algorithm processing and product generation) to prepare for the start of a new volume scan.

Supplemental Elevation Scans. Additional elevation scans, not included in standard VCP definition, may be included in the VCP definition at the time of execution. The RDA shall (3) support the ability to scan elevations in any sequence (order) defined in the VCP definition. The elevation sequence, including supplemental scans, does not have to be monotonically increasing in elevation. Additionally, individual elevations may be scanned multiple times, non-sequentially, within a volume scan.

When Dynamic Scanning is implemented with the following scanning strategies, the volume scan time may be different depending on the combinations of scanning strategies used.

3.7.2.2.1.2.2 Scanning Strategy 1

One scanning strategy shall (1) provide, for a volume scan time not to exceed five minutes, a sample of the coverage volume. This sample shall (2) consist of 14 unique elevation scan levels, from zero degrees to +20 degrees in elevation, with at least the lowest six degrees of elevation having no gaps between the one-way pattern 3 dB points of adjacent elevation scans. This scan strategy is designated Scan Strategy 1.

3.7.2.2.1.2.3 Scanning Strategy 2

Another scanning strategy shall (1) provide a sample of the coverage volume of nine unique elevation angles from zero degrees to +20 degrees in elevation, for a volume scan time not to exceed six minutes. The lowest four degrees of elevation shall (2) have no gaps between the one-way pattern 3 dB points of adjacent elevation scans. This scan strategy is designated Scan Strategy 2.

3.7.2.2.1.2.4 Scanning Strategy 3

Another scanning strategy shall (1) provide a sample of the coverage volume of five unique elevation angles from zero degrees to +5 degrees in elevation, for a volume scan time not to exceed ten minutes. Two versions of this scanning strategy shall (2) be available, one employing short pulse and the other employing long pulse. This scan strategy is designated Scan Strategy 3.

3.7.2.2.1.2.5 Scanning Rate

The scanning rate may vary with elevation, but should be consistent with reasonable dwell times (T_b) for the required system performance for reflectivity and radial velocity data. Additional scan strategies are specified in Appendix I.

3.7.2.2.1.2.6 Scanning Strategy 4

Another scanning strategy shall (1) provide a sample of the volume coverage of nine unique elevation angles from zero (0.0) degrees to +20.0 degrees in elevation for a volume scan time not to exceed five minutes forty five seconds (345.0 seconds). The lowest five degrees of elevation shall (2) have no gaps between the one-way pattern 3 dB points of adjacent elevation scans. This scan strategy is designated Scan Strategy 4.

3.7.2.2.1.2.7 Scanning Strategy 5

Another scanning strategy shall (1) provide a sample of the coverage volume of 14 unique elevation angles from zero (0.0) degrees to +20.0 degrees in elevation for a volume scan time not to exceed 4 minutes 42 seconds (282.0 seconds). The elevation angles shall (2) maintain constant height uncertainty at all ranges to 460 km and for all heights to 70,000 ft MSL of storm features. This scan strategy is designated Scan Strategy 5.

3.7.2.2.1.3 Spot Blanking

The RDA shall (1) provide the capability to inhibit transmission of radiated power for up to five (5) separate areas within any volume coverage pattern. Blanking is defined as a reduction in power of the radiated beam. These selectable areas shall (2) be capable of being defined by an upper and lower elevation angle and a beginning and ending azimuth angle with a selection resolution of one degree. These data shall (3) be capable of being entered only through controlled zone definition adaptation parameters at the RDA.

The design of the spot blanking shall (4) not impose any undue physical or electronic constraints on the transmitter and receiver assemblies of the RDA. Spot Blanking shall (5) be done in such a manner as to minimize any disruption to the meteorological content of the radar data, as well as any clutter suppression invoked at the time, to the data outside of the area selected for blanking.

3.7.2.2.2 Sample Interval

The following sections define the azimuth and range sample interval requirements.

3.7.2.2.2.1 Azimuth Sample Interval

The RDA shall (1) provide a configurable azimuth sample. The azimuth sample interval shall (2) be configurable to greater than 0.953 degrees and less than 1.0555 degrees centered at the half degree or greater than 0.477 degrees and less than 0.528 degrees centered on the quarter and three-quarter degree.

3.7.2.2.2.2 Range Sample Interval

The RDA shall (1) provide a configurable range sample interval of not greater than 250 km or 1 km, from 1 km to 460 km, including the effects of range averaging, for reflectivity measurements. The RDA shall (2) provide a range sample interval of not greater than 250 meters for mean radial velocity and spectrum width measurements from a range of 1 km to 300 km.

3.7.2.2.2.3 Effective Beam Width

For half-degree azimuth sample (3.7.2.2.2.1) mode of operation, the effective beam width shall (1) be no larger than 1.10 degrees (half power point). For one degree azimuth sample (3.7.2.2.2.1) mode of operation, the effective beam width shall (2) be no larger than 1.50 degrees (half power point).

3.7.2.2.3 Accuracy and Precision

The following sections define range accuracy and precision requirements for mean radial velocity, spectrum width, reflectivity, differential reflectivity, correlation coefficient and differential phase. The RDA shall (1) provide the capability of operating over a range of dwell times compatible with nominal scan rates of from 0.5 rpm to 3.2 rpm.

3.7.2.2.3.1 Mean Radial Velocity and Spectrum Width

For the range of possible dwell times (T_D), the RDA shall (1) provide estimates of mean radial velocity (V) and spectrum width (W) with the following accuracy and precision for all Azimuth Sample Intervals specified in 3.7.2.2.2.1:

TABLE 3-14 MEAN RADIAL VELOCITY AND SPECTRUM WIDTH

	Accuracy	Precision (Note 2)
Mean Radial Velocity	For a true spectrum width (Note 1) of 4 ms^{-1} the standard deviation in the estimate of the mean radial velocity shall (1a) be less than or equal to 1.0 ms^{-1} including quantization errors, for signal-to-noise ratio (SNR) greater than 8 dB.	0.50 ms^{-1}
Spectrum Width	For a true spectrum width (Note 1) of 4 ms^{-1} the standard deviation in the estimate of the spectrum width shall (1b) be less than or equal to 1.0 ms^{-1} including quantization errors, for SNR greater than 10 dB.	0.50 ms^{-1}
Note 1: True spectrum width is defined as one standard deviation of the meteorological phenomenon spectrum width within a sampled volume for which the indicated accuracy applies. Stated accuracy values apply for Nyquist velocities of the system. (Gaussian spectrum assumed.) Note 2: Precision is defined as the quantization, the smallest resolvable increment. This is based upon the method of encoding and the number of bits used to represent the data. Note 3. Significant biases introduced by the computational technique should be minimized.		

During short pulse operations, spectrum widths of 1 to at least 10 ms⁻¹ shall (2) be calculated. Spectrum widths larger than 10 ms⁻¹ shall (3) be displayed as being 10 ms⁻¹.

3.7.2.2.3.2 Reflectivity (Z), Differential Reflectivity

For the reflectivity provided by the RDA, from 2 to 230km, all estimates shall (1) meet the following accuracy and precision requirements when averaged over 1 km for all Azimuth Sample Intervals as specified in 3.7.2.2.1:

TABLE 3-15 REFLECTIVITY (Z)

	Accuracy	Precision (Note 2)
Reflectivity(Z)	For a true spectrum width of 4 ms ⁻¹ the standard deviation in the estimate of the reflectivity shall (1a) be less than or equal to 1 dB at SNR ≥ 10 dB (including only the error due to meteorological signal fluctuations).	+0.5 dB
Differential Reflectivity	For a spectrum width of 2 ms ⁻¹ , correlation coefficient ≥ 0.99, and dwell time of 50 ms, the standard deviation in the estimate of the differential reflectivity shall (1b) be less than 0.4 dB for SNR ≥ 20 dB.	0.06 dB
Note 1: Accuracy is defined as one standard deviation (SD) of the estimate, including effects of precision. Note 2: Precision is defined as the quantization, the smallest resolvable increment. This is based upon the method of encoding and the number of bits used to represent the data.		

Automatic calibration shall (2) reduce system hardware drift errors to less than 1 dB.

3.7.2.2.3.3 Correlation Coefficient, Differential Phase

The correlation coefficient estimates shall (1) meet the following accuracy and precision requirements:

TABLE 3-16 Correlation Coefficient, Differential Phase

	Accuracy	Precision (Note 2)
Correlation Coefficient	For a true spectrum width of 2 ms ⁻¹ , correlation coefficient ≥ 0.99 , and dwell time of 50 ms, the standard deviation in the estimate of the correlation coefficient will be less than 0.006 for SNR ≥ 20 dB.	0.0033
Differential Phase	For a true spectrum width of 2 ms ⁻¹ , correlation coefficient ≥ 0.99 , and dwell time of 50 ms, the standard deviation in the estimate of the differential will be less than 2.5° for SNR ≥ 20 dB.	0.35
Note 1: Accuracy is defined as one standard deviation (SD) of the estimate, including effects of precision. Note 2: Precision is defined as the quantization, the smallest resolvable increment. This is based upon the method of encoding and the number of bits used to represent the data. Note 3: Accuracy and precision requirements apply to the 1 degree sample interval. No requirements are stated for the 0.5 degree sample interval.		

3.7.2.2.3.4 Range

The error in range placement of base data output from the RDA due to timing instabilities shall (1) be less than 50 meters.

3.7.2.2.4 Minimum Detection Capability and Dynamic Range

The RDA shall (1) provide minimum detection capability in both the H and V channels at least a 0 dB signal-to-noise ratio for a -8.5 dBZ_e target at 50 km while operating in simultaneous transmission, horizontal and vertical (SHV) polarimetric mode.; Provide minimum detection capability in either the H or the V channel of at least a 0 dB signal to noise ratio for a -12.5 dBZ target at 50 km while operating in either of the single polarization modes (H or V).; Provide an effective dynamic range at least 93 dB for both reflectivity and velocity data, where Dynamic Range is defined to as the range from the noise level to the 1db compression point; Not use RF Sensitivity Time Control (STC).

3.7.2.3 Antenna

3.7.2.3.1 Antenna Assembly

The antenna assembly shall (1) include items such as a mounting pedestal, drive mechanism(s), rotary joint, transmit and receive feed horn capable of processing both Horizontally polarized signals and Vertically polarized signals, reflector, radome and other items needed to comply with the mechanical and electrical requirements described in this document. The antenna shall (2) be designed to be mounted inside a radome atop a tower. The tower is available in 5 meter height increments with a maximum height of 30 meters. The antenna shall (3) meet all requirements herein while operating under the environmental conditions defined in Section 3.2.6.

3.7.2.3.2 Antenna Performance

The antenna performance requirements are derived from the RDA performance requirements contained in Section 3.7.2.2 and the following specific requirements apply to both Horizontally and Vertically polarized signals:

- a. The antenna type shall (1) be single pencil beam.
- b. The antenna shall (1a) provide both linear horizontal polarization and linear vertical polarization.
- c. The antenna pattern for both polarizations shall (2) provide a median gain of ≤ -10 dBi.
- d. The antenna shall (3) provide a half power beamwidth in azimuth and elevation and for both polarizations of ≤ 1 degree.
- e. The main lobes of H polarized radiation and V polarized radiation down to 20 dB below the peaks shall (3a) be within 1 dB of each other.
- f. The horizontally polarized signal shall (3b) be within 0.1 degree of true horizontal, and the vertically polarized signal shall (3c) be within 0.2 degree of true vertical.
- g. Sidelobe Levels:
 - (1) In any plane, the first sidelobe shall (4) be less than or equal to -25 dB relative to the peak of the beam.
 - (2) In the region between ± 2 and ± 10 degrees, the sidelobe level shall (5) lie below the straight line connecting -25 dB at ± 2 degrees and -34 dB at ± 10 degrees.
 - (3) Between ± 10 degrees and ± 180 degrees, the sidelobe envelope shall (6) be less than or equal to -40 dB relative to the peak of the beam.
 - (4) Beyond ± 10 degrees, the peak of an individual sidelobe may be averaged with the peak of the two nearest side-lobes, one on either side; or with the peaks of the four nearest side-lobes, two on either side, provided that the level of no individual sidelobe exceeds the -40 dB level by more than 6 dB and that the combined angular extent of all peaks exceeding -40 dB does not exceed 30 degrees.
- h. Cross Polarization Radiation:
 - (1) In any plane, the cross polarization radiation shall (7) be at or below the level specified for the sidelobe and back-lobe levels of the co-polarized radiation within the angular regions applicable to these lobes.
 - (2) Within the main lobe angular region, the cross polarization component shall (8) be at least 25 dB below the peak of the main beam of the co-polarized radiation.
- i. Beam location reporting error shall (9) be ≤ 0.15 degrees.

3.7.2.3.3 Antenna Design

The antenna assembly shall (1) be designed to facilitate removal and replacement of any portion except the antenna reflector, radome, and the pedestal. The antenna assembly shall (2) be constructed to be disassembled and reassembled for maintenance and repair at the top of the

antenna tower using tools supplied. The antenna assembly design shall (3) include provisions for routine inspection of critical internal pedestal and drive mechanism parts without disassembly. The design shall (4) allow any maintenance activity to be completed within the repair times defined in Section 3.2.4 (Maintainability). Provision shall (5) be made to introduce calibrating test signals into both the Horizontal and Vertical receiving channels as close to the feedhorn as practicable.

3.7.2.3.4 Radome

The RDA shall (1) include a rigid radome. The radome shall (2) protect the antenna and pedestal from hazards, hazardous weather, direct wind forces (to reduce antenna structural requirements), and severe weather conditions (to ensure severe weather performance and operation during heavy snow conditions). The radome shall (3) have a surface which precludes rain sheeting and facilitates the prevention of icing and snow buildup. The radome shall (4) incorporate venting. Means for closing the radome vents shall (5) be provided.

3.7.2.3.4.1 Radome Service Conditions

The radome shall (1) withstand all of the environmental conditions defined in Section 3.2.6.

3.7.2.3.4.2 Radome Radiation Considerations

The radome design shall (1) minimize the effect of radome attenuation, beam distortion, polarization degradation, and sidelobe perturbations due to the radome. These effects shall (2) be minimized for any look angle through the radome used in the operational scanning sequences. The effects of the radome shall (3) be included in meeting the RDA performance requirements. The procedures for performing antenna measurements shall (4) be based on the applicable details of Appendix A, Section 50 of MIL-STD-469.

3.7.2.3.5 Pedestal and Antenna

The pedestal and antenna shall (1) meet the requirements of the following sections.

3.7.2.3.5.1 Pedestal Requirements

The following pedestal requirements will be met:

- a. The pedestal type shall (1) be elevation over azimuth.
- b. The pedestal shall (2) provide continuous 360 degree rotation in azimuth (both directions) and from at least -1 degree to +60 degrees in elevation.
- c. The pedestal shall (3) provide a variable rotation rate between zero degrees per second and 30 degrees per second.
- d. The pedestal shall (4) provide an acceleration of at least 15 degrees per second squared (both azimuth and elevation).
- e. The pedestal shall (5) provide a deceleration of at least 15 degrees per second squared (both azimuth and elevation)
- f. The absolute value of the positioning error, including effects of angular encoding precision, shall (6) be less than or equal to 0.2 degrees (both azimuth and elevation).
- g. The angular encoding precision shall (7) be less than or equal to 0.1 degrees (both azimuth and elevation).

3.7.2.3.5.2 Antenna Mounting

The antenna pedestal shall (1) be designed for installation within the radome.

3.7.2.3.5.3 Antenna Scanning Control

A programmable control of antenna scanning shall (1) be provided. Any changes to the tilt sequencing and scan rates of the set of scan strategies shall (2) only be accomplished through changes issued in adaptation data released by the WSR-88D ROC.

3.7.2.4 Transmitter

The WSR-88D radar equipment shall (1) be governed by the requirements set forth in Section 5.3, Radar Spectrum Engineering Criteria (RSEC) of the Manual of Regulations and Procedures for Federal Radio Frequency Management. (These requirements apply to promote an efficient use of the RF spectrum and are not intended to restrict new design approaches required for attaining performance goals). RSEC provisions of Subsection 5.3.3 (Criteria D, subparagraphs 2, 3, 6, and 7) are applicable to all WSR-88D Systems operating anywhere in the 2.7 - 3.0 GHz band. In addition, WSR-88D Systems designated as operating in high density areas shall (2) be subject to the RSEC provisions of Subsection 5.3.3, subparagraphs 9, 9a, and 9b.

3.7.2.4.1 Transmitter Interlock

An interlock shall (1) be provided which automatically halts RF emissions when an antenna stoppage occurs for a period of thirty seconds or more. Means shall (2) be provided for manually overriding this interlock at the maintenance position. RF emissions shall (3) be halted five minutes (± 10 seconds) from the time that the interlock override is initiated by the maintainer.

3.7.2.4.2 Transmitter Recovery Time

The time for the transmitter to recover from primary power feed interruptions and outages shall (1) be based on the required klystron heater warm-up time. The recovery time, after heater voltage is restored, shall (2) be as listed in Table 3-16.

TABLE 3-16 TRANSMITTER RECOVERY TIMES

HEATER OFF TIME (MINUTES)	RECOVERY TIME
less than 0.5	None
From 0.5 to 5.0	Equal to heater off time
Greater than 5.0	12.0 Minutes

3.7.2.4.3 Redundant Transmitter

Any redundant transmitter configuration (3.2.5, 3.2.1.5.1, and 3.2.1.5.2) shall (1) include a four port waveguide switch and a high power RF dummy load. The waveguide switch shall (2) be configured to permit maintenance of one transmitter while the second transmitter is fully operational into the antenna. There is no requirement for both transmitters to be operating at the same time while on backup power.

Transmitter RF emissions shall (3) be disabled if the waveguide switch is not positively engaged. The high power RF dummy load shall (4) be capable of operating continuously at full transmitter power without liquid or forced air cooling. The redundant transmitters shall (5) operate at different frequencies separated by a minimum of 30 MHZ. Each transmitter at FAA sites shall (6) communicate with the FAA RMS in accordance with the NEXRAD RDA/RMS ICD.

3.7.2.4.4 Pulse Repetition Frequency

The RDA design tradeoffs between the maximum unambiguous range, velocity estimation requirements, and volumetric scanning rate shall (1) determine the set of PRFs for the transmitter. Requirements for the automatic control of PRF are in Section 3.7.1.2.

3.7.2.4.5 Transmitter Maintenance Control and Protection

The design includes the following transmitter control and protection circuits.

3.7.2.4.5.1 Protective Circuitry

Protective circuitry shall (1) be provided to adequately protect major transmitter components from damage. This circuitry shall (2) include, but not be limited to, sensing, control and visual indication (with memory) at the RDA maintenance position of malfunctions in the following parameters.

Status messages for these parameters shall (3) be sent to the RDA HCI, the RPG interface and the FAA RMS interface:

- a. Modulator Module Failure
- b. Waveguide Arcing
- c. HV OverCurrent
- d. Modulator Overload
- e. Waveguide Air Pressure
- f. Waveguide Air Humidity
- g. RF Output Tube Temperature
- h. Coolant Flow
- i. High Voltage Standing Wave Ratio

3.7.2.4.5.2 Maintenance Controls

Principal maintenance and tuning controls shall (1) be provided on the interior front panel of the transmitter/modulator so that optimum operation may be obtained in a minimum of time. The capability shall (2) be provided to select the pulse-width and PRF for maintenance purposes. The capability shall (3) be provided for manual control of the RDA for maintenance purposes. Transmitter control and status shall (4) be provided to FAA RMS in accordance with the NEXRAD RDA/RMS ICD.

3.7.2.5 Receiver

The receiver unit shall (1) be a stable, sensitive, wide dynamic range receiver providing the flexibility to implement a variety of interference rejection filters. The radar equipment emissions shall (2) conform with requirements established in the Manual of Regulations and Procedures for Federal Radio Frequency Management, RSEC Subsection 5.3.3 (Criteria D), subparagraphs 2, 3, 6 and 7. In addition, the radar shall (3) be capable of complying with subparagraphs 9 and 9c when operating in designated high density areas. The receiver shall (4) be capable of producing both Horizontal and Vertical polarization RF and IF data.

3.7.2.6 Signal Processing Capability

Doppler radar output data shall (1) be processed to provide, at a minimum, estimates of reflectivity, mean radial velocity, and spectrum width, differential reflectivity, correlation coefficient and differential phase in accordance with the requirements defined in Section 3.7.2.2. In addition, the processor shall (2) be capable of estimating reflectivity and velocity at SNRs down to and including -

3 dB. The Z, V, and W estimates shall (3) be discarded when the corresponding SNR is less than a set threshold value.

Independent threshold values shall (4) be used for each variable (reflectivity, mean radial velocity, spectrum width, differential reflectivity, correlation coefficient and differential phase). The capability shall (5) be provided to use a separate set of threshold values for each scan strategy and each scan rate used. All threshold values shall (6) be adaptable at the RPG in increments of 0.125 dB, over a range of SNR's of at least -12 dB to +20 dB. Outbound radial velocities shall (7) be defined as positive velocities. Inbound radial velocities shall (8) be defined as negative velocities.

3.7.2.7 Ground Clutter Suppression

The objective of the ground clutter suppression capability is to improve the measurement of weather return parameters in the presence of clutter. Clutter suppression is defined as the ratio of clutter input to clutter output, normalized to filter noise gain. Clutter suppression performance is described by the improvement in the ability to measure weather return parameters from a Gaussian random process weather model in the presence of two different clutter models. Clutter model A is a Gaussian random process with a Gaussian spectrum centered at zero mean velocity. The clutter root mean square (rms) spectrum width is the rms of 0.1 ms^{-1} plus the rms spectrum width resulting from the antenna rotation rate at the lowest two elevation angles. Clutter Model B represents a scattering echo from a point target and consists of a complex wave form with an amplitude envelope which is an approximation of the two-way antenna pattern.

3.7.2.7.1 Ground Clutter Suppression Requirements

The RDA shall (1) provide ground clutter suppression capability of at least 30 dB in both the H and V channels. The RDA shall (2) provide ground clutter suppression capability for the Doppler channel which meets the requirements shown in Table 3-17 for mean radial velocity (V) and spectrum width (W). In areas where the clutter suppression is not applied, there shall (3) be no degradation in the weather return parameter measurement accuracy with respect to sections 3.7.2.2.3.1 and 3.7.2.2.3.2.

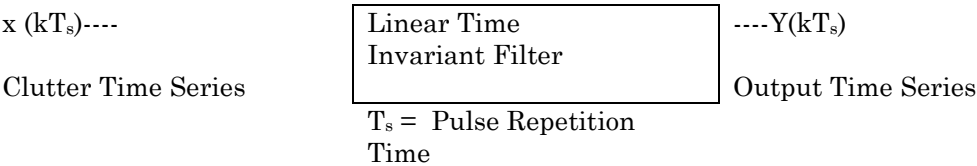
For testing purposes, the following shall (4) apply:

- a. Ground clutter suppression performance will be measured at a nominal frequency of 2.85 GHz carrier.
- b. Ground clutter suppression requirements are met when implemented for the lowest two elevation scans on both scan strategies 1 and 2 (see Appendix I).

TABLE 3-17 CLUTTER SUPPRESSION REQUIREMENTS FOR MEAN RADIAL VELOCITY AND SPECTRUM WIDTH ESTIMATES

Minimum Usable Mean Radial Velocity, $V_{\min}$ in ms^{-1} (see Section 3.7.2.7.2)	Required Ground Clutter Suppression Capability (I) in dB (Spectrum Width = 4 ms^{-1})	
	Clutter Model A	Clutter Model B
2	20	20
3	28	30
4	50	50

If clutter suppression is accomplished by a linear time invariant filter followed by a conventional weather return parameter estimation process such as a pulse pair algorithm, the clutter rejection performance is the ratio of filter input power to filter output power for each of the two clutter models, as illustrated in Figure 3-3. The window time duration shall (5) correspond to the appropriate weather parameter estimation interval.



Clutter Model	Algorithm for Computing Clutter Suppression Capability: I
A	$I = 10 \log_{10} [\langle x(kT_s) ^2 \rangle / \langle y(kT_s) ^2 \rangle]$
B	$I = 10 \log_{10} [\max \langle x(kT_s) ^2 \rangle / \max \langle y(kT_s) ^2 \rangle]$

$\langle \rangle$ represents the value of the time average over a moving window as the antenna scans past a point target. The scan rate assumed should be that used on the lowest two elevations.

$| \ |$ represents absolute value.

Figure 3-3 Clutter Suppression Capability Characterization for Linear Time Invariant Clutter Filtering

If clutter suppression is accomplished by a frequency domain filter, then a minimum number of time series samples are required to process each resulting clutter corrected estimate. For contiguous surveillance scans, a minimum of 16 samples shall (6) be provided by the scan strategy. For contiguous Doppler scans, a minimum of 32 samples shall (7) be provided by the scan strategy. *(Note: these are the minimum numbers of samples used in prior studies to certify the Gaussian Model Adaptive Process (GMAP) clutter filter used in the Open RDA.)*

3.7.2.7.2 Clutter-related Estimate Errors

In the presence of clutter, weather return parameter estimate errors are due to the following two components:

- a. An error results from undesired effects on the weather returns due to the introduction of the clutter suppression device.

Bias error contributions by the clutter suppression device to the reflectivity estimates shall (1) be less than or equal to the values given in Table 3-18. This bias is measured with an input signal-to-clutter ratio (SCR) of at least 30 dB. It applies for all mean radial velocities (V) and shall (2) be systematic, i.e., predictable from mean radial velocity and spectrum width parameters of the Gaussian random process weather model.

Bias and standard deviation contributions by the clutter suppression device to the mean radial velocity and spectrum width estimates shall (3) each be less than 2 ms⁻¹ for mean radial velocities greater than or equal to the minimum usable mean radial velocity, Vmin, for the following conditions:

- (1) Clutter suppression capability selected from Table 3-17.
- (2) Input SNR of at least 20 dB.
- (3) Input signal-to-clutter ratio of at least 30 dB.
- (4) A spectrum width of 4 ms⁻¹.

b. An additional error results from the clutter residue (which is a function of the signal-to-clutter ratio at the output of the filter).

With the clutter suppression device operating and a spectrum width equal to or greater than 4 ms⁻¹, the expected bias error in the reflectivity estimate due to the clutter residue shall (4) be less than 1.0 dB for an output signal-to-clutter ratio of 10 dB.

- (1) The contribution of the clutter residue to the bias error for the mean radial velocity estimate shall (5) be equal to or less than 1 ms⁻¹ for the following conditions:
 - (a) Output signal-to-clutter ratio of 11 dB.
 - (b) Clutter model A (see Section 3.7.2.7).
 - (c) $V = (V_a/2)$ (where V_a is the Nyquist velocity).
 - (d) $W = 4 \text{ ms}^{-1}$.
- (2) The contribution of the clutter residue to the bias error for the spectrum width estimate shall (6) be equal to or less than 1 ms⁻¹ for the following conditions:
 - (a) Output signal-to-clutter ratio of 15 dB.
 - (b) Clutter model A.
 - (c) $V = (V_a/2)$.
 - (d) $W = 4 \text{ ms}^{-1}$.

TABLE 3-18 MAXIMUM ALLOWABLE BIAS IN REFLECTIVITY ESTIMATES DUE TO THE CLUTTER SUPPRESSION DEVICE

Weather Spectrum Width (W) (ms ⁻¹)	Maximum Allowable Bias in Reflectivity Estimate (dB)
1	10
2	2
³ 3	1

3.7.2.8 Control Functions

The RDA includes all hardware and software necessary for operational control and monitoring of the RDA functional area. The RDA control capability supports performing the transitions between

System States as described in this section. The RDA control capability shall (1) also support the running of off-line diagnostics as defined in Section 3.7.6.1.

The RDA HCI shall (2) provide the capability to shutdown in a controlled manner the following parts of the system:

- a. The RDA
- b. The wideband communications links.

3.7.2.8.1 RDA System States

The RDA System States shall (1) be as follows:

- 1. Power-off. All power is removed from radar equipment.
- 2. Halt. Power is applied to all radar equipment except for the RDA Processors.
- 3. Idle. Power is applied to all radar equipment. Operating System software is running on all processors. No application software is running.
- 4. Ready. Application software is executing. Three Ready Modes exist (see Appendix B, Definitions).

3.7.2.8.2 RDA Transitions

RDA Transitions are processes that occur to move from one state to another state within the RDA system. They are initiated when selected by the operator or automatically when specific conditions are present in the RDA system.

3.7.2.8.2.1 Cold Startup Transition

The Cold Startup Transition of the RDA functional area shall (1) provide the capability to start up the RDA either initially or after a failure requiring maintenance action to be taken at the RDA. The cold startup transition shall (2) provide progress and other information related to the status of the Cold Startup transition. After cold startup, the WSR-88D shall (3) allow the Operator to initiate or inhibit data transfer from the RDA to the RPG. (See Appendix B for Cold Startup definition.)

Following the cold startup, data available within the RDA shall (4) be provided to the MSCF, RDA HCI and the FAA RMS interface for test and adjustment purposes. It shall (5) be possible to initiate sending data to the RPG, if it was inhibited prior to or following the completion of the cold startup procedure, without repeating the entire cold startup procedure.

3.7.2.8.2.2 Terminate transition

Terminate shall (1) transition the system from the Ready System state to the Idle System state. This process shall (2) stop all application software at each processor in the RDA system. Terminate shall (3) cause the RDA to stop sending data to the RPG. Terminate shall (4) result in notification to all users.

3.7.2.8.2.3 Applications Load transition

Applications Load shall (1) transition the system from an Idle System state to the Ready System state. This process shall (2) load the application software for each processor and bring the system to the Ready system state.

3.7.2.8.2.4 Shutdown transition

Shutdown shall (1) transition the system from an Idle System state to the Halt System state. Power shall (2) be removed from all RDA processors.

3.7.2.8.2.5 Restart transition

Restart shall (1) transition the system from the Ready System state through Idle and Halt back to the Ready System state. This action will correspond to a system reboot. (See Appendix B for Restart information) Restart shall (2) be initiated by exception handling functions. Restart shall (3) be capable of being initiated on Operator command. The restart transition shall (4) be capable of being performed by one person.

WSR-88D shall (5) allow the operator to initiate restart from:

- a. RPG HCI
- b. RDA HCI,
- c. FAA RMS interface

Restart shall (6) be initiated by automatic or manual equipment substitution action if redundant or standby equipment is provided. A restart action shall (7) place the RDA into a standby state, then reinitialize the RDA applications software, and then place the RDA into an operational state. The RDA shall (8) return to its default state after the transition is complete if the restart was initiated from the RDA HCI or FAA RMS.

If the restart is the result of exception handling, the RDA shall (9) go to Standby after the restart transition is complete. The RDA shall (10) return to its state prior to restart after the transition is complete if the restart was initiated from the RPG. All unit adaptation and control information shall (11) be stored in such a manner that required response times can be met for a restart up to at least one hour after power interruption.

3.7.2.8.2.6 Startup transition

Startup shall (1) transition the system from the Halt System state to the Ready System state.

3.7.2.8.2.7 Power Up transition

Power Up shall (1) transition the system from the Power-off System State to the Halt System state.

3.7.2.8.2.8 Maintenance transition

Maintenance shall (1) transition the system from the Halt System state to the Idle System state.

3.7.2.8.2.9 Power Down transition

Power Down shall (1) transition the system from the Halt System state to the Power-off System state.

3.7.2.8.3 Shutdown

A shutdown of the RDA shall (1) return it to a state from which an RDA startup may be performed. A controlled shutdown of the RDA shall (2) notify RPG of the shutdown. An RDA shutdown shall (3) stop sending data to the RPG. A controlled shutdown of the RDA shall (4) also result in notification of the shutdown to all users.

3.7.2.8.4 Warm Startup

The warm startup function shall (1) provide both local control (RDA) and remote control. (See Appendix B for detailed Warm Startup information.) The warm startup transition shall (2) be capable of being initiated by the Operator. The warm startup transition shall (3) be capable of being

performed by one person. WSR-88D shall (4) allow the operator to initiate warm startup from the following:

- a. RDA HCI,
- b. FAA RMS Interface

3.7.2.8.5 Control Commands

Control commands shall (1) be accepted from the RPG, RDA HCI, or FAA RMS. See Sections 3.7.1.6 and 3.7.3.4 for additional requirements related to RDA control commands.

3.7.2.9 Monitoring and Error Detection Capability

The capability shall (1) be provided to send RDA status, including a positive indication that the RDA is operating within all specification parameters, to the RDA HCI, RPG, MSCF and the FAA RMS interface. The RDA equipment shall (2) include built-in test equipment to provide a combination of on-line and off-line status monitoring and self test to support system operation and fault isolation to the degree required to meet the maintenance concept as described in Section 3.4.1. All alarms shall (3) drive indicators on individual units, where appropriate, and at the maintenance position in the RDA.

3.7.2.10 Display Functions

The RDA shall (1) include a capability to display radar status data, calibration data, performance data, and adaptation data for purposes of system monitoring, maintenance, and adjustment. The design shall (2) include displays, meters, and indicators as necessary for maintenance. The RDA shall (3) support a local Real Time Display of base data. All RDA text messages shall (4) use descriptive English language text.

3.7.2.11 Calibration

The RDA shall (1), during each volumetric scan sequence, automatically perform the calibration measurements defined in this section and automatically correct the measurement of base data in the subsequent volume scan. The calibration data and parameters shall (2) be included in the RDA status data which accompanies base data for applications by the users at all levels of data processing. RDA performance data shall (3) be incorporated into the digital information message traffic provided to the RDA HCI, the RPG and the FAA RMS, and shall (3a) be displayable at the RDA maintenance interface.

The calibration measurements include but are not limited to the following:

- a. A measurement of the average transmitted power level shall (4) be monitored.
- b. A calibration of both the H and V reflectivity channels from the channel's first active device through the signal processor shall (5) be monitored.

This calibration shall (6) be timed to correspond to regions of space devoid of both ground clutter and weather echoes. The calibration shall (7) be performed over the entire linear range of performance. The calibration signals shall (8) undergo the same averaging as the weather echoes.

- c. The mean radial velocity and spectrum width processing performance shall (9) be monitored.

This monitoring function shall (10) be timed to correspond to regions of space devoid of both ground clutter and weather echoes. This monitoring function shall (11) include at least one known value of

mean radial velocity and spectrum width at one range cell. The system shall (12) be capable of expansion to monitor new hardware in receiver channel, to include a second receiver for dual polarization and processing parameters from a digital receiver.

3.7.2.12 Setup Mode

The RDA shall (1) include the capability to measure and monitor the following parameters:

- a. Transmitter frequency
- b. Peak transmitted power
- c. Transmitted pulse shape
- d. Transmitted pulse width
- e. Pulse repetition interval
- f. Receiver calibration over entire dynamic range
- g. Mean radial velocity and spectrum width computation over entire Nyquist interval
- h. Complex spectrum in at least one range cell that is controllable over the entire system PRF with the number of samples being variable from 32 to 512 in power of 2 increments.

The measurements and status of these parameters shall (2), when available to the RDA application software, be sent to the RDA HCI, the RPG and the FAA RMS interface.

3.7.2.13 Archive Level I Interface Connections

The RDA shall (1) provide interface connections to allow Archive Level I data to be obtained. The Archive Level I data shall (2) be the output of the receiver (digital time-domain data). Synchronization, calibration, and other data required to properly interpret the archive data, including date, time, antenna position, etc., shall (3) be available at the Archive Level I interface connection.

RADAR PRODUCT GENERATION (RPG) SEGMENT

3.7.3 Radar Product Generation Functional Area

The Radar Product Generation (RPG) functional area uses base data received from the Radar Data Acquisition (RDA) functional area to generate a set of products which contains information on meteorological phenomena including, but not limited to, the following:

- a. Precipitation
- b. Wind and Wind Shear
- c. Tornadoes
- d. Fronts and Fine Lines
- e. Tropical Cyclones
- f. Mesocyclones
- g. Thunderstorms
- h. Turbulence
- i. Icing Conditions
- j. Hail
- k. Freezing/Melting Level
- l. Hydrometeor Discrimination
- m. Biological Scatterer Identification
- n. Data Quality Assessment

The products are then distributed to the users (see Figure 3-2).

The RPG shall (1) be capable of accepting, storing, and processing base data for differing numbers of elevation scans and scan rates, including those scan strategies defined in Appendix I. The RPG shall (2) provide for real-time distribution of base data to multiple users. It shall (3) be possible to perform RPG functions in either a fully automated mode or manually through the MSCF or through commands received from the FAA RMS interface.

In a fully automated mode (with/without the FAA RMS interface the RPG shall (4) be capable of being operated unattended. The RPG functional area capabilities are detailed in the following sections. All capabilities will not be provided at every site. Tables and/or equipment stands shall (5) be provided to support all display, data entry, and control equipment. All display, data entry, access to RDA and RPG control, and RDA, RPG, and PUP status monitoring and error detection will be handled through the MSCF control position and the FAA RMS interface.

3.7.3.1 Product Generation

Base data shall (1) be used to detect and measure meteorological phenomena based on supplied algorithms. The algorithms are defined in the NEXRAD Algorithm Report, R400-AR401. Algorithms shall (2) be implemented in an easily changeable manner, based on a system design to be specified.

The RPG shall (3) produce the following types of products (Defined in Appendix B.) as described in Appendix E:

- a. Base Products
- b. Derived Products
- c. Alphanumeric Products

d. Derived Data Array Products

In addition to product data, each product shall (4) include the following descriptive information: the standard annotations set, identification of range obscured data, and appropriate other meteorological information. All times displayed or included in products shall (5) be expressed as Coordinated Universal Time (UTC). All distances to be numerically displayed or included in products shall (6) be expressed in nautical miles, all altitudes in feet, and all speeds in knots. Presentation of base products and derived products to operational personnel assumes the use of a color graphic display. Presentation of alphanumeric products does not require this capability.

3.7.3.1.1 Base Products

Non Polarimetric Products

Base products shall (1) provide reflectivity estimates for each elevation angle to a range of 460 km (Note the RDA is not required to provide data above 70,000 ft MSL. See Section 3.7.2.2.1.1 for base data coverage requirements). Base products shall (2) provide mean radial velocity and spectrum width estimates for each elevation angle to a range of 230 km. The highest resolution provided in a base product shall (3) be equal to the resolution available from the base data.

Polarimetric Products

Polarimetric base products shall (3a) provide differential reflectivity, differential phase, specific differential phase and correlation coefficient estimates for each elevation angle to a range of 300 km. The highest resolution provided for differential reflectivity, differential phase, and correlation coefficient shall (3b) be equal to the resolution available from the base data.

These products shall (4) be formatted to facilitate display as a vertical projection onto the earth's surface, e.g., 100 km at 20 degrees will be displayed at approximately 94 km. For a given coverage area (see Table E-1a), each elevation scan shall (5) be displayed at the same geographic scale relative to the earth's surface.

3.7.3.1.2 Derived Products

Derived products shall (1) provide users with access to large volumes and varieties of data in easily displayable formats. These large volumes of data shall (2) be concentrated into easily interpretable products to identify specific meteorological phenomena.

3.7.3.1.3 Alphanumeric Products

Alphanumeric products shall (1) provide weather information in a form suitable for display. This information shall (2) include alert messages, supplemental and summary data, and free text messages.

3.7.3.1.4 Derived Data Array Products

Derived data array products shall (1) provide information in a form suitable for transmission to and processing in a system external to the WSR-88D System. These products are not in a displayable format and can include intermediate outputs from the meteorological algorithms.

3.7.3.2 Product Storage

The RPG shall (1) provide the capability for on-line storage of all products. The RPG shall (2) accept input from the MSCF for defining and changing the product mix and retention time (up to a maximum of six hours) for products to be retained in the on-line RPG storage. Products stored within the RPG on-line storage shall (3) not be capable of being altered from a product user interface.

The RPG on-line storage of products shall (4) be capable of storing base data for up to 1 hour and other products for 6 hours as defined by the full load product mix in Table E-1a. When load shedding of products stored on-line is required, the shedding shall (5) be performed by deleting the oldest product(s) with the least amount of retention time and replacing it with the newest product produced.

3.7.3.3 Product Distribution

Each RPG shall (1) make products available for distribution to the users as defined in the following sections. Upon receipt of the appropriate message from a user, the RPG shall (2) transmit a list of products available for distribution to that user. The number of sites which an RPG must be capable of distributing products shall (3) be limited only by the limits shown in Section 3.2.1.5. Adaptation data inputs from the RPG HCI or FAA RMS interface shall (4) control the number and types of products distributed through any product distribution user port. Characteristics of external interfaces are defined in section 3.7.1.8, Interface Requirements.

3.7.3.3.1 Distribution to Class 1, 98 and/or 99 Users

For each User, the RPG shall (1) provide those products requested by that user, including both products to be sent on a routine basis and products requested on a onetime basis. Products generated because alert criteria were met shall (2) be transmitted to the requester prior to all other products defined for transmission to that requester. Different Users may request different product sets for routine receipt from an RPG. The distribution of products shall (3) satisfy the users' frequency and priority requirements within the constraints of available communications bandwidth.

If the culmination of requested products from a line exceeds the narrowband communication line capacity during a volume scan time, the RPG shall (4) distribute products based on transmission priority. The transmission priority value shall (5) provide control of the product transmission sequence to the user so that a onetime product selection may be interleaved with or transmitted prior to products defined for routine transmission. If no transmission priority value is included with the onetime product selection, then the product shall (6) be transmitted prior to the products defined for routine transmission to the user.

All products stored on-line within the RPG shall (7) be available to any of these user types. Any product which has not been generated prior to the receipt of a request (e.g., cross section) shall (8) be generated and distributed to the requester. The RPG shall (9) support on-line requests for changes and additions to the set of products routinely sent to one of these user types. It shall (10) also support on-line changes to their frequency or priority. In the event a product request cannot be met, the RPG shall (11) transmit a response message describing the reason why.

3.7.3.3.1.1 Distribution to a Collocated Legacy Class 1 User

This section is no longer applicable to this specification.

3.7.3.3.2 Distribution to Class 2 Users

Each RPG shall (1) be capable of providing products to any Class 2 User in response to a onetime request (i.e., dialup or WAN). The specific products available for distribution to any Class 2 User shall (2) be controlled by adaptation data through the RPG HCI or FAA RMS interface.

3.7.3.3.3 Deleted

3.7.3.3.4 Distribution of Base Data to Users

The RPG shall (1) distribute the base data stream (unaltered from the RDA) through the Base Data Distribution Interfaces.

3.7.3.4 RDA and RPG Control

Control capabilities will provide for operation of the WSR-88D System. These include, but are not limited to:

- a. RPG Startup
- b. RPG Restart
- c. RDA Restart and Warm Startup
- d. User Interface Connection Control
- e. Controlled Shutdown
- f. RDA Mode Control
- g. RPG Reconfiguration
- h. Level II Control
- i. Product Generation Control
- j. Product Storage Control
- k. RPG Adaptation Control
- l. RDA Backup Power Control
- m. Communication Interface Configuration
- n. RPG and RDA Redundant Channel Control
- o. Volume Coverage Pattern Control
- p. Clutter Suppression Control

3.7.3.4.1 RPG Startup

An RPG startup capability shall (1) be provided to initiate the operation of the RPG without any concern for the current state of the remainder of the WSR-88D System. RPG startup shall (2) be used to initiate RPG operation or after a RPG system failure of longer than one hour. It shall (3) also be used after system failures which make it impossible to recover any data present at the time of the failure.

RPG startup shall (4) also be used when the RPG restart function (see Section 3.7.3.4.2) fails to operate properly. RPG startup response time is specified in Section 3.2.1.3. During RPG startup, it shall (5) resume its operational mode.

The RPG startup capabilities (via the operating system and/or applications software) shall (6) include:

- a. Verification of the configuration.
- b. Initialization of RPG equipment, firmware, and software.
- c. Initialization of all clocks and internal timers.
- d. Establishment of the interface with the RDA (if operating).
- e. Establishment of the interfaces with all applicable users.

f. Notification to the users that the system is again available.

g. Establishment of the FAA RMS (if configured).

3.7.3.4.2 RPG Restart

An RPG restart capability shall (1) be provided to restore the previous state of the RPG after a failure from which recovery is possible without maintenance action. RPG restart shall (2) also be capable of being performed automatically or manually. An RPG restart shall (3) be performed when the RPG experiences non-catastrophic failures, (Defined in Appendix B. (See Catastrophic Failures)) power interruptions of less than one minute, and other such failures. All unit adaptation and control information shall (4) be stored in such a manner that required response times can be met for a restart up to at least one hour after power interruption.

The RPG restart response time is specified in Section 3.2.1.3. The RPG restart capabilities shall (5) include:

- a. Automatic reboot of the system software (if necessary).
- b. Restoration to the operating state (e.g., scan mode, product menus, adaptation data, etc.) existing when service was interrupted.
- c. Re-initialization of all clocks and internal timers.
- d. Reestablishment of the interface with the RDA (if operating).
- e. Reestablishment of interfaces with all applicable users.
- f. Verification of the configuration.
- g. Notification to the users that the system is again available.
- h. Notification to users of potentially degraded data.
- i. Establishment of the FAA RMS interface (if configured).

If the approved design includes a distributed RPG architecture that facilitates restart of individual subcomponents (for example a communication server, product server, control workstation, etc.), then a restart capability for each RPG subcomponent shall (6) be provided to restore the subcomponent to an operational state after a controlled shutdown or a non-catastrophic failure of that subcomponent.

A restart of an RPG subcomponent shall (7) include the following:

- a. Re-initialization of all subcomponent clocks and internal timers.
- b. Verification of the subcomponent configuration.
- c. Reestablishment of the available user interfaces with the RPG.
- d. Notification to the users that the subcomponent is again available.

e. Automatic redistribution of processing load, if the subcomponent is a processor.

3.7.3.4.3 RDA Restart and Warm Startup Control

An RDA restart and warm startup capability shall (1) be provided from the RPG and FAA RMS interface. It shall (2) include those restart and warm startup functions defined in Section 3.7.2.8.2. It shall (3) also include reestablishment of the interface of the RDA with its RPG (if operating).

3.7.3.4.4 Interface Connection Control

The capability shall (1) be provided to connect, terminate, or prevent connection of any RPG interface.

3.7.3.4.5 Controlled Shutdown

The capability shall (1) be provided to shut down in a controlled manner the following parts of the system:

- a. The RDA as stated in Section 3.7.2.8.3.
- b. The communications links with communications interfaces.
- c. The RPG, or its subcomponents.

The controlled shutdown of a communications interface shall (2) not affect other RPG interfaces. The controlled shutdown of the RPG shall (3) include the controlled shutdown of all internal and external communications interfaces, notification of the shutdown to all connected users, and the return of the RPG to a condition from which an RPG startup is required to resume operation. If the approved design includes a distributed RPG architecture that facilitates restart of individual subcomponents (for example a communication server, product server, control workstation, etc.), then a controlled shutdown capability for each RPG subcomponent shall (4) be provided to place the subcomponent into a state from which the subcomponent can be restarted without accomplishing an RPG Startup procedure.

A controlled shutdown of an RPG subcomponent shall (5) include notification to all connected users. A controlled shutdown of one subcomponent (e.g., a communication server, product server, control workstation) shall (6) not affect the operational state of any other subcomponent.

3.7.3.4.6 Mode Control

3.7.3.4.6.1 RDA Mode Control

The capability shall (1) be provided for the RPG to change operational modes of the RDA. This shall (2) include all modes defined in Section 3.7.1.6.4.7.1.

3.7.3.4.6.2 RPG Mode Control

The RPG shall (1) be capable of operational mode. When entering the operational mode, the following capabilities shall (7) also be provided:

- a. Establishment of interfaces with all applicable users.
- b. Verification of the operational mode configuration.

c. Notification to dedicated users that the system is available.

3.7.3.4.7 RPG Reconfiguration

The capability shall (1) be provided to manually reconfigure RPG equipment. This capability is provided by the RPG HCI and the FAA RMS interface. If the RPG is configured with more than one processor, the capability shall (2) be provided to control the shifting of loads between the processors. At redundant sites, the RPG shall (3) provide the capability to have resources (e.g., the storage capacity, processing capacity) manually reallocated between redundant RPGs.

3.7.3.4.8 Level II Control

The capability is provided as described in Section 3.7.1.6.4.9 to control and monitor data archive through the RPG HCI and the FAA RMS interface.

3.7.3.4.9 Product Generation Control

3.7.3.4.9.1 Routine Product Generation

The RPG shall (1) routinely generate a predefined set of products. In addition, the RPG shall (2) generate those products that have been selected by Class 1 Users for routine receipt. In addition, the RPG shall (3) generate products at the Class 1 User or MSCF specified frequency, on a per product basis.

3.7.3.4.9.2 Alert Product Generation

The RPG shall (1) be capable of generating one product for each of the alert categories (See Appendix J) for which alert conditions are met in any of the Class 1 Users' selected alert areas. When a product is generated due to the detection of an alert condition, if the product is an elevation based product, the product shall (2) be generated for the elevation angle corresponding to the maximum value exceeding the alert condition.

If the product is a window product, it shall (3) be centered at the coordinates of the meteorological phenomenon causing the alert condition. The product shall (4) be automatically sent to the Class 1 User(s) whose alert areas are affected and who have requested the automatic product distribution. The product selected for generation for each alert category shall (5) be an adaptable parameter at the RPG.

3.7.3.4.9.3 One-time Product Generation

The RPG shall (1) generate products in response to onetime product requests from Class 1 Users.

3.7.3.4.10 Product Storage Control

The capability shall (1) be provided to select or change the set of product types to be stored in the on-line storage. In addition, the capability shall (2) be provided to specify, on a per product basis, the frequency at which the product is stored and the storage time period.

3.7.3.4.11 RPG Adaptation Control

The RPG shall (1) store and use the most recent RPG adaptation parameter values, as appropriate to operations. The RPG shall (2) provide the capability to use the baseline adaptation data. A multiple read/multiple write storage device with removable media shall (3) be provided for the backup of all adaptation data. This device may also be used to support other similar archive, logging, backup, or recording functions.

3.7.3.4.12 RDA Backup Power Control

The capability shall (1) be provided by the RPG HCI to perform a systematic, controlled switch over and recovery of operations between commercial and backup power for all WSR-88D equipment located at the radar site.

3.7.3.4.13 Communication Interface Configuration

It shall (1) be possible to configure any narrowband port to emulate any of the three Legacy classes, up to 24 serial ports (fully expanded), by adaptation data alone.

3.7.3.4.14 RDA and RPG Redundant Channel Control

The capability to control the state of the FAA and NWS redundant channels is specified in Section 3.7.1.6.4.15.

3.7.3.4.15 Volume Coverage Pattern (VCP) Control

The capability to select the operational VCP from the set of defined local and remote (adaptation data) VCP suites shall (1) be provided (Note: "Local" scan strategies are stored in the RDA. "Remote" scan strategies are stored in the RPG). The capability shall (2) be provided to modify VCP parameters by elevation scan where appropriate. The parameters shall (3) include, but are not limited to Doppler PRF selection, SNR, sector definition, and velocity measurement increment. The capability to display the current operational VCP shall (4) be provided. The display shall (5) include but not limited to Doppler PRF selection, SNR, sector definition, base data resolution and velocity measurement increment.

3.7.3.4.16 Clutter Suppression Control

The capability to control the application of clutter filtering to include clutter suppression region definition shall (1) be provided. A minimum of 4 files for defining clutter suppression regions shall (2) be stored in adaptation data. Each file shall (3) allow for a minimum of 15 clutter suppression regions.

3.7.3.5 RDA and RPG Status Monitoring and Error Detection

3.7.3.5.1 Status Monitoring

The capability shall (1) be provided to monitor the status of the RDA, the RPG, and each communications interface. This status monitoring capability shall (2) include operational modes, conditions, and parameters, as well as error conditions. Alarm messages received shall (3) be considered status messages for the purposes of further dissemination.

Status shall (4) be provided to the RPG HCI and the FAA RMS interface: (a) when requested; (b) periodically, as specified; and (c) when there is a change in the status of the RDA, the RPG, or any of the WSR-88D communications interfaces. A status message shall (5) be provided to any communications interface: (a) when requested, (b) upon connection, and (c) when there is a change in the status of the RDA or RPG.

3.7.3.5.2 Storage of Status Messages

The capability shall (1) be provided to detect and store on-line all status and error messages. When stored on-line, a descriptive English language message shall (2) be provided for each status or error message. Status and error messages shall (3) be stored on-line for at least 24 hours.

A hardcopy device for the recording of all status and error messages shall (4) be provided. A multiple read /multiple write storage device with removable media shall (5) be provided for the backup of all status and error messages. This device may also be used to support other similar archive, logging, backup, or recording functions.

3.7.3.6 Archive Level III

This section is no longer applicable to this specification.

3.7.3.7 Data Ingest

3.7.3.7.1 Bias Table Ingest and Processing

Hydrometeorological processing requires rain gage vs. radar bias data for quality control of rainfall estimates. Based on criteria as defined by the Precipitation Adjustment algorithm, the WSR-88D shall (1) ingest a table of rain gage vs. radar bias data determined over various memory timespans.

3.7.3.8 Alert Processing

The RPG shall (1) generate alert outputs based upon automatically identified meteorological phenomena occurring or projected to occur in alert areas defined by each Class 1 User. Each alert area shall (2) be definable within the range of coverage (See Appendix J) on a grid divided into squares of no larger than 16 X 16 km.

3.7.3.8.1 Alert Categories

The RPG shall (1) be capable of generating alerts for all categories of phenomena identified in Appendix J. The categories of Group "a." are ones that have different data for each grid square. The coarsest resolution product available may be used to generate these data. The categories of Group "b." are based on volume data and use the algorithm outputs.

The categories of Group "c." are a subset of "b." but shall (2) alert based on levels or phenomena projected to occur in an alert area according to the forecast motion of the storm cell centroid(s). For each alert area, each Class 1 User may choose up to 10 categories to alert on. The capability for additional categories shall (3) be provided in the amounts listed in Appendix J.

3.7.3.8.2 Alert Criteria

There shall (1) be one set of thresholding criteria for each alert category as outlined in Appendix J. The values used in the thresholding criteria shall (2) be part of the adaptation data. All criteria shall (3) be set up as greater than or equal to the value selected. When an alert condition has been detected, an alert message shall (4) be transmitted to the appropriate user.

3.7.3.8.3 Alert Message

The alert message shall (1) contain at a minimum the alert area, location, date and time, and a brief description of the alert criteria. For those alerts that are storm cell related (i.e., Groups b and c, Appendix J, except for VAD and rainfall accumulation) the storm cell ID shall (2) also be provided. Upon receipt of the appropriate message from a Class 1 User, the RPG shall (3) transmit the alert threshold values to that user.

3.7.3.9 RPG HCI

See Section 3.7.1.6 for additional requirements related to the RPG HCI.

The RPG HCI capabilities will provide for operation of the WSR-88D System. These capabilities include, but are not limited to, the following:

- a. Mode Control
- b. Spot Blanking Control

3.7.4 Principal User Processing Functional Area

This section is no longer applicable to this specification.

FACILITIES SEGMENT

3.7.5 Facilities Functional Area

The Facilities functional area includes construction or modification of buildings and towers, provision and installation of Real Property Installed Equipment (RPIE) such as power generation and air conditioning equipment which are needed in support of the RDA, RPG, and Support functional areas.

At each radar site, the required WSR-88D equipment, structures, and WSR-88D support equipment include, utilities and utility extensions, footings and foundations, ancillary equipment shelter(s) and associated footings and foundations, towers, environmental control, backup power, fuel storage and associated piping, grounding, and wideband communications link to user site including any wideband repeater installations.

Towers constructed at selected sites shall (1) be of structural steel. Towers at FAA facilities will be designed and constructed so as to minimize fall hazards in accordance with FAA O Safety and Health Program, Chapter 10, 29 CFR 1910, Occupational Safety and Health Standards (General Industry), and 29 CFR 1926, Safety and Health Regulations for Construction, FAA Order 3900, 19B, FAA Occupational.

Provisions for a commercial T-1 (1.54 Mb/s) service may be required at certain sites.

Provisions to house the RPG equipment when collocated at the radar site shall (3) also be provided, including the fully expanded RPG, the redundant configuration of Sections 3.2.1.5.1 and 3.2.1.5.2, and the combination of both.

An automatic fire detection capability for the WSR-88D indoor environment equipment located at the radar site (excluding radome interior) shall (4) be provided. Equipment supplied satisfies the requirements with respect to space, electrical, environment, and architecture as specified in Sections 3.7.5.1 through Section 3.7.5.4, respectively.

3.7.5.1 Space

Equipment to be located in an operator work area at a principal user site comprising all equipment required to perform any operator function, including all PUP functional area equipment, and a display terminal for the MSCF and RDA HCI, shall (1) not require a horizontal area greater than 12 square meters.

3.7.5.2 Electrical

The equipment installed in the WSR-88D facilities shall (1) satisfy electrical requirements in the following subsections.

3.7.5.2.1 Power Characteristics

The power characteristics for facilities will satisfy the following:

Steady State Voltage 120/208 VAC $\pm$ 10%
 (3 Phase, 4 wire)

Steady State Frequency 60 Hz $\pm$ 5%

Each WSR-88D System shall (1) be designed to be compliant with IEEE Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits, with respect to lightning, transient suppression, and motor starting disturbances caused by the air conditioning system when both the air

conditioning system and other radar equipment are operating in parallel and receiving power from a standby engine/generator. To accommodate overseas installations, the capability to convert alternate primary power sources to the standard primary power defined above shall (2) be provided. Optional provisions shall (3) be made to accommodate the following primary voltage and frequency configurations:

TABLE 3-20 PRIMARY VOLTAGE AND FREQUENCY CONFIGURATIONS

	single phase volts AC	three phase volts AC	frequency in Hertz	
a	120	208	60	(typical U.S.)
b	127	220	50	
c	220	380	50	
d	230	400	50	
e	240	415	50	
f	277	480	50	
g	100/200	-	50	
h	100/200	-	60	
i	110/220	-	60	

A single frequency converter shall (4) be provided to support a collocated RDA/RPG. A separate converter shall (5) be provided to support the PUP. When connected to the 50 Hz power through the frequency converter, all other aspects of the WSR-88D system shall (6) remain unchanged. Each WSR-88D System shall (7) withstand a power outage without any damage. It shall (8) be capable of automatic restarting and rapid recovery after power is restored.

3.7.5.2.2 Lighting

Radome obstruction lighting shall (1) be provided as per specification AC 70/74601K. Otherwise, WSR-88D facilities shall (2) not require any special lighting effects other than incandescent lighting with dimmer control.

3.7.5.2.3 Cables

Cables used in an outdoor environment shall (1) be designed for direct earth burial and shall (2) be armored to be gopher resistant. The cable shielding, grounding, and routing shall (3) be in accordance with paragraph 5.1.1.2.5 of MIL-STD188-124B, using the guidelines of MIL-HDBK-419/2. The limits on electromagnetic emissions and susceptibility shall (4) be as stated in Section 3.2.6.11 of this document.

3.7.5.2.4 Electromagnetic Interference Shielding

The WSR-88D system shall (1) include sufficient electromagnetic interference shielding and filtering to meet the requirements stated in Section 3.2.6.12. The requirements and practices to be followed regarding shielding effectiveness, screening, seams/joints/apertures, gaskets, and conductive surface coating shall (2) be as given in paragraph 5.3.2 of MIL-STD-188-124B, using the guidelines of MIL-HDBK-419/2. The guidelines for attenuation requirements, selection of suppression components, and mechanical considerations as described by the Electronic Industries Association (Filtering Electronic Equipment, Bulletin No. 9) shall (3) be used.

3.7.5.2.5 Internal and External Cableways

Cableways shall (1) be in accordance with National Electrical Code (NEC) requirements. Electromagnetic interference requirements for cableways are specified in Section 3.2.6.12.

3.7.5.2.6 Grounding

The radar equipment shall (1) operate with a facility grounding system whose resistance complies with the National Fire Protection Association (NFPA), National Electrical Code and NFPA 70AE standards for Electrical Safety in the Workplace. In addition, DOT Standard FAA-STD-019E and FAA G-2100H, Electronic Equipment General Requirements; and ANSI/IEEE Standard 1100-2005 Recommended Practice for Powering and Grounding Electrical Equipment shall (1a) be used. The grounding connections and grounding methods shall (2) be as required in paragraph 5.1.1.3 of MIL-STD-188-124B, using the guidelines of MIL-HDBK-419/2. The resistance of the radar equipment to earth shall (3) be 10 ohms or less.

3.7.5.3 Environmental

The radar equipment shall (1) operate with a facility grounding system whose resistance complies with the National Fire Protection Association (NFPA), National Electrical Code and NFPA 70E standards for Electrical Safety in the Workplace.

In addition, DOT Standard FAA-STD-019E and FAA G-2100H, Electronic Equipment General Requirements; and ANSI/IEEE Standard 1100-2005 Recommended Practice for Powering and grounding Electrical Equipment shall (2) be used.

3.7.5.3.1 Personnel

Working environment and safety requirements for operating and maintenance personnel are stated in Section 3.3.6.

3.7.5.3.2 Equipment

Equipment should be compatible with the environment in which it will operate and not cause any deleterious effects on the natural environment. To provide visual compatibility with the environment, provisions shall (1) be made for the selection (on a case-by-case basis) of optional alternate colors of: the radome, the tower, and any other major outdoor facility item such as a separate shelter, backup power generator, etc.

The choice of alternate colors to be provided shall (2) be in accordance with color numbers 30219, 34127, and 34258 of FED-STD-595A. For FAA safety reasons, provisions shall (3) be made for optional alternate colors of the tower structure (radome not included) in accordance with paragraphs 8, 9, and 10.b of AC 70/7460-1K.

3.7.5.3.3 Electromagnetic Environment

The 2.72.9 GHz band, nationally, is a Government-exclusive frequency band intended for the primary use of the Aeronautical Radio-navigation Service (air traffic control radars) and the Meteorological Aids Service (weather radars). The WSR-88D System operates in the Communication Electronics equipment environments within CONUS as well as at DOD overseas bases. This will require preplanning and EMC analysis to minimize the impact of interfering signals.

To ensure that the WSR-88D System will not be susceptible to electromagnetic interference within the allocated frequency band, the following steps are taken:

- a. The antennas shall (1) be capable of being located at the sites designated by the Government.
- b. The design shall (2) take into account the presence of a composite interference signal profile. A composite interference signal profile (composite of contributing PRFs, scan rates, pulse widths, frequencies, etc.) can be based on equipment in the high equipment density areas within CONUS.

c. The design shall (3) provide the flexibility to incorporate the recommendations of EMC analysis including one or more of the following items:

- (1) Inserting filters in the receiver equipment,
- (2) Providing adjustable transmitter and receiver frequencies in the allocated band
- (3) Augmenting the signal processing software capability,
- (4) Incorporating special rejection circuits for interference, and
- (5) Providing adjustable antenna coverage volume.

3.7.5.4 Architecture

Architectural requirements for installed equipment are detailed in the following subsections.

3.7.5.4.1 Clearances

Consoles and cabinets shall (1) be of a modular design with no unit exceeding 0.8 m x 1.2 m in plan; the vertical dimension shall (2) not exceed 2 m. Minimum clear working space will be maintained in accordance with 29 CFR 1910.303.

3.7.5.4.2 Access

Access to all enclosures for cabling and ventilation shall (1) be designed to allow installation on both standard and raised floor facilities. A cable access capability shall (2) be provided through the rear of the cabinet near the base, and also through the bottom surface of the cabinet. Ventilation openings shall (3) be limited to the front, rear, and top surfaces of the cabinet.

3.7.5.4.3 Human Factors

Human factors requirements for consoles and other equipment are in Section 3.3.7.

3.7.5.4.4 Telecommunications

3.7.5.4.4.1 RDA Intercom and RDA/RPG Voice Communications

The RDA shelter shall (1) include intercom capability. At a minimum, the intercom capability shall (2) provide for voice communication between the radome interior and the RDA shelter interior. The intercom system shall (3) operate independent of primary power. All equipment, cabling, and connections shall (4) be provided.

Options for UHF/VHF radios, (minimum 3002 voice grade for domestic usage) 4-wire dedicated leased line, or (minimum 3002 voice grade for domestic usage) 2-wire dialup service shall (5) be provided in the RDA shelter to facilitate communications for the primary maintaining agency and operating personnel.

3.7.5.4.4.2 RDA Data Communications

The RDA facilities, as a minimum, shall (1) support connection and equipment space, as appropriate, for either private hard-wire, private T-1, commercial T-1, private fiber optic, or private MLOS connection to allow base data transfer to the RPG.

The characteristic circuit requirements for the RDA are as follows:

- a. Commercial facilities for the base data access and control link to the RPG shall (2) support at a minimum AT&T compatible framed T-1 service at 1.544 Mbps. The same signal format is required for the private hard-wire, private T-1, private fiber optic, or private MLOS facilities.

3.7.5.4.4.2.1 Microwave-Line-of-Sight System - This Section is not Applicable

3.7.5.4.4.2.2 Microwave Repeaters - This Section is not Applicable

3.7.5.4.4.2.3 MLOS Facilities - This Section is not Applicable

3.7.5.4.4.3 RPG Data Communications

A communications demarcation frame with the capacity to terminate 100 pairs of telephone cable shall (1) be provided for support of voice grade circuits used for serial modem connections at each RPG. The facility communications demarc shall (1a) have sufficient capacity to support telephone provided equipment for one digital wide area network (WAN) link per agency (e.g. DoD, NWS, & FAA) and for each base data user. An appropriate multi-conductor cable with sufficient capacity to connect the maximum number of product user ports as defined in Table 3-4b shall (2) be provided from the RPG modems to the communications demarcation frame. Both ends of this cable shall (3) be appropriately marked to allow future expansion of ports.

Provisions for intrusive line testing shall (4) be included. Intrusion shall (5) allow on-line monitoring as well as the ability to break the line(s) for look-back and look-forward testing. The point of intrusive testing shall (6) be at communications/equipment racks or cabinets and shall (7) provide for easy connection of test equipment via a patch panel or other similar equipment. Appropriate commercial/private telecommunications circuits will be provided and cross-connected, as required.

There are a number of possible RPG configurations. The number and types of communication circuits required for a specific RPG site depends on the mix of product user ports served by that site, the proximity of users to the specific site, and the proximity of the RPG to the RDA. Table 3-4b specifies the minimum deployed and maximum number (full load) for each legacy data user type and the base data distribution ports.

The proximity of the RDA and right of way governs choice of base data access telecommunication facilities. The facilities, as a minimum, shall (8) support connection and equipment space, as appropriate, for either private hard-wire, private T-1, commercial T-1, or private fiber optic connection.

The characteristic circuit requirements for the RPG support are as follows:

- a. Commercial facilities for the base data access and control link shall (9) be the same as that defined for the RDA in section 3.7.5.4.4.2.
- b. Base Data Users access shall(10) be via WAN link. Base data from DoD/FAA RPGs may be first collected at an NWS facility prior to external base data user access.
- c. Class 98 shall (11) support either LAN interface or shared WAN access dedicated link interface.
- d. Class 1 (14.4 Kbps minimum) and 99 shall (12) support a commercial 4-wire dedicated (or private dedicated hard-wired) connection of minimum specification 3002 voice grade.
- e. Legacy Class 2 (14.4 Kbps minimum) shall(13) support a switched commercial 2-wire dial line of minimum specification 3002 voice grade.
- f. Deleted

g. The RPG to MSCF interconnections shall (15) support either a commercial 4-wire dedicated (or private dedicated hard-wired) connection of minimum specification 3002 voice grade or a shared WAN access dedicated link interface.

SUPPORT SEGMENT

3.7.6 Support Function Area

The Support functional area includes an off-line diagnostics capability, adaptation data maintenance, map generation, computer performance monitoring, test data generation capability, test software, and CM.

3.7.6.1 Off-Line Diagnostics

An off-line diagnostic capability shall (1) be provided at each maintenance and logistics support facility and within each WSR-88D System. This capability shall (2) permit isolation of faults to the LRU or group of three LRUs. Fault isolation shall (3) be manually initiated, with all manual actions requested via prompts. In addition to isolating equipment failures, the off-line diagnostic capability shall (4) verify the correct operation of the equipment after the failure has been corrected.

3.7.6.2 Support Software

The following support software capabilities are provided.

3.7.6.2.1 Adaptation Data Maintenance

Adaptation data is that data which is installation-dependent, or parameters that the system is required to use that may vary according to operational needs such as the latitude, longitude, and elevation of a site and the geographic information used to generate maps applicable to that site.

Adaptation data maintenance software includes the following capabilities.

- a. The capability to generate adaptation data for the entire system shall (1) be provided. It shall (2) be possible to generate any subset of the adaptation data. The adaptation software shall (3) provide the capability to store the generated data on a mass storage device.
- b. The capability to accept input from an operator's console or an input file (which specifies which adaptation data to generate and how to generate it) shall (4) be provided.
- c. The capability to add, modify, or delete data from the adaptation data base shall (5) be provided.
- d. It shall (6) be possible to identify a particular set of adaptation data and copy it from a mass storage device to removable storage media.
- e. The capability to make local changes to the adaptation data base for a particular facility shall (7) be provided.
- f. It shall (8) be possible to merge a local adaptation data base with a more general adaptation data base and produce a single composite data base.
- g. All changes and additions made to the adaptation data base shall (9) be recorded.

3.7.6.2.2 Map Generation

High resolution maps, including but not limited to geo-political, navigation, topography, watershed, rivers and river basins, shall (1) be provided to support positioning and analysis of WSR-88D product data. The capability shall (2) be provided to produce and maintain these maps for the RPG and PUP functional areas.

3.7.6.2.2.1 Background Map Contents

Background maps shall (1) include:

a. Political Boundaries outlining states and counties, and annotating counties. States and counties will include the same detail in both LOW and HIGH resolution. Counties will be annotated (named) only in HIGH resolution.

b. Geographic Boundaries

(1) State Boundaries - including international treaty lines, national boundary lines (both land and water) and state lines (both land and water). The detail will be the same for both LOW and HIGH resolution.

(2) County lines - including county lines (both land and water), corporate limits and small independent cities. The detail will be the same for both LOW and HIGH resolution.

c. Range Rings

Low resolution - polar rings and radii spacing shall (2) be determined as a function of the product with which they may be associated.

A status line shall (3) indicate the spacing of the polar rings and radii.

High resolution - Radii spaced every 10 degrees of azimuth, and polar circles drawn every 10 nm from the RDA position.

There shall (4) be tick marks every 1 nm along each radius, with emphasis on every fifth tick mark.

d. Airports

All commercial and military airports as defined by the USGS Digital Line Graph data. The detail will be the same for both LOW and HIGH resolution.

e. Navigation Aids

The detail will be the same for both LOW and HIGH resolution.

- (1) VHF Omni-Directional Range (VOR)
- (2) VHF Omni-Directional Range/Tactical Air Navigation (VORTAC)
- (3) VHF Omni-Directional Range/Directional Measuring Equipment (VORDME)
- (4) Non-Directional Beacon (NDB)
- (5) Instrument Landing System (ILS)

f. Radar Data Acquisition Sites

Acronym and icon for all other active RDA sites within 250 nm range. The detail will be the same for both LOW and HIGH resolution.

g. City and Town Names

Low resolution - Five large cities spaced at least 50 nm from one another, and within 125 nm range from the site.

High resolution - three cities/towns for each county (borough, parish, census area, etc.) in each of the fifty states, and within 250 nm range of the site.

h. Special Use Airspace

Vectorial detail will be the same for both LOW and HIGH resolution.

Textual information will appear only on HIGH resolution.

- (1) Prohibited Areas
- (2) Restricted Areas
- (3) Military Operation Areas

A PUP icon indicating the location of each PUP located on a Military installation.

- (4) Warning Areas (International Waters)

i. Rivers

Rivers and streams (perennial and intermittent), centerline of water bodies, canals (navigable and other), U.S. coastlines including the Great Lakes, and permanent water bodies (lakes, reservoirs and islands).

The level of detail included in LOW and HIGH resolution is a function of data sources and software/hardware design and will be determined by the government.

j. River Basin Boundaries (Watersheds)

The level of detail included in LOW and HIGH resolution is a function of data sources and software/hardware design and will be determined by the Government.

k. Airways Including Military Refueling Zones and Training Routes

The detail will be the same for both LOW and HIGH resolution.

l. Grid Overlays

Vectorial detail will be the same for both LOW and HIGH resolution.

Vectorial detail will be the same for both LOW and HIGH resolution.

Textual information will appear only on HIGH resolution.

m. Highways

Interstate highways, major limited access highways (U.S., state and other) minor U.S. limited access highways, U.S. non-limited access highways, and state secondary highways.

The level of detail included in LOW and HIGH resolution is a function of data sources and software/hardware design and will be determined by the government.

n. County Names

A six character abbreviation for each county (borough, parish, census area, etc.) within 250 nm of the site shall (5) be included in the high-resolution map.

This applies only to sites in the 50 states.

County names will be displayed only in HIGH resolution.

o. Latitude/Longitude Tick Marks (Overseas)

For sites outside the 50 states, the background map shall (6) contain a small tick mark (plus sign) every 5 degrees of latitude and longitude on low-resolution and every degree on high-resolution.

3.7.6.2.3 Computer Performance Monitoring

A capability shall (1) be provided to record information concerning the utilization of the various system resources and to format this information into a form suitable for output such as peak and average memory or CPU utilization.

3.7.6.2.4 Test Data Generation

Test data generation software shall (1) be provided for use in developing, testing, and maintaining the WSR-88D operational software and in training WSR-88D operators, users, and maintenance personnel. The data generated shall (2) be stored in a transportable, machine readable form. The test data generation software shall (3) provide the capability to generate simulated radar data, including noise returns and echoes, and to process [Shall (4)] inputs which specify the characteristics of the test data to be generated.

3.7.6.2.5 Test Software

Test software shall (1) be provided to verify the performance of the operational software.
The test software:

1. shall (2) be operable in an interactive mode with prompting for all required inputs.
2. shall (3) be operable in an automatic mode requiring minimum operator action.
3. shall (4) generate formatted output products that show the responses associated with each input and the cumulative results for each test.
4. shall (5) include data samples that are both within and outside the range specified for the function under test.
5. shall (6) be subject to the same quality control procedures as the operational software.

3.7.6.2.6 Configuration Management

Software tools and appropriate hardware platforms shall (1) be available to support hardware and software configuration management (CM).

4 QUALITY ASSURANCE

This Quality Assurance Section establishes the Quality Assurance portion of Test and Evaluation (T&E) requirements for the WSR-88D System.

4.1 General

Quality Assurance Testing includes all of the examinations and tests to be performed in order to ascertain that the product, material or process to be offered for acceptance conforms to the requirements in Section 3 of this Specification.

The verification of the requirements for all Configuration Items (CIs) and Computer Program Configuration Items (CPCIs) allocated for meeting the requirements of this System Specification will be in accordance with the philosophy, techniques and locations as specified in the WSR-88D Test and Evaluation Master Plan (TEMP). The verification of the requirements for all critical system functions identified in Table 3-5 of this specification will be described in the System Test Plan.

4.1.1 Responsibility for Tests

Unless otherwise specified by the Government, responsibility for all planning, conducting, and documentation of the tests shall be as defined in the TEMP.

4.2 Quality Conformance Inspections

Quality conformance verification of all CIs and CPCIs shall include those activities which verify the requirements in this document.

4.2.1 Classification

The TEMP will address the verification of all requirements in this document by the use of Inspection, Analysis, Demonstration, and Test (I, A, D, and T). These four test methods are defined below.

Test methods are defined as specified herein.

4.2.1.1 Inspection

This verification method is based on a visual examination to prove that specified requirements have been met.

This method is applicable to the verification of the assembly and of configuration of hardware items and assembled software by comparison with appropriate documentation.

4.2.1.2 Analysis

The verification method is based on proving, by induction or deduction, that specified requirements have been met. This relies upon a technical analysis and evaluation of data, equations, graphs, circuit and timing diagrams, coding, or computer simulations.

This method of verification is used where the nature of the requirement best lends itself to this method; e.g., reliability predictions. This method is also generally applicable where the design uses hardware or software which is available off-the-shelf and for which data are readily available. Additionally, analysis is applicable for new designs where a more comprehensive method will be used for verification in a subsequent program phase. Analysis may also be used in lieu of alternate methods of verification where significant cost saving would result.

4.2.1.3 Demonstration

This verification method does not normally involve the use of special instrumentation or simulated inputs that specified requirements have been met. This method will be based on one of the following:

- An operation, movement, or adjustment which relies on observation of an action or series of actions and the consequent results (e.g., on a CRT display or printout) to decide on the success of demonstration.

or

- A statistical review of a number of demonstration examples performed under specified conditions to determine compliance at a specified confidence level; e.g., reliability and maintainability demonstrations.

4.2.1.4 Test

This verification method normally involves the use of special instrumentation, simulated inputs, or a combination of both to prove that specified requirements have been met.

This method is generally applicable where specified external stimuli (hardware or software) and/or external devices are required to produce and/or measure (record) predictable results; e.g., for verification of response times. Also, is applicable where special conditions are involved, such as environmental, interference, susceptibility, or noise.

APPENDIX A – DELETED

APPENDIX B – DEFINITIONS

1 Definition Of Terms

1.1 Adaptation Data

Generally, adaptation data are defined as modifiable parameters. These parameters may consist of system configuration, algorithm parameters, telephone numbers (auto dial), or other like data. Such data may be generated and/or modified at either a centralized location or locally within an RDA, RPG or PUP.

1.2 Alarm

An alarm is an attention getting notification of a system, equipment or software failure or an out of tolerance operating conditions that is provided to the operator.

1.3 Alert

An alert is the attention getting mechanism associated with receipt of an alert message at a Class 1 user indicating the initial detection of a significant meteorological parameter or phenomenon.

1.4 Alphanumeric Products

Those products generated within the RPG functional area that present, in an alphanumeric format, some representation of information in base products, derived products, or other alphanumeric products whether manually or automatically generated (e.g., a free text message).

1.5 Availability (Achieved) (A_a)

The probability that when used under stated conditions in an ideal support environment without consideration of administrative and logistics delay times, a system will operate satisfactorily at any given time.

1.6 Availability (Inherent) (A_i)

The probability that when used under stated conditions in an ideal support environment without consideration for preventive maintenance actions, a system will operate satisfactorily at any given time.

1.7 Availability (Operational) (A_o)

A_o covers all segments of time that the system is intended to be operational and capable of performing its mission functions. A_o is used to support design goals, operational testing assessment, and life cycle costing.

1.8 Availability (Service) (A_s)

A_s covers all segments of time that the system is intended to be operational and capable of performing its mission functions except for Preventive Maintenance down time. A_s was defined by the WSR-88D tri-agencies as the mission performance effectiveness measure to be used for the WSR-88D.

1.9 Base Data

Those digital fields of reflectivity, mean radial velocity, spectrum width, differential reflectivity, differential phase, and correlation coefficient data in spherical (i.e., rho, theta, phi) coordinates provided at the finest resolution available from the RDA to the RPG and to the base data users. These data resolutions are specified in Sections 3.7.2.2.2.1 and 3.7.2.2.2.2.

1.10 Base Products

Those products generated within the RPG functional area that present some representation of each base data field. This representation may not necessarily be either in full resolution or depict the full area of coverage. Base Products can be used by a user to generate a graphic display or perform further processing.

1.11 Channel, Dual

The minimum equipment configuration needed to provide 100% redundancy (excluding the antenna assembly) in the capability to perform the single channel functions.

1.12 Channel, Single

The minimum equipment configuration needed to perform RDA key functions 1, 2 and 3 as defined in Table 3-5.

1.13 Class 1 User

A Class 1 User is capable of requesting any set of up to 31 products for routine receipt from an associated RPG via a dedicated TELCO circuit or WAN. The Class 1 User may specify the frequency at which a product is to be sent (up to the frequency at which the product is generated) and its priority. The Class 1 User may request a change at any time to the set of products routinely sent from the associated RPG, or to their frequency or priority. A Class 1 User may request selected products on a one-time basis from any associated RPG. For any one-time product selection the Class 1 User may include a transmission priority value.

A Class 1 User may be capable of defining up to two alert areas. A Class 1 User may be capable of selecting threshold criteria for up to 10 of the alert categories defined in Appendix J for each of the two alert areas. A Class 1 User may be capable of specifying if the designated product will be sent to that position when its alert criteria have been met.

1.14 Class 98 User

A Class 98 User is capable of requesting any set of up to 160 products for routine receipt from an associated RPG via a dedicated LAN connection. The Class 98 User may specify the frequency at which a product is to be sent (up to the frequency at which the product is generated) and its priority. The Class 98 User may request a change to any time to the set of products routinely sent from the associated RPG, or to their frequency or priority. A Class 98 User may request selected products on a one-time basis from any associated RPG. For any one-time product selection the Class 98 User may include a transmission priority value.

A Class 98 User may be capable of defining up to two alert areas. A Class 98 User may be capable of selecting threshold criteria for up to 10 of the alert categories defined in Appendix J for each of the two alert areas. A Class 98 user may be capable of specifying if the designated product will be sent to that position when its alert criteria have been met.

1.15 Class 99 User

A Class 99 User is capable of requesting any set of up to 50 products for routine receipt from an associated RPG via a dedicated TELCO or WAN connection. The Class 99 User may specify the frequency at which a product is to be sent (up to a frequency at which the product is generated) and its priority. The Class 99 User may request a change at any time to the set of products routinely sent from the associated RPG, or to their frequency or priority. A Class 99 User may request selected products on a one-time basis from any associated RPG. For any one-time product selection the Class 99 User may include a transmission priority value.

A Class 99 User may be capable of defining up to two alert areas. A Class 99 User may be capable of selecting threshold criteria for up to 10 of the alert categories defined in Appendix J for each of the two alert areas. A Class 99 user may be capable of specifying if the designated product will be sent to that position when its alert criteria have been met.

1.16 Cold Startup

Cold Startup is a transition to bring the radar from a Power-Off System state to the Standby Operating state; an Operator must manually intervene to start the process to power on all equipment including the processors. This causes the system to load the Operating Systems, initialize, start the RDA application software, and initialize the radar hardware. The radar enters a Startup transition which includes (but not limited to) establishing the communications link with the RPG, commanding the pedestal to the park position, establishing the controlling channel, initializing the maintenance display interface, and performing initial radar calibrations.

1.17 Collocated

The data signaling rate and interconnecting cable characteristics, using the EIA-RS-422A transmission standard, determine the distance that WSR-88D equipment can be considered collocated. The RDA is collocated with a RPG when the two are separated by not greater than 80 meters of cable. The RPG is collocated with its PUP when the communication rate is 56 Kbps.

1.18 Commercial -Off-The-Shelf (COTS)

A commercial product made for commercial use. It is bought exactly as found in the civilian market, and allowed to flow with the changes and updates the vendor provides to his commercial customers.

1.19 Configuration Item (CI)

A CI is any individual item of equipment or combination of equipment items needed to perform some portion of the operational WSR-88D functional areas (i.e., RDA, RPG, or PUP). Furthermore, an independent CI is defined to have the following characteristics:

- a. To fail without adversely affecting the performance capability of another CI.
- b. To be able to have power turned on/off without an adverse effect on the performance capability of another CI.
- c. The capability to have all external data, power, and/or control lines removed or restored without an adverse effect on the performance capability of another CI.

1.20 Co-polar Cross Correlation Coefficient

Correlation between horizontal and vertical signals which provides a measure of scatterers' orientation and shape.

1.21 Correlation Coefficient

The same as co-polar cross correlation coefficient.

1.22 Dedicated User

This terminology refers to a user connected continuously via a dedicated TELCO circuit, LAN or WAN to the RPG. The user is connected continuously and is a single user on a designated RPG port.

1.23 Derived Data Array Products

Those products generated automatically within the RPG functional area that present, in a non-displayable format, some combination of intermediate meteorological algorithm output, Base

Products and derived products. These products are intended only for transmission to systems external to the WSR-88D System for further processing.

1.24 Derived Products

Those products generated within the RPG functional area that represent either some combination of Base Products or a base product that has been enhanced or otherwise changed by the use of automated processing techniques. Derived products can be used for graphic display of weather radar information or for further processing by a user.

1.25 Differential Phase

Phase difference between horizontal and vertical signals which is a measure of total path propagation time difference in the medium traversed. For rain, this is a measure of total liquid water along the path.

1.26 Differential Reflectivity

The ratio of horizontal reflectivity to vertical reflectivity. This parameter provides a measure of hydrometeors' relative dimensions.

1.27 Dual Polarization

The process whereby both horizontally and vertically polarized transmitted and received signals are used to enhance the ability of the Nexrad system to generate hydrometeorological products.

1.28 Dwell Time

The time interval of target illumination over which samples are taken to make the required base data estimates.

1.29 Effective Beam Width

Effective Beam Width is the azimuthal resolution of a combination of the one-way antenna pattern, the azimuthal extent of the processed receiver samples, and any applied data windowing. The Effective Beam is computed by the convolution of the one-way antenna pattern with the square of the data window as defined over the azimuthal extent of the sample.

1.30 Expansion Capability

The capability to easily add CIs or LRUs to any WSR-88D System for the purpose of increasing its data acquisition, processing, and/or product distribution capability. However, this addition will not require:

- a. New CIs or new LRUs within a CI that are not part of the logistics system. A new CI that is form, fit and functionally compatible with an existing CI is permitted provided it complies with items 3-5 of this definition.
- b. Modification of an existing LRU or CI (the addition of cables between pre-existing connectors on existing CIs is not considered a modification. Also, the addition of LRUs to a CI will not be considered a modification if such LRU additions are made by means of pre-existing wired connectors within a CI, e.g., the addition of a memory board to a CPU where one or more "slots" for such boards have been reserved).
- c. The writing of new software (operational or maintenance).
- d. The modification of any existing operational or maintenance software (changes made to the constants or parameter values are not considered modifications).

e. Addition or modification to any depot ATE capability.

1.31 FAA NIMS proxy agent/concentrator Interface (RMS)

The interface to the NAS Infrastructure Management System (NIMS). Proxy agent/concentrator will be used to allow National Airspace (NAS) subsystems, using proprietary and non-standard management functions, to communicate with the NIMS Manager as specified on DOT FAA Interface Control Document, NIMS/Managed Subsystem using the Simple Network Management Protocol Version 1 (SNMPv1), NAS-IC-51070000-1, dated 07/22/97.

1.32 Facility

The building(s) or enclosure(s) or that portion of the building(s) or enclosure(s) that houses some specific WSR-88D equipment.

1.33 Failure

Failure is the loss of any of the system functions as defined in Table 3.5. Along with system hardware failures, it also includes software failures, operator errors, and publication or other documentation errors that cause mission loss.

1.34 Hydrometeor Discrimination

Polarimetric measurements are sensitive to particle size, orientation, shape, phase (liquid or solid) and density. Particular hydrometer types have characteristic signatures and the ensemble of measurements and derived products can be used to determine the dominant scatterers within the radar sample volume.

1.35 Interface

The total software and hardware, excluding Data Communication Equipment, in the WSR-88D system necessary to communicate between two CIs within the WSR-88D System or between the WSR-88D System and another system.

1.36 Interface Connection

A capability within the system (e.g., back-plane slot) which, without any other changes to the system, but provided with the software and hardware, will serve as an Interface.

1.37 Line (Lowest) Replaceable Unit (LRU)

An essential support item which is removed and replaced at the field level to restore the end item to operational ready condition

1.38 Master System Control Function (MSCF)

The MSCF provides operator access to all RDA and RPG control, data entry, status monitoring, and error detection functions. The MSCF includes a human/machine interface in the form of a graphical user interface.

1.39 Maximum Time To Repair (MAXTTR)

The maximum corrective maintenance downtime within which 90% of all corrective maintenance actions can be accomplished (excluding supply and administrative downtime).

1.40 Mean Time Between Failure (MTBF)

The Mean Time Between Failure (MTBF) is defined as the statistical mean time between system failures requiring maintenance action.

1.41 Mean Time To Repair (MTTR)

The statistical mean of the distribution of repair times for individual failures. The summation of active repair times during a given period of time divided by the total number of malfunctions during the same time interval. This definition applies to components as well as the whole system.

1.42 Median Gain

That level over an angular region at which the probability is 50% that the observed or measured gain at any position of the antenna will be less than or equal to that level.

1.43 MSCF

The MSCF software executing at the MSCF display/control position.

1.44 National Facility

Those principal users located at one of the following locations: SPC, AFGWC, TPC, or the NCEP.

1.45 Network Site

A Department of Commerce WSR-88D site in the conterminous United States that continuously collects, collates, and makes available radar data and products in support of the National Weather Radar Network.

1.46 NEXRAD Program

The summation of the procurement, testing and fielding of all hardware, software, facilities, communications, logistics, training, and staff, together with operations, training, and maintenance procedures. The program includes Network sites, Non-network sites, and Supplemental sites.

1.47 Non-Network Site

A non-CONUS Department of Defense site or a non-CONUS Department of Transportation site.

1.48 Operate

A sub-state of Ready Operational Mode where the RDA is sending base data to the RPG.

1.49 Operational Facility

Any local, regional, or national government facility which uses or supports the use of WSR-88D radar data in real-time in support of an operational mission.

1.50 Other User

Any user not identified as a principal user.

1.51 PC PUP

A SCO Unix-based meteorological software application. A Class 4 other-user dedicated interface commonly connected to a DOD PUP.

1.52 Port

An Interface, plus all required Data Communications Equipment (cables, modems, etc.) needed to communicate with another device, either within or outside of the WSR-88D System.

1.53 Preview Area

An area on the alphanumeric or graphic displays for the viewing of operator entered data before physical modification of the display occurs.

1.54 Principal User

The National Weather Service, Air Weather Service, Naval Meteorology and Oceanography Command, or Federal Aviation Administration.

1.55 Principal User Site

Site of the equipment necessary to perform the principal user function. May be collocated with another site (i.e., a radar site).

1.56 Product

A discrete collection of meteorological data and descriptive header information processed by the RPG and typically displayable as an graphical image or in tabular format.

1.57 Radar Operations Center

The Triagency organization tasked with providing hardware, software, operations, logistics, and training support to the WSR-88D program.

1.58 Radar Site

Site of a specific WSR-88D antenna and RDA equipment. May be collocated with another site (i.e., a principal user site).

1.59 RDA Default State

The RDA Default State will include Operational RDA mode, RDA control, Standby state, all moments (Reflectivity, Velocity, and Spectrum Width) enabled, VCP 21, automatic calibration enabled, Spot-Blanking Enabled (if Installed) and channel Non-Controlling if redundant configuration.

1.60 RDA HCI

The RDA HCI is the application software that provides the interface between the RDA and the user/operator.

1.61 RDA System States (see diagram that follows)

The RDA Startup and Shutdown processes include a number of system states. They are as follows:

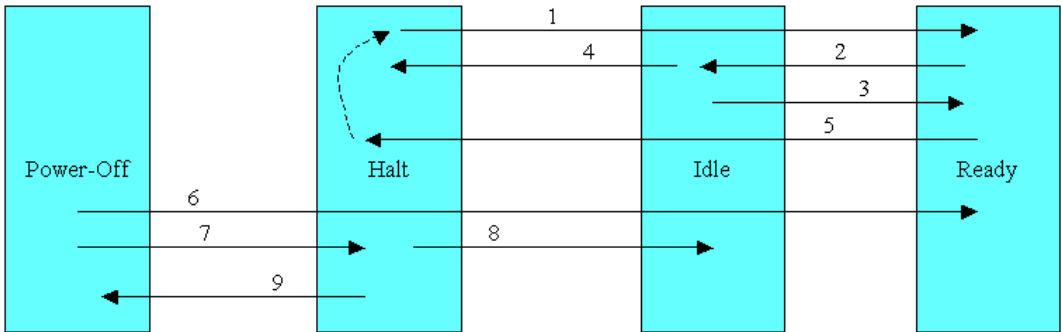
Power-off is a system state where no power is applied to radar equipment.

Halt is a system state where no power is applied to computer processors and the synchronizer.

Idle is a system state with the following characteristics:

- Power applied to all radar equipment
- Operating System software is running on all processors
- No Application Software running

RDA Startup/Shutdown System States



Transitions

- | | |
|----------------------|-----------------|
| 1. Startup | 6. Cold Startup |
| 2. Terminate | 7. Power Up |
| 3. Applications Load | 8. Maintenance |
| 4. Shutdown | 9. Power Down |
| 5. Restart | |

1.62 Ready

Ready is a system state with the following characteristics:

- Power applied to all radar equipment
- Applications software loaded & executing
- Three Ready modes exist
- Operational
- Off-Line Maintenance

Operational is a mode of Ready where the RDA is capable of sending base data to the RPG.

Test is a mode of Ready.

Off-line Maintenance is a mode of Ready.

Standby is a state of Operational mode where the RDA is operational but is not sending base data to the RPG.

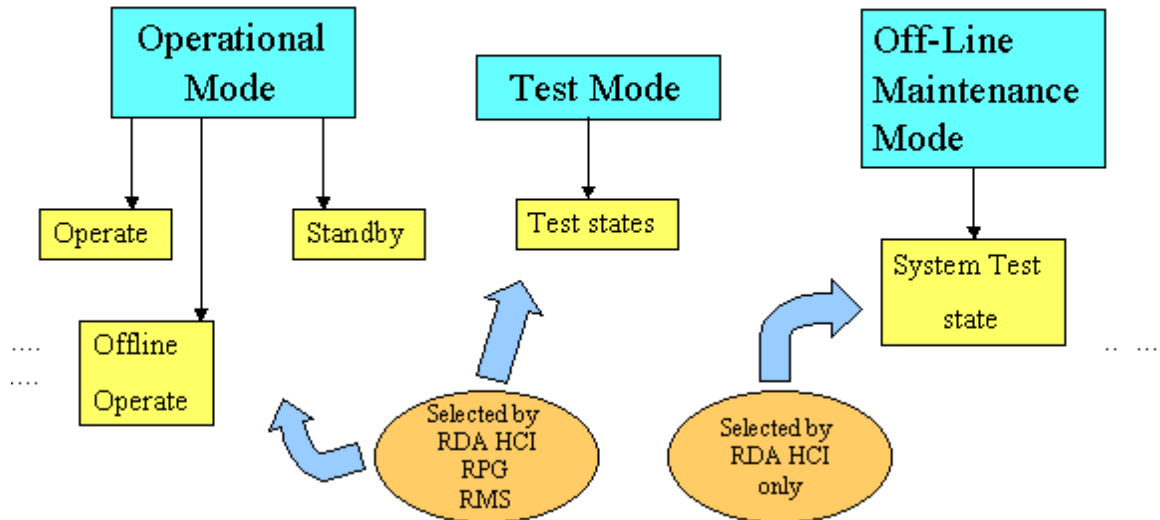
Operate is a state of Operational mode.

Off-line Operate is a state of Operational mode.

Test States are states of Test mode.

System Test States are states of Off-line Maintenance mode.

Ready System State



1.63 Regional Facility

Those principal users located at one of the following locations: RFCs or ARTCCs (CWSUs).

1.64 Remote Monitoring SubSystem (RMS)

The interface to the NAS Infrastructure Management System (NIMS). RMS will be used to allow National Airspace (NAS) subsystems, using proprietary and non-standard management functions, to communicate with the NIMS Manager as specified on DOT FAA Interface Control Document, NIMS/Managed Subsystem using the Simple Network Management Protocol Version 1 (SNMPv1), NAS-IC-51070000-1, dated 07/22/97.

1.65 Repair Time

The repair time for configuration items includes the total time required to detect and isolate the failure, to replace the CI or the failed component, check out the repair, and return the item to full operational status (includes the time to restore the system, load adaptation data, restore data base, etc., to its condition prior to the failure).

1.66 Response Time

The time interval between completion of all needed manual actions and completion of the results of those actions. For example, if a user calls up one product for display, the response time is the time interval between making the last needed keystroke to call up the product and completion of the display of the resulting product (not just the start of display of a product).

1.67 Restart

The purpose of a restart is to return the radar to a normal operational state. A restart can begin manually with the Operator locally at (a) the RDA maintenance position (or (b) FAA RMS) or remotely via RPG HCI (at (c) the RPG or (d) the MSCF) issuing a Restart command, or automatically when a hardware or software condition causes the processor to reboot, when one of the RDA software tasks pauses, under certain error conditions (i.e. timeouts), and upon prime power recovery following

a power failure. During the restart process, the operating system is reloaded and application software is reloaded. At the completion of the restart process, the RDA will be returned to the state present at the time the restart was initiated (if initiated remotely via the RPG HCI), or to a default state (if initiated locally from the RDA).

1.68 Scan Period

The time required for a radar to complete a full 360 degree rotation in azimuth.

1.69 Site

The geographic location of some specific WSR-88D equipment.

1.70 Specific Differential Phase

Differential phase change per unit range between the horizontal and vertical signals which is a measure of the difference in propagation time. For rain, this is a measure of liquid water over range increments.

1.71 Standard Annotation Set

Each product includes a standard set of annotations to uniquely identify that product. The set of standard annotations to each product include: product name, RDA ID, date and time of volume scan, radar location (latitude/longitude), radar height (above MSL), and weather mode.

1.72 Standby

State of the Operational mode where the RDA is operational but is not sending base data to the RPG.

1.73 Startup

The process of initializing an RDA, RPG, or PUP without regard to its previous state.

1.74 Supplemental Site

A Department of Defense site in the CONUS.

1.75 Support Facility

A facility that provides some form of support primarily for the NEXRAD Program. This support may be in such areas as training, logistics, supply, or testing.

1.76 System Fault

A system fault is any malfunction which degrades WSR-88D performance below its specifications. Malfunctions arising from accidents, failures of test instrumentation or monitoring equipment are not considered a system fault for the purpose of defining availability requirements.

1.77 Throughput Time

The time interval measured from the completion of the base data necessary for the generation of a product until the product is displayed at an associated PUP for base, derived and alphanumeric products, or received at an associated PUP for derived data array products. This time interval includes transmission time but excludes RPG communications port queue time.

1.78 User

Any system user.

1.79 Volume Scan

The process of completing all elevation angle scans in a specific scanning sequence.

1.80 Volume Scan Time

The time interval between the start of one volume scan and the start of the next volume scan.

1.81 WAN

A physical or logical communications network that uses such devices as telephone lines, satellite dishes, or radio waves to span a larger geographic area than can be covered by a LAN.

1.82 Warm Startup

Warm startup is a transition that begins with the radar in Standby or offline Operate states, and the Operator issuing an Operate command to bring the radar to the Operate state. During warm startup, the radar channel is checked to make sure it is the controlling channel and the pedestal is initialized; the end result of warm startup, the radar is in an Operate state.

1.83 WINPUP

A Windows-based meteorological software application. A Class 4 other-user dedicated interface commonly connected to a DOD PUP.

1.84 WSR-88D System

The combination of one RDA, one RPG, all associated Class 1 Users, and the necessary communications (external interfaces) to fully implement the operational capability. System configuration requirements are in Section 3.2.1.5.

APPENDIX C – DELETED

APPENDIX D – HYDROMETEOROLOGICAL PROCESSING FUNCTIONS

The following items are Mode Selection Function (MSF) requirements. The purpose of the MSF is to collect data necessary for automated processing of volume coverage pattern usage and to preserve remnant features of the Precipitation Detection Function (PDF) while removing the PDF.

1 Mode Selection Function

1.1 Functional Description

In order to perform four basic functions, the Mode Selection Function processes for each volume the Hybrid Scan Reflectivity which is output from the Precipitation Preprocessing Subsystem (PPS). 1.) It detects whether sufficient, significant reflectivity above an aerial extent has occurred within a 230km radius of the WSR-88D radar and derives a system Recommended Mode status appropriate to the situation. 2.) It maintains Mode Status Messages put into the RPG Status Log. 3.) It will automatically update to a default VCP when a Recommended Mode status changes and the update has been enabled by the operator. 4.) Finally, when a conflict in MSF Recommended Mode and VCP type has been detected for an adaptable length of time, the MSF will override mode status settings and switch to the appropriate VCP. MSF rules and exceptions are below.

1.1.1 MSF Reflectivity Detection

The Mode Selection Function shall calculate for each volume the area (km²) of reflectivity values (dBZ) equal to or exceeding a threshold value (dBZ) using as input the Hybrid Scan Reflectivity (HSR). The reflectivity threshold value shall be a locally adaptable parameter.

MSF Recommended Mode shall be assigned by comparing calculated area to an Area Threshold value (km²). The Area Threshold value shall be locally adaptable. MSF Recommended Mode shall be either Precipitation Mode (often called Mode A) if the calculated area is equal to or exceeding the Area Threshold or Clear Air Mode (often called Mode B) if the calculated area is less than the Area Threshold.

MSF will operate when the RPG is in control of the RDA.

As with the PDF, the MSF allows an operator to change or download a VCP for one volume; however, after that volume is completed, actions may or may not be the same.

Mode selection categories shall be expandable for future enhancements.

1.1.2 Mode Automation Status

The Mode Automation Status shall be updated in the Mode Automation Status window every volume. Operators can open the Mode Automation Status window from either the VCP and Mode Control window or the Mode Conflict button from the Main HCI.

The Mode Automation Status window shall contain: 1) Current Mode and beginning date/time, 2) Recommended Mode and beginning date/time of recommended change, 3) Date and time status was last updated, 4) Delay time before switching to Clear Air Mode or “NA” if not applicable, 5) Conflict status message showing whether current mode agrees with recommended mode, 6) Adaptable Parameter settings and MSF values (Reflectivity Threshold, Area Threshold, Area Detected, Area Exceeded).

MSF Parameters of the Mode Selection Algorithm will be modifiable by selecting Modify Parameters from the Mode Automation Status window or from Algorithms in the RPG Products window. URC LOCA parameters in the Mode Selection Algorithm will be: 1) Precipitation Mode Threshold, 2) Precipitation Mode Areal Coverage Threshold, 3) Clear Air Select Time, 4) Mode Conflict Duration, and 5) Ignore Mode Conflict Duration.

Mode status information shall be reported in the RPG Status Log.

Availability of mode status information shall be expandable for future enhancements.

1.1.3 Mode and Volume Coverage Pattern (VCP) Conflict

The MSF shall keep track of mode status and VCP conflicts for the purpose of system override when a continuous conflict occurs for an extended adaptable time.

Status of a conflict (or agreement) between MSF Recommended Mode and current VCP shall be provided to the operator in the RPG Status Log.

If no conflict between mode status and VCP exists, a status message will indicate no conflict. If a VCP that is assigned to run in a particular mode status is running while in an alternate mode status, a status message will indicate the conflict to the operator.

A mode status message in the Main HCI RPG status log area shall be provided once every volume as categorically filterable information. Sample agreement or conflict status messages are:

“Sep 13,05[14:53:08]>>MSF STATUS: 536km² > 30.0dBZ, VCP 21, no mode conflict”

“Aug 23,05[07:48:56]>>MSF STATUS: 3km² > 30.0dBZ, VCP 21, recommend CLEAR AIR mode”

The system status message should have no highlighted color when no conflict exists. If a conflict exists, the message shall have a yellow background. Variables in the mode status message are date, time, area of reflectivity above threshold, the reflectivity threshold, current VCP number, and text indicating recommended mode or text indicating no mode conflict.

A mode status message in the Main HCI RPG status log area shall also be provided to record operator changes to the MSF for the purpose of monitoring network operations.

If a conflict has occurred continuously for an adaptable time period (8 to 48 hours), the MSF shall provide a highly visible text message (Main HCI) of pending override beginning an hour before and once every volume. The MSF will resolve mode status and VCP conflict by invoking the correct VCP for the detected mode status on the next volume after that time period has been reached.

The MSF override function (i.e., the Mode Conflict Duration of 8 to 48 hours) shall be capable of being disabled by URC authority.

1.2 Processing Environment

It is assumed that the Precipitation Processing Subsystem (PPS), particularly the Radar Echo Classifier (REC), provides the Hybrid Scan Reflectivity as output without loss of data quality or quantitative accuracy.

It is further assumed that several pre-processing steps will be applied to the reflectivity data before the data is used by the PPS.

The MSF assumes that HSR data as input shall have been converted to equivalent reflectivity factor data, hereafter referred to as simply HSR Factor data, using the standard assumptions in the classical radar equation. In addition, it is assumed that the clutter mitigation technique, particularly the REC, makes HSR Factor data as good in quality as Reflectivity Factor data.

1.3 Acquisition

Reflectivity Bins are intrinsic system parameters acquired directly from the radar.

HSR Factor data are acquired from Reflectivity Factor data which are acquired from the NEXRAD base data with clutter filtering and other quality control procedures being used to the extent possible.

Elevation Angles are system adaptation parameters which may change with changing scanning strategies.

The Multiplicative ZR Coefficient and the Power ZR Coefficient are system adaptation parameters. The Mode Automation Status table shows two adaptable parameters used in the Mode Selection algorithm; these adaptable parameters are the Precipitation Mode Reflectivity Threshold and the Precipitation Mode Areal Coverage. Other Mode Selection algorithm adaptable parameters are the time delay before switching to Clear Air Mode (Mode B Selection Time, an adaptable parameter ranging from 20 to 60 minutes), binary decisions of automatic or manual switching for both Clear Air and Precipitation modes (Precipitation Switching, Clear Air Switching), a binary decision to ignore mode conflict (Yes, No), and a length of time in hours until the system overrides a mode conflict (Mode Conflict Duration). The Mode Status Message is an output of the MSF maintained continuously for access by other algorithms as needed.

The radar ID is a site adaptation parameter.

The Time Stamp is acquired from the WSR-88D system clock in UTC.

The Last Mode Detected Time is computed by the Mode Selection Function.

1.4 Computation

1.4.1 Notation

A_{HSR}=Area of Hybrid Scan Reflectivity detected

Z=Reflectivity Factor value in dBZ at a particular range (r) and azimuth (az)

r=Range to nearest point of a particular Reflectivity Bin (km)

az=Azimuth of a particular Reflectivity Bin (degrees)

dr=Range extent of a particular Reflectivity Bin (km)

da=Azimuthal width of a particular Reflectivity Bin (degrees)

Z_t=Precipitation Mode Reflectivity Threshold (dBZ)

T_A=Area Threshold in km² in the Mode Selection Threshold table

1.4.2 Symbolic Formula

First Set $A_{HSR} = 0.0$.

For All Reflectivity Bins (i.e., all r and az)

If $Z(r,az) > Z_t$

Then $A_{HSR} = A_{HSR} + \frac{\pi da}{360} [(dr)^2 + 2r dr]$

This computation proceeds by processing the completed Hybrid Scan Reflectivity for that volume. The Precipitation Mode Reflectivity Threshold (Z_t) in the computation is defined directly from the Reflectivity Threshold value listed in the table entry.

If $A_{HSR} > T_A$, assign the MSF Recommended Mode indicated by the Mode Threshold Table. When the Area Detected exceeds the Area Threshold, Precipitation Mode is the MSF Recommended Mode.

If Precipitation Mode is the Current Mode, it remains in effect until the Area Detected is less than or equal to the Area Threshold and that condition exists for a period of time defined in the Clear Air Mode Time Delay adaptation parameter (20 to 60 minutes). An automatic switch to Clear Air Mode follows on the next volume scan if Auto Clear Air Mode is set to "Auto." If Auto Clear Air Mode is set to "Manual," Clear Air Mode is the MSF Recommended Mode as detected but the system takes no action until a conflict between mode status and VCP reaches an adaptable timeout limit (8 to 48 hours).

If Clear Air Mode is the Current Mode, it remains in effect until the Area Detected exceeds the Area Threshold (e.g., becomes superseded by Precipitation Mode). An automatic switch to Precipitation Mode follows on the next volume if Auto Precipitation Mode is set to "Auto." If Auto Precipitation Mode is set to "Manual," Precipitation Mode is the MSF Recommended Mode but the system takes no action until a conflict between mode status and VCP reaches a timeout limit.

1.4.3 Logical Processes

Users control automatic mode switching. When the radar automatically changes the current mode, the default VCP for that mode is downloaded for the next volume scan.

The operator settings for each mode do not affect mode detection but do determine whether a default VCP may or may not automatically become the current mode.

Operators will control mode automation by selecting automatic or manual switching for each mode. For each mode, one state, either "Auto" or "Manual" may be selected. Four combinations exist: 1) Clear Air Switching is Manual, Precipitation Switching is Auto; 2) Clear Air Switching is Auto, Precipitation Switching is Manual; 3) Clear Air Switching is Manual, Precipitation Switching is Manual; and, 4) Clear Air Switching is Auto, Precipitation Switching is Auto.

How the radar functions will depend on how an operator toggles the Auto or Manual radio buttons for each mode. However, if the radar detects a conflict in recommended mode and current mode for a lengthy period of time (an adaptable parameter of 8 to 48 hours), the radar will override manual settings to resolve the conflict. This 8 to 48 hour wait period is a system override that can be completely disabled.

For combination #1, when Clear Air Switching is Manual and Precipitation Switching is Auto, the radar mode switching will behave similar to earlier software releases.

- If the Area Detected becomes less than the Area Threshold, a timer (20 to 60 minutes) begins. After the timer value is exceeded, the Recommended Mode becomes Clear Air.
- If the Recommended Mode becomes Clear Air, the current Precipitation VCP will not automatically update to a Clear Air VCP. An operator must manually change to a Clear Air VCP or wait until the 8 to 48 hour adaptable parameter (Mode Conflict Duration) is exceeded.
- If the Recommended Mode becomes Precipitation Mode, the default Precipitation VCP will automatically be used with the next volume scan.

For combination #2, when Clear Air Switching is Auto and Precipitation Switching is Manual, the radar mode switching WILL NOT automatically switch to a Precipitation VCP.

- If the Recommended Mode becomes Clear Air, the default Clear Air VCP will automatically be used with the next volume scan.
- If the Recommended Mode becomes Precipitation, the current Clear Air VCP will remain in use until an operator manually changes or downloads a Precipitation VCP or until the 8 to 48 hour adaptable parameter is exceeded.

For combination #3, when Clear Air Switching is Manual and Precipitation Switching is Manual, the radar mode switching is disabled.

- Any Recommended Mode will have no effect on the use of Precipitation or Clear Air VCPs until the 8 to 48 hour adaptable parameter is exceeded.

For combination #4, when Clear Air Switching is Auto and Precipitation Switching is Auto, the radar mode switching will be automatic.

- If the Area Detected becomes less than the Area Threshold, a timer (20 to 60 minutes) begins. After the timer value is exceeded, the Recommended Mode becomes Clear Air.
- If the Recommended Mode becomes Clear Air, the default Clear Air VCP will automatically be used with the next volume scan.
- If the Recommended Mode becomes Precipitation, the default Precipitation VCP will automatically be used with the next volume scan.

1.5 Distribution

The Mode Status Message is updated in the RPG Status Log every volume.

1.6 Definitions

Reflectivity Bins	Grid cells for Reflectivity Factor data being input to this function. Data are required from all elevation angles specified by the Elevation Angles.
Elevation Angles	Elevation angles from which Reflectivity Factor derived from the Hybrid Scan data are being input to the Mode Selection Function. The elevations used will vary as Hybrid Scan Reflectivity varies.

Reflectivity Factor Equivalent reflectivity factor data (in dBZ)

Mode Threshold Table Table of threshold values for mode status detection. The table is composed of four columns:

Reflectivity Threshold, Area Threshold, Area Detected, and Area Exceeded. The table has only one row and the content is determined by adaptation data.
An example table follows:

	Reflectivity Threshold (dBZ)	Area Threshold (km2)	Area Detected (km2)	Area Exceeded
	30.00	80	186	Yes

Special MSF Terms.

Area Threshold Total area (km2) used to compare the area of detected HSR factor data of the specified Precipitation Rate before MSF recommended Mode can be determined in the Mode Threshold Table.

Mode Status Message The Mode Status Message is a status message generated each time the MSF is run and is composed of the following items:
date, time, area of reflectivity above threshold, the reflectivity threshold, current VCP number, and text indicating recommended mode or text indicating no mode conflict.

Time Stamp Time of occurrence of Mode Status Message in UTC.

Last Mode Detected Time Time Stamp when precipitation was last detected. This will be updated every full volume reflectivity scan.

MSF Recommended Mode As detected by radar, either Precipitation Mode (Mode A) or Clear Air Mode (Mode B) are based on the area of reflectivity from the Hybrid Scan above a threshold value.

Current Mode As currently being used by the radar, either Precipitation Mode (Mode A) or Clear Air Mode (Mode B) regardless of the MSF Recommended Mode

1.7 Limitations

If any scan mode status in the WSR-88D system fails to meet the requirements of Table D-1, there can be no assurance that the Mode Selection Function will detect mode; i.e., the NEXRAD site may completely fail to recognize a correct mode status for a given scenario. Since automated VCP switching assumes the correct mode status is detected by this function, a VCP in use may not specifically apply to weather conditions if this function fails to detect mode status. The failure to detect mode status will have no effect on data requirements for the Precipitation Processing Subsystem during precipitation periods. In addition, if any scan mode does not meet the requirements of Table D-1, there can be no assurance that the category of mode status (A or B) as assigned by this function is representative of current conditions. Mode Selection Function controls allow an operator to flexibly control VCPs and override automation.

**TABLE D-1 MINIMAL SCAN MODE REQUIREMENTS NECESSARY FOR DETECTING
MODE**

<u>Scan Characteristics</u>	<u>Requirements</u>
Range	230 km or more
Range Resolution	2 km or less
Azimuthal Coverage	360 °
Azimuthal Resolution	2 ° or less
Tilts and Repetition	Lowest tilts required by Precipitation Processing Subsystem (see PREPROCESSING algorithm) at least every 10 minutes to generate the Hybrid Scan Reflectivity.

1.8 Future Developments

Due to the limitation cited above, the Mode Threshold Table will require fine tuning for best performance at each site using actual WSR-88D data.

A future plan is to expand Mode Categories and include additional Volume Coverage Patterns (VCP) to make automated VCP usage more effective to NEXRAD operations.

APPENDIX E – WSR-88D PRODUCTS

This Appendix lists the WSR-88D products required. Table E1 provides a summary for each product of: the area of coverage; product resolution; frequency of product generation for full load sizing; throughput time requirement; response time requirement for onetime product requests; and the frequency of product archival for full load sizing. The final form of any product is subject to user agency approval.

1. Reflectivity

This product will provide the reflectivity data displayable as an image and formatted as a data array. (Defines the form of presentation on a graphic display; not necessarily the form of transmission.) For the image version, variations of the product will be organized to provide various areas of coverage and display resolutions, while the data array version will provide the highest resolution available for the entire radar coverage area. On Super-Resolution azimuth scans, variations of the data array version of the product shall be organized to provide super-resolution and legacy-resolution coverage and resolution requirements. Both versions will be limited to the lowest 70,000 feet AGL of the atmosphere. The product will be generated for any elevation scan based on user requirements. Each scan will be updated once per volume scan time. (Defined in Appendix B.) For the image, each product will be available for both 8 and 16 reflectivity data levels, while 256 reflectivity data levels will be provided in the data array version. Each product will include a standard set of annotations, data level code, elevation angle, product resolution, and maximum data value (dB).

Super-Resolution refers to the following three spatial resolution and coverage enhancements: 0.5deg azimuth sampling, 0.13nmi range sampling of Reflectivity, 162nmi coverage area for Mean Radial Velocity and Spectrum Width. The relevant distinction for Reflectivity is the increased coverage area and range sample interval. “Super-Resolution” was introduced in 2008 (Build 10) and was optionally generated during “Split Cuts” processing (the lowest 2 or 3 elevation angles) as specified in the Volume Coverage Pattern definition.

2. Mean Radial Velocity

This product will provide the mean radial velocity data both displayable as an image and formatted as a data array. For the image version, variations of the product will be organized to provide various areas of coverage and display resolutions, while the data array version will provide the highest resolution available for the entire radar coverage area. On Super-Resolution azimuth scans, variations of the data array version of the product shall be organized to provide super-resolution and legacy-resolution coverage and resolution requirements. Both versions will be limited to the lowest 70,000 feet AGL of the atmosphere. The product will be generated for any azimuth scan at a single elevation angle based on user requirements. Each scan will be updated once per volume scan time. For the image version, each product will include both 8 and 16 mean radial velocity data levels, while 256 velocity data levels will be provided in the data array version. Each product will include a standard set of annotations, data level code, elevation angle, maximum data values detected (knots, positive and negative), product resolution, and identification of range obscured data.

Super-Resolution refers to the following three spatial resolution and coverage enhancements: 0.5deg azimuth sampling, 0.13nmi range sampling of Reflectivity, 162nmi coverage area for Mean Radial Velocity and Spectrum Width. The relevant distinction for Mean Radial Velocity is the increased coverage area and reduced azimuth sample interval.

3. Spectrum Width

This product will provide the radial velocity spectrum width data both displayable as an image and formatted as a data array. For the image version, variations of the product will be organized to provide various areas of coverage and display resolutions, while the data array version will provide the highest resolution available for the entire radar coverage area. Both versions will be limited to the lowest 70,000 feet AGL of the atmosphere. The image version of the product will be generated for any azimuth scan at a single elevation angle based on user requirements. The data array version of the product will be generated for any super-resolution azimuth scan at a single elevation angle based on user requirements. Each scan will be updated once per volume scan time. For the image version, each product will be available for 8 spectrum width data levels, while 256 spectrum width data levels will be provided in the data array version. Each product will include a standard set of annotations, data level code, elevation angle, maximum data value detected (knots), product resolution, and identification of range obscured data.

Super-Resolution refers to the following three spatial resolution and coverage enhancements: 0.5deg azimuth sampling, 0.13nmi range sampling of Reflectivity, 162nmi coverage area for Mean Radial Velocity and Spectrum Width. The relevant distinction for Spectrum Width is the increased coverage area and reduced azimuth sample interval.

4. Deleted

5. Deleted

6. Composite Reflectivity

This product will provide composite reflectivity data displayable as an image. For each geographical resolution element this product will provide the highest reflectivity value above the resolution element available from any elevation angle scan of a volume scan. This product will be available in two versions, one with the effects of anomalous propagation removed and one without. This product will be updated once per volume scan time. The product will be available for both 8 and 16 reflectivity data levels. The product will include a standard set of annotations, data level code, maximum data value (db) and its position, and product resolution. In addition, storm information generated by the various meteorological algorithms, e.g., the Storm Cell Identification and Tracking or Tornado Vortex Signature Algorithms, will be included as graphic overlays. When selected, the algorithm generated information will be provided for all identified storm cells. When these additional overlays are selected, they will be provided routinely until deselected. The information in these optional overlays will be updated once per volume scan time.

7. Deleted

8. Echo Tops

This product will provide the echo tops information displayed as an image. This product will be produced from the output of the Echo Tops Algorithm. The product will be updated once per volume scan time. The product will be available for 16 data levels referenced to mean sea level. The product will include a standard set of annotations, product resolution, data level codes, maximum data value detected (Height in feet, MSL) and its location.

9. Deleted

10. Deleted

11. Deleted

12. Velocity Azimuth Display Winds

This product will provide the Velocity Azimuth Display Algorithm derived wind speed and direction. Two versions of the product will be produced. For the first version, the wind speed and direction at up to 30 altitudes will be computed and displayed as wind barbs on a height scale.

The specific altitude levels at which VAD winds are computed and reported will be site adaptable in one thousand foot increments. The specific altitudes at which winds are requested will be used to derive the specific slant range and elevation angle for the VAD algorithm analysis. The use of lookup tables relating a specific slant range and elevation angle to a specific wind reporting altitude for computational efficiency is permitted. To facilitate this, off-line processing may be used to generate new lookup tables when new scanning sequences are invoked. The specific method of computing slant range and elevation angle is subject to user agency approval.

Wind speed and direction will be reported to the highest altitude level at which sufficient signal is available for processing by the VAD algorithm. This product will be produced in a tabular format of alphanumeric values and as a stand-alone graphic product. The graphic product will contain the current and up to 10 previous height plots (user selectable) displayed simultaneously on a time versus height scale. This version of the product will include annotations for the product name, radar ID, radar position, radar elevation above MSL, time and date of volume scan, and maximum wind speed and associated direction of most current plot. Altitude levels will be shown referenced to mean sea level. The alphanumeric format will contain all wind data derived by the VAD algorithm for the current volume scan. Upon user request, all site adaptable parameter identified as inputs to the algorithm used to generate data for this product will be available at the alphanumeric display.

A second version of the VAD winds product will be produced upon user request. This version will present the VAD wind analysis for a specific altitude. The data displayed will consist of the Doppler velocity data used to compute the VAD wind and the best fit function used to determine the horizontal wind speed and direction. The data will be presented to the user as a graphical plot of the actual Doppler velocity and its associated reflectivity and the best fit function on an orthogonal axis of speed versus azimuth. The vertical axis presented will be speed, scaled as necessary to allow all data to be displayed. The Nyquist velocity for the specific scan will be graphically annotated on the display if the range of the scaled vertical axis is sufficient. The horizontal axis will be azimuth, scaled from 0-360 degrees with true north as 0/360 degrees. This product will be available for any wind value included in the most recent time-height cross-section version of the VAD Winds Product.

This version of the product will include a standard set of annotations, VAD slant range, elevation angle, wind speed and direction, altitude of winds, NIMS proxy agent/concentrator (RMS) error, and best fit function in the form

$A_1 + V \sin(I-I)$ (See Algorithm Report)

13. Deleted

14. Cross Section (RCS, VCS)

This product will provide a vertical cross section of reflectivity or mean radial velocity data displayable as an image for a user selected vector. This vector will be defined by the operator using two endpoints, up to 230 km apart, and at any orientation and location within 230 km of the radar. This product will be produced by mapping the nearest value in range along a radial, to a point in the plane of the vertical cross section defined by the intersection of the plane and the radial. The product will be produced by linearly interpolating vertically between the mapped values. The

product will be available for both 8 and 16 data levels defining the intensity range and velocity range data levels. The product will include a standard set of annotations, and maximum data value and location(s). The location of the line segment and the end points (az/ran) will also be indicated.

15. Deleted

16. Storm Relative Mean Radial Velocity (SRM, SRR)

This product will provide mean radial velocity for: (a) a small geographic area centered upon or near an identified storm of interest with the storm motion removed, or (b) the entire area of radar coverage (to 230km) with the average storm motion removed. This product will be produced upon request for any azimuth scan at any elevation angle. The product will be generated as a displayable image by removing the radial (velocity component away from the radar antenna) component of storm motion from the mean radial velocity values.

The radial component of storm motion will be computed using the storm motion value computed for the identified storm cell by the Storm Cell Tracking Algorithm, the vector average of all currently identified storms or a value input by the user. The value of storm motion used to adjust the mean radial velocity values will be user selectable at the time of product request or default to the vector average of all currently identified storms if not selected. Each product will contain 16 data levels for storm-adjusted mean radial velocity. Each product will include annotations for the product name, radar ID, time and date of scan, elevation angle, storm motion, coordinates of product center, radar position, radar elevation above MSL, and radar operational mode.

17. Vertically Integrated Liquid

This product will provide vertically integrated liquid values displayed as an image. The output of the VIL Algorithm will be used to produce this product. The product will be updated once per volume scan time. The product will be available for 16 data levels. Each product will include a standard set of annotations, maximum data value (VIL value) and its location, and data code levels.

18. Storm Tracking Information

This product will provide information concerning the past, present and future positions of each identified storm cell. This product will be generated from the output of the Storm Cell Identification and Tracking algorithms. It will be produced in a tabular format of alphanumeric values, as a standalone graphic product, and in a format for generating graphic overlays to other products. This product will be updated once per volume scan time. Each product will include a standard set of annotations and total number of identified storms for which tracking is available. Upon user request, all site adaptable parameters identified as inputs to the algorithm(s) used to generate data for this product will be available at the alphanumeric display.

19. Hail Index

This product will provide, for each storm identified by the Storm Cell Identification and Tracking algorithm, the Probability of Hail, the Probability of Severe Hail and the Maximum Expected Hail Size. The hail probabilities and size shown for each storm cell will be generated by the Hail Algorithm. This product will be produced in a format that can be used to generate an alphanumeric tabular type of display, an alphanumeric annotation to other products, or a graphic overlay. This product will be updated concurrently with the storm structure product. This product will include a standard set of annotations. Upon user request, all site adaptable parameters identified as inputs to the algorithm(s) used to generate data for this product will be available at the alphanumeric display.

20. Mesocyclone (M, MRU, MD, DMD)

The Mesocyclone (M) product will provide information about identified shear and mesocyclone features. This product will be generated from the output of the Legacy Mesocyclone Detection Algorithm. This product will be generated in a format that can be used to generate an alphanumeric tabular display for any identified feature or all simultaneously, a graphic display or a graphic overlay to other products. This product will be updated once per volume scan time. If on a particular volume scan there is no output from the Legacy Mesocyclone Detection Algorithm (i.e., no features of any type are identified), a version of the product will be produced that exhibits the negative condition. This product will include a standard set of annotations. Upon user request, all site adaptable parameters identified as inputs to the algorithm(s) used to generate data for this product will be available at the alphanumeric display.

“A Mesocyclone Rapid Update (MRU) version of this product will be generated once per elevation scan time to provide updated Legacy Mesocyclone Detection Algorithm information. Current Legacy Mesocyclone Algorithm data at an elevation scan will be based on the elevations that have been completed thus far in the current volume scan. This information will be combined with Mesocyclone and Storm Track Algorithm information from the previous volume scan to form the MRU product.

The average motion of all SCIT storm cells from the previous volume scan will be used to derive a forecast position of each previous feature at the current volume scan time. In feature type order, the forecast position of each feature from the previous volume scan will be matched to the closest feature from the current volume scan, within a search radius defined by SCIT algorithm adaptation data. Current 3D features which are not matched to a feature from the previous volume scan, will be assigned the status of New. If previous volume scan data are unavailable, all features will be reported as new. Current features will inherit the attributes of the matched previous feature (associated storm ID, feature type, maximum tangential shear, height of maximum tangential shear, top height, base azimuth, base range, base height, azimuth diameter, range diameter).

The position attributes (base azimuth, range, and height) of a previous feature matched to a current feature will be updated to the current detection. If the top height of the matched feature is higher, the feature top height will be updated. The position attributes of a previous feature not matched to a current feature, will be set to the extrapolated forecast position. The status of unmatched previous features will be assigned to Extrapolated. Strength attributes will be updated if they increase in magnitude.

The strength attributes are feature type and maximum tangential shear. If the maximum tangential shear is updated, the radial and azimuthal diameters and the height of the maximum tangential shear will also be updated. Features with increasing strength attributes will be assigned the status of Increasing. All other matched features will be assigned the status of Persistent. Attribute data updated with current volume data will be identified. At the end of the volume scan extrapolated features will be removed.

This product will be generated in a format that can be used to generate an alphanumeric tabular display, a graphic display or a graphic overlay to other products. On alphanumeric displays, the status (Persistent, Increasing, New, or Extrapolated) of each feature status will be reported. In the graphic symbol display, features status will be reported as either extrapolated or current. Current features include all features with a status of Increasing, Persistent, or New. If on a particular elevation scan there is no output (i.e., no features of any type are identified), a version of the product will be produced that exhibits the negative condition. This product will include annotations for the product name, radar ID, date and time of volume scan, elevation angle, radar position, radar elevation above MSL, and radar operational mode.”

"The MD version of this product will provide information about circulation features generated from the output of the new Mesocyclone Detection Algorithm. This product will provide information concerning the past and future positions of each tracked circulation feature. This product will be generated in a format that can be used to generate an alphanumeric tabular display for an identified feature or all simultaneously, a graphic display or a graphic overlay to other products. This product will be updated once per volume scan time. If on a particular volume scan there is no output from the Mesocyclone Detection Algorithm (i.e., no features of any type are identified), a version of the product will be produced that exhibits the negative condition. This product will include annotations for the product name, radar ID, date and time of volume scan, radar position, radar elevation above MSL, and radar operational mode."

The DMD version of this product will be generated to support generation of interactive user displays at an external system (e.g., AWIPS). This product will be updated once per elevation scan time. The end-of-volume edition will contain complete algorithm data for the volume scan. The elevation editions will contain the algorithm data which has been updated since the previous volume edition product. A detection status will be reported for circulation features that are topped or extrapolated. If on a particular elevation scan there is no output from the Mesocyclone Detection Algorithm (i.e., no features of any type are identified), a version of the product will be produced that exhibits the negative condition. This product will include annotations for the product name, radar ID, date and time of volume scan, elevation angle, radar position, radar elevation above MSL, and radar operational mode.

21. Tornado Vortex Signature (TVS, TRU)

This product will provide information regarding the existence and location of any identified Tornado Vortex Signature (TVS). This product will be produced from the output of the TVS Algorithm. This product will produce an alphanumeric tabular display of the algorithm output data for each identified TVS signature and generate a graphic overlay of the TVS signature information when such is identified. This product will be updated once per volume scan time. This product will include a standard set of annotations. Upon user request, all site adaptable parameters identified as inputs to the algorithm(s) used to generate data for this product will be available at the alphanumeric display.

A Tornado Vortex Signature Rapid Update (TRU) version of this product will be generated once per elevation scan time to provide updated Tornado Detection Algorithm (TDA) information. Current Tornado Detection Algorithm data at an elevation scan will be based on the elevations that have been completed thus far in the current volume scan. This information will be combined with TDA, MDA, and Storm Track Algorithm information from the previous volume scan to form the TRU product.

The average motion of all tracked MDA features from the previous volume scan will be used to derive a forecast position of each TDA feature from the previous volume scan at the current volume scan time. If the average motion from MDA is unavailable, the average motion of all SCIT storm cells from the previous volume scan will be used. In priority rank order, the forecast position of each feature from the previous volume scan will be matched to the closest feature from the current volume scan, within a search radius defined by SCIT algorithm adaptation data.

Priority rank order places TVS types before ETVSs, and within each type features are ordered by Low-level Delta Velocity, from greatest to least. Current TVS/ETVS features which are not matched to a feature from the previous volume scan, will be assigned the status of New. If previous volume scan data are unavailable, all features will be reported as New. Current features will inherit the

attributes of the matched previous feature (associated storm ID, feature type, low-level delta velocity, maximum delta velocity and its height, average delta velocity, maximum shear and its height, base and top heights, depth, and base azimuth and range).

The position attributes (base azimuth, range, and height) of a previous feature matched to a current feature will be updated to the current detection. The position attributes of a previous feature not matched to a current feature, will be set to the extrapolated forecast position (base azimuth and range only). The status of unmatched previous features will be assigned to Extrapolated.

Strength attributes of a previous feature matched to a current feature will be updated if they increase in magnitude. The strength attributes are feature type, low-level delta velocity, and depth. The attributes maximum delta velocity, maximum shear, and average delta velocity are also updated to the current value if they increase in magnitude, but their increase will not trigger a status change to Increasing. If the maximum delta velocity and maximum shear are updated, their heights will also be updated. Features with increasing strength attributes will be assigned the status of Increasing. All other matched features will be assigned the status of Persistent.

Attribute data updated from current volume data will be identified. At the end of the volume scan, the extrapolated features will be removed. This product will be generated in a format that can be used to generate an alphanumeric tabular display, a graphic display or a graphic overlay to other products. In each type of display, features will appear in priority rank order. On alphanumeric displays, the status (Persistent, Increasing, New, or Extrapolated) of each feature status will be reported.

In the graphic symbol display, features status will be reported as either extrapolated or current. Current features include all features with a status of Increasing, Persistent, or New. If on a particular elevation scan there is no output (i.e., no features of any type are identified), a version of the product will be produced that exhibits the negative condition. This product will include annotations for the product name, radar ID, date and time of volume scan, elevation angle, radar position, radar elevation above MSL, and radar operational mode.

22. Storm Structure

This product will provide, for each identified storm, a history of storm cell attributes. This product will be produced from and contain values that are output by the Storm Cell Identification and Tracking and Hail Core Aloft Algorithms. This product will be updated once per volume scan time. Storm Cell Attributes for the current volume scan will be represented in a displayable tabular alphanumeric format and will include a standard set of annotations. The history of storm cell attributes (Trend Data) will be represented in a non-displayable tabular format. Upon user request, all site adaptable parameters identified as inputs to the algorithm(s) used to generate data for this product will be available at the alphanumeric display.

23. Layer Composite Reflectivity

This product will display as an image the maximum or average composite reflectivity for a given layer. This product will be available in two versions, one with the effects of anomalous propagation removed and one without. All layer products will be on a geographically based Cartesian grid, centered on the radar. Up to three layers will be available for layer products at each site; the depth of each layer will be controlled via adaptation data. No layer will be less than 6,000 ft thick. Each layer product will be updated on a volume scan basis.

Eight data levels will be available for each product, including one for data below the minimum threshold and one for no data. The default values for the reflectivity will be the standard NWS DVIP levels, as defined in FMH #7. The data level thresholds will be controlled via adaptation data.

It will be possible to select either a maximum value or the average reflectivity for the grid box, but not both simultaneously. The range of coverage for each product will be controlled via adaptation data; the default is a 460 km X 460 km square centered on the radar.

A three-dimension (3D) grid box will be defined as the vertical projection of the geographically based Cartesian grid square through the appropriate layer. All radar resolution volumes whose centers are contained in a given 3D grid box will be included in the computation for the 3D grid box. All 3D boxes whose centers fall in a radar resolution volume will include the radar resolution volume in the computation for that 3D grid box. A center that falls on a boundary will be considered to fall on both/all sides of the boundary. The product resolution will be 4 km x 4 km.

Each product will contain a standard set of annotations, layer definition, data level code, maximum data value detected and its location.

24. Deleted

25. User Alert Message

This product will provide a brief message containing descriptive information pertaining to all alerts identified in the current volume scan. The message will contain the location (azimuth/range), alert category, threshold exceeded, exceeding value, date and time, storm ID# (if appropriate), , and storm speed and direction of movement. The product will be updated once per volume scan. The product will include a standard set of annotations.

26. Radar Coded Message

The Radar Coded Message (RCM) product will be produced automatically up to 2 times per hour based on time of day specified in minutes after the hour. This product will also be produced upon one-time request from the designated user using data from the last completed volume scan. For a continually scanning radar, the product will be produced from the last completed volume scan prior to the scheduled time.

27. Free Text Message

This product will provide a capability for defining an alphanumeric message for onetime transmission to specific WSR-88D interface(s). The product will be generated by an operator through the MSCF or collocated PUP for transmission to the RPG's users or by the PUP operator for transmission to the PUP's users. Messages will be designated to be transmitted to one or several operator specified WSR-88D interfaces. Product will include a standard set of annotations.

28. Surface Rainfall Accumulation

These products will provide 1-hour and 3-hour rainfall accumulation maps displayed as an image. The 1-hour map will be updated every volume scan time and the 3-hour map will be updated at the top of each hour. These products will be available for 16 precipitation accumulation data levels. Each product will include a standard set of annotations, data level codes, hybrid scan date and ending time (or clock hour as appropriate) of the precipitation accumulation integration, maximum data value and its location, the applied bias based on gage reports , and the effective gage-radar pair sample size associated with the bias estimate.

29. Storm Total Rainfall Accumulation

This product will provide rainfall accumulation data, both displayed as an image formatted as non-displayable, digital data. The product format and content of the image version (STP) will be the same as the surface Rainfall Accumulation Product except the time period will be a variable and will equal the period of continuous rainfall in the radar area of coverage. This product will include annotations for product name, radar ID, maximum data value detected, radar position, times and dates of the beginning and end (TSavgcur* or clock hour as appropriate) of the rainfall rate integration, radar position, radar elevation above MSL, radar operational mode, the mean-filed bias in the radar estimate of the precipitation rate, and the effective (Gage-Radar Pair) sample size associated with the bias estimate.

The digital version of this product (DSP) will be available for 256 data levels at each sample bin location of a 1 degree by 2 kilometer polar grid, extending to a range of 230 kilometers (124 nautical miles). The digital version is provided to support processing external to the NEXRAD system. It will be updated every volume scan and will include the same annotations as the graphical version of the product (listed above).

30. Hourly Digital Precipitation Array

This product will provide hourly running total digital radar-rainfall estimates in an array format (not display oriented) to support processing performed external to the WSR-88D System. This product will be updated once per volume scan time. The product will be available for 256 data levels for each array element. Each product will include a standard set of annotations, data level codes, hybrid scan date and ending time (or clock hour as appropriate) of the precipitation accumulation integration, maximum data value and its location, the applied bias based on gage reports, and the effective gage-radar pair sample size associated with the bias estimate.

31. Supplemental Precipitation Data

This product will provide selected supplemental data generated or collected during the operation of the precipitation processing subsystem algorithms. This product will also provide the complete table of Gage-Radar Bias information determined, externally to the NEXRAD RPG, over various memory timespans and used within the RPG (at the option of the operator) to apply a mean-field adjustment to the radar-estimated rainfall products. This product will be produced in a tabular alphanumeric format.

This product will be updated once per volume scan time. This product will include a standard set of annotations and maximum data value and its location.

32. User Selectable Rainfall Accumulation

This product will provide a rainfall accumulation map displayed as an image, for a user selected accumulation period. The product format and content will be the same as the Surface Rainfall Accumulation and Storm Total Rainfall Accumulation products, except the accumulation period will be of variable duration (in whole clock hours), ranging from a beginning to an ending time specified by the user. The product will usually be generated by request, but may also be generated routinely for limited, designated periods. One of these periods will be the Default Period, which will span a 24 hour period from 1200Z of the previous day to 1200Z of the present day. The default version of the product may be generated any time following the completion of the first volume scan to start after 1200Z each day, up to 6 hours later. The period of accumulation will be defined by the operator via two parameters: the Ending Hour (ranging from 0 to 23Z), and the Time Span (ranging from 1 to 24 whole clock hours prior to the Ending Hour). No more than 30 hours of rainfall data prior to the most recent clock-hour will be required to generate this product.

If a requested product cannot be generated due to an error condition, a message will be displayed explaining why, and the available hours of precipitation accumulation in the precipitation data base will be listed.

This product will include a standard set of annotations, maximum data value detected and its current position, data level codes hybrid scan beginning and end times and dates of the precipitation accumulation integration, the applied bias (based on gage reports) for each hour in the product, an indication whether each hour is included in the precipitation accumulation, and the effective gage-radar pair sample size associated with the bias estimate.

33. Hybrid Scan Reflectivity

This product will provide radar-reflectivity values for the composite Hybrid Scan, assembled from the lowest four elevation angles, in digital and image versions. The digital version will yield a representative reflectivity value at every sample bin location of a 1 degree by 1 kilometer polar grid, out to a range of 230 kilometers (124 nautical miles). This digital data is provided to support processing performed by external systems and is not display oriented. This product will be updated once every volume scan time. This product will be available for 256 data levels at each sample bin location. The image version of the product will be available for 16 reflectivity data levels and will be updated every volume scan. Each product will include a standard set of annotations, the hybrid scan data and time of the scan, and the maximum data value (db) and its location.

34. Clutter Filter Control

This product will provide for display of Clutter Filter and Bypass Maps as a radial image, for the user selected elevation segment. The product, while considered a volume product, is not generated every volume scan. The product will be generated upon detection of updated Clutter Filter Map and/or Bypass Map data received from the RDASC, or in response to user request(s) when product is not in the RPG Product Database. Up to five elevation segments are available for selection. (See Algorithm Report)

The product will include annotations for elevation segment number (1 through 5), Bypass Map generation date/time, Clutter Filter Map generation date/time, and all other standard annotations.

35. ITWS Digital Base Velocity

This product will provide mean radial velocity in a digital array format to support processing external to the NEXRAD system. This product will be generated for each azimuth scan (elevation) based upon user requirements. This product will consist of a one degree by 1 kilometer polar grid out to a range of 115 kilometers (62 nautical miles). This product will provide 256 data levels (including 254 mean radial velocity data levels and indications for data "below the Signal-to-Noise threshold" and "Range Folding") for each array element. The product will be formatted as a Graphic Product as specified in the ICD for the RPG/Class 1 User, Document Number 2620001, using Digital Radial Data Array Packets, Figure 3-11c. In addition to the standard contents of the Graphic Product, this product will include the minimum data value (in meters per second), the increment (in meters per second), the number of data levels, the elevation (in degrees), the maximum negative velocity detected (in knots) and the maximum positive velocity detected (in knots).

36. Clutter Likelihood Reflectivity

The Clutter Likelihood Reflectivity product will provide a displayable image of the percentage probability that the radar is detecting ground clutter in the reflectivity radial samples. The processing will be based on the output of the Radar Echo Classifier (REC) algorithm using the Anomalous Propagation /Clutter target logic. The product will be generated for any azimuth scan at a single elevation angle based on user requirements. Each product will be available in an 11-level

display. A separate data level code will be used to represent sampled bins below SNR threshold. The product will display the percentage likelihood of clutter for each reflectivity bin, from the radar to 124 nautical miles range. The product will be a radial format display with a range resolution of 0.54 nautical miles. Each product will include annotations for the product name, radar ID, time and date of scan, data level code, elevation angle, radar position, radar elevation above MSL and radar operational mode. Upon user request, all site adaptable parameters identified as inputs to the algorithm used to generate data for this product will be available to the user.

37. Clutter Likelihood Doppler

The Clutter Likelihood Doppler product will provide a displayable image of the percentage probability that the radar is detecting ground clutter in the Doppler radial samples. The processing will be based on the output of the Radar Echo Classifier (REC) algorithm using the Anomalous Propagation /Clutter target logic. The product will be generated for any azimuth scan at a single elevation angle based on user requirements. Each product will be available in an 12-level display. A separate data level code will be used to represent sampled bins below SNR threshold. A separate data level code will be used to represent bins where the Doppler data is undermined due to range folded data. The product will display the percentage likelihood of AP/clutter for each Doppler bin, from the radar to 124 nautical miles range. The product will be a radial format display with a range resolution of 0.54 nautical miles. The value displayed for a given 0.54 nautical mile bin will be the maximum of the four corresponding Doppler bins. Each product will include annotations for the product name, radar ID, time and date of scan, data level code, elevation angle, radar position, radar elevation above MSL and radar operational mode. Upon user request, all site adaptable parameters identified as inputs to the algorithm used to generate data for this product will be available to the user.

38. SuperOb

This product will contain 8 bit precision radial velocity data as an average of all base velocity bins within discrete sampling areas over a predetermined time span. The product, while considered a volume product, will not be generated every volume scan. The time span over which the velocity data is averaged will be adaptable between 10 minutes and 3 hours with the default value being 1 hour. The product will be generated at the end of the time span.

Each elevation scan will be divided into multiple sampling areas, or SuperOb cells. The SuperOb cells are defined by an adaptation data supplied range size and azimuth size. A maximum range will also be defined to limit the range extent of the averaging.

Velocity data will be averaged within the SuperOb cells for each elevation of the volume coverage pattern. If the volume coverage pattern changes during the time span, thereby causing a change in elevation angles, a tolerance will be established so that elevation differences beyond the tolerance will cause velocity data for the new elevation(s) to be averaged separately. Elevation differences within the tolerance value will cause velocity data from the new elevation(s) to be included in the averages for the nearest elevation.

A minimum number of samples for a usable average will be defined in adaptation data. If a SuperOb cell does not meet this minimum number of samples within the time span, an average will not be computed for that cell.

At the end of a time span, the resulting product will supply the following information for each SuperOb cell in each elevation: average radial velocity, root mean square of the average radial velocities, azimuth, latitude, longitude, height, and time deviation. The time deviation is the mean number of seconds deviation from the base time of all the sample times included in the cell. The

base time is defined as the center of the time span of data collection. The latitude, longitude, height, and azimuth values are also determined from the mean of all samples in the SuperOb cell.

39. High Resolution Vertically Integrated Liquid

This product will provide vertically integrated liquid water values for a volume scan at every sample bin location of a 1 degree by 1 kilometer polar grid, out to a range of 460 kilometers (248 nautical miles). These digital data are provided to support processing performed by external systems and are not display oriented. Input to this product is provided by the intermediate linear buffer edited for artifacts and anomalous propagation by the Data Quality Assurance algorithm. This High Resolution Vertically Integrated Liquid product will be updated one time upon completion of each full volume scan. This product will be available with 254 data levels at each sample bin location with an additional 2 values reserved for “no data” or “flagged data”. Each product will include a standard set of annotations, maximum data value (VIL value), number of radials in volume edited for artifacts, and data code level information.

40. User Selectable layer Composite Reflectivity Maximum

This product will be provided on a 1 degree by 1 kilometer polar grid, out to a range of 230 kilometers (124 nautical miles). For each resolution element, this product will provide the maximum reflectivity value above the resolution element from the sample bins intersecting the layer from any elevation angle scan of a volume scan. At most ten layers will be available per volume. The altitude limits of each layer will be specified by user product request. No layer will be less than 1000 ft thick. Each layer product will be updated on a volume scan basis. Sixteen data levels will be available for each product, including one for data below minimum Signal-to-Noise threshold and one for unavailable data. Data is unavailable at a given range bin due to gaps between elevation scans or if the given range bin lies above the layer or below the layer. Each product will contain annotations for product name, radar ID, date and time of volume scan, maximum data value, radar position, radar elevation above MSL, and radar operational mode.

41. High Resolution Enhanced Echo Tops (HREET)

This product will provide the echo tops information to support processing performed by external systems.

Reflectivity data used to generate this product is edited for artifacts and anomalous propagation by the Data Quality Assurance algorithm. The HREET algorithm also removes meteorologically implausible spurious tops at or above 45,000 feet. The High Resolution Enhanced Echo Tops product will be updated once per volume scan time. This product will report echo tops at a vertical resolution of 1000 feet referenced to mean sea level on a 1 degree by 1 kilometer polar grid to a range of 345 kilometers (186 nautical miles). Echo top data will be flagged to indicate the “topped” condition, meaning echo tops may be higher than indicated. A Topped condition results when either the reflectivity is greater than the threshold of 18 dBZ at the top elevation scan, or when the value immediately above an 18+ dBZ value is bad and no other values at higher elevations are above threshold. Additionally, two values are reserved for “no data” and “bad data”. A sample bin is flagged “bad data” if the reflectivity was edited by the DQA algorithm. The value “no data” means there are no reflectivity data above threshold at the location of the sample bin. The flagged conditions cannot be “topped”.

Each product will include a standard set of annotations, maximum data value (echo top value), number of radials in volume edited for artifacts, data code level information, and the reflectivity (dBZ) threshold value.

42. One Hour Snow Accumulation (OSW, OSD)

These products shall provide radar algorithm estimated 1-hour snow water equivalent and snow depth accumulation images. The products shall be updated every volume scan. The products shall be available for 16 data levels. Each product shall include annotations for the product name, radar ID, date and ending time of the accumulation, the type of range/height correction (such as algorithm derived or static) has been applied to the accumulation, maximum data value, azimuth and range of the maximum value, radar position, radar elevation above MSL, and radar operational mode.

43. Storm Total Snow Accumulation (SSW, SSD)

These products shall provide radar algorithm estimated storm total snow water equivalent and snow depth accumulation images. The products shall be updated every volume scan. The products shall be available for 16 data levels. Each product shall include annotations for the product name, radar ID, Starting and ending date and time of the accumulation, the type of range/height correction (such as algorithm derived or static) has been applied to the accumulation, maximum data value, azimuth and range of the maximum value, radar position, radar elevation above MSL, and radar operational mode.

44. User Selectable Snow Accumulation (USW, USD)

These products shall provide radar algorithm estimated storm total snow water equivalent and snow depth accumulation images for a user selected accumulation period. The product format and content shall be the same as the storm total snow accumulation products, except the accumulation period shall be of variable duration and in whole top-of-the-hour clock hours. The products shall be updated at the top of the every hour (as requested). The accumulation period shall be definable by the requester via two parameters: the Ending Hour (ranging from 0 to 23Z) and the Time Span (ranging from 1 to 30 whole clock hours prior to the Ending Hour.) The product will usually be generated by request, but may also be generated routinely for designated periods. The default accumulation period shall span a 24 hour period from 1200Z of the previous day to 1200Z of the present day. The products will be available for 16 data levels. Each product will include annotations for the product name, radar ID, dates and times of the starting and end (clock hour) of the snow accumulation, the type of range correction (such as algorithm or static derived) has been applied to the accumulation, maximum data value, azimuth and range of the maximum value, radar position, radar elevation above MSL, and radar operational mode.”

45. Archive III Status Product (ASP)

The ASP product contains an adaptable number of hours of RPG status log messages. The product (by default) is generated every eight hours at 00:00, 08:00, and 16:00 GMT. The product contains all RPG status log messages since the last product was generated.

This product shall include annotations for the product name, radar ID, date and time of volume scan, elevation angle, radar position, radar elevation above MSL, and radar operational mode.

46. Gust Front MIGFA (GFM)

This product shall provide convergence boundary detections (such as gust fronts) with 10 and 20 minute forecast positions. The product extends to 70 kilometers (38 nautical miles) range from the radar. This product is provided to support processing performed by external systems and is not display oriented. Input to MIGFA is a merger of the 0.5 and 1.5 degree elevation cuts for all VCPs but VCP 12. For VCP 12, the merger is for the 0.5 and 1.3 degree elevation cuts. This product will be updated once every volume scan time. Each product will include a standard set of annotations and a count of the number of detections in the volume. The product shall be formatted as a Generic Data Product as specified in the ICD for Class 1 Users (Document Number 2620001?) using Packet 28 as described in Appendix E.”

Each product shall include annotations for product name, radar ID, time and date of volume scan, number of detections in the volume, radar position, radar elevation above MSL, and the radar operational mode.

47. NEXRAD Turbulence Detection Algorithm (NTDA)

This product will provide values of in-cloud atmospheric turbulence (eddy dissipation rate, EDR) and associated confidence values (EDC) for each elevation tilt in a polar radial format. The product extends to 230 kilometers (124 nautical miles) range from the radar. These data are provided for external use to support the detection of conditions hazardous to aviation. The product will be generated for each elevation at its completion. This product will provide 64 data levels for EDR and 8 data levels for confidence.

48. Differential Reflectivity

This product shall provide the polarimetric differential reflectivity data displayable as an image and formatted as a data array. Both versions shall be limited to the lowest 70,000 feet Above Radar Level (ARL) of the atmosphere. The product shall be generated for any azimuth scan at a single elevation angle based on user requirements. Each scan shall be updated once per volume scan time. For the image version, the product shall be available for 16 data levels, while 256 data levels will be provided in the data array version. Each product shall include annotations for the product name, radar ID, time and date of scan, data level code, elevation angle, maximum and minimum data value (dB), radar position, radar elevation above MSL, and radar operational mode.

49. Correlation Coefficient

This product shall provide the polarimetric correlation coefficient data displayable as an image and formatted as a data array. Both versions will be limited to the lowest 70,000 feet ARL of the atmosphere. The product shall be generated for any azimuth scan at a single elevation angle based on user requirements. Each scan shall be updated once per volume scan time. For the image version, the product shall be available for 16 data levels, while 256 data levels will be provided in the data array version. Each product shall include annotations for the product name, radar ID, time and date of scan, data level code, elevation angle, maximum and minimum data value, radar position, radar elevation above MSL, and radar operational mode.

50. Specific Differential Phase

This product shall provide the polarimetric specific differential phase data displayable as an image and formatted as a data array. Both versions will be limited to the lowest 70,000 feet ARL of the atmosphere. The product shall be generated for any azimuth scan at a single elevation angle based on user requirements. Each scan shall be updated once per volume scan time. For the image version, the product shall be available for 16 data levels, while 256 data levels will be provided in the data array version. Each product shall include annotations for the product name, radar ID, time and date of scan, data level code, elevation angle, maximum and minimum data value (deg/km), radar position, radar elevation above MSL, and radar operational mode.

51. Hydrometeor Classification

This product shall provide derived hydrometeor classification data displayable as an image and formatted as a data array. All versions will be limited to the lowest 70,000 feet ARL of the atmosphere. The HC and DHC products shall be generated for any azimuth scan at a single elevation angle based on user requirements. Each azimuth scan shall be updated once per volume scan time. The HHC product shall be a data array version generated once per volume scan time and consist of a hybrid of the azimuth scan versions where the displayed hydro classification for each range bin is the value used by the Quantitative Precipitation Estimation (QPE) algorithm. The

displayable image version shall be generated for 16 hydrometeor categories. The data array versions shall be generated for all hydrometeor categories generated by the Hydrometeor Classification Algorithm. Each product shall include annotations for the product name, radar ID, time and date of scan, data level code, elevation angle, radar position, radar elevation above MSL, and radar operational mode.

52. Melting Layer

This product shall provide contours of the melting-layer/radar-beam intersections and shall be produced by contouring the output of the Melting Layer Detection Algorithm. This product shall be displayable alone or as an overlay on elevation based products. At 1 degree azimuth angle increments, it shall depict the range where the radar beam is centered on the top and bottom of the melting layer and the range where the beam starts and stops intersecting the melting layer. The ranges for each of these four melting-layer/radar-beam transitions will be used to construct four contours.

The product shall be generated for any azimuth scan at a single elevation angle based on user requirements. Each scan shall be updated once per volume scan time. Each product shall include annotations for the product name, radar ID, time and date of scan, data level code, elevation angle, radar position, radar elevation above MSL, and radar operational mode.

53. One-Hour Accumulation

The One-Hour Accumulation (OHA) product shall be a polar grid of precipitation accumulation over the last hour. The product will be displayable as a graphical image. The accumulations are computed using the dual polarization Quantitative Precipitation Estimation (QPE) algorithm. The product format is similar to the Storm Total Accumulation product. The product shall be updated every volume scan. The product will contain 16 data levels. Each product shall include annotations for the product name, radar ID, time and date of scan, data level code, radar position, radar elevation above MSL, and radar operational mode.

54. Digital Accumulation Array

The Digital Accumulation Array (DAA) product shall be a polar grid of digital high-resolution one-hour precipitation accumulation. The product will span the last hour regardless of when the Precipitation Accumulation Initiation Function (PAIF) indicated that precipitation began. The product will be displayable as a graphical image. The accumulations are computed using the dual polarization Quantitative Precipitation Estimation (QPE) algorithm. The product shall be updated every volume scan. This product shall not have a bias applied (within the RPG).

This product will contain 256 data levels on an automatically adjustable linear scale based on the maximum accumulation in the grid over the user specified time span. Data level 0 will correspond to either no accumulation or NO_DATA, and data levels 1 through 255 will indicate accumulations scaled to 0.01 inches. Color levels can be determined by the display system. The product is updated every volume scan. Each product shall include annotations for the product name, radar ID, time and date of scan, data level code, radar position, radar elevation above MSL, and radar operational mode.

55. Storm Total Accumulation

The Storm Total Accumulation (STA) product shall be a polar grid of precipitation accumulation since the Precipitation Accumulation Initiation Function (PAIF) indicated that precipitation has begun. The product will be displayable as a graphical image. The beginning of the precipitation and the accumulations are computed using the dual polarization Quantitative Precipitation Estimation (QPE) algorithm. If there has not been any precipitation in the last hour, the polar grid shall be blank. The product format and content is similar as to the One-Hour Accumulation product except

this product also has an alphanumeric layer. This product shall be appended with an alphanumeric layer including the adaptable parameter values, bias information (for future implementation), and other supplemental (precipitation status) data.

56. Digital Storm Total Accumulation

The Digital Storm Total Accumulation (DSA) product shall be a polar grid of digital high-resolution precipitation accumulation since the Precipitation Accumulation Initiation Function (PAIF) indicated that precipitation has begun. The product will be displayable as a graphical image. The beginning of the precipitation and the accumulations are computed using the dual polarization Quantitative Precipitation Estimation (QPE) algorithm. The product shall be updated every volume scan. If there has not been any precipitation in the last hour, the polar grid shall be blank. The product is appended with an alphanumeric layer including the adaptable parameter values, bias information (for future implementation), and other supplemental (precipitation status) data.

This product will contain 256 data levels on an automatically adjustable linear scale based on the maximum accumulation in the grid over the duration of a storm event. Data level 0 will correspond to either no accumulation or NO_DATA, and data levels 1 through 255 will indicate accumulations scaled to 0.01 inches. Color levels can be determined by the display system.

57. Digital User-Selectable Accumulation

The Digital User-Selectable Accumulation (DUA) product shall be a polar grid of high-resolution precipitation accumulation over a user-selected accumulation period. The product will span the user-selected period regardless of when the Precipitation Accumulation Initiation Function (PAIF) indicated that precipitation began. The product will be displayable as a graphical image. The accumulations are computed using the dual polarization Quantitative Precipitation Estimation (QPE) algorithm. The accumulation period shall be of variable duration as specified by the user and defined via two parameters: the Ending Time (Z) in *hh:mm* format, and the Time Span in *hh:mm* format (ranging from a minimum of 15 minutes to 24 hours prior to the Ending Time). The product can be generated by request or routinely for designated periods and times. By default, the DUA product shall be generated at 1200 Z for a time span of 24 hours and every volume scan for time spans of 1 and 3 hours.

This product will contain 256 data levels on an automatically adjustable linear scale based on the maximum accumulation in the grid over the user specified time span. Data level 0 will correspond to either no accumulation or NO_DATA, and data levels 1 through 255 will indicate accumulations scaled to 0.01 inches. Color levels can be determined by the display system.

58. Digital One-Hour Difference

The Digital One-Hour Difference product shall be a polar grid of digital accumulation differences. The differences are computed by subtracting the latest one-hour-estimate computed by the legacy PPS from the one-hour estimates computed by the QPE. Neither estimate will have any bias applied. This product will be generated every volume scan whether or not precipitation has been detected.

This product will contain 256 data levels on an automatically adjustable linear scale based on the difference with the maximum magnitude (positive or negative). The 0 data level always indicates NO_DATA. The remaining data levels will be scaled at equal steps centered at 0 difference, for the 128th data level. The minimum step will be 0.01 inches. The difference with the maximum magnitude (greatest positive or negative value) is used to determine the scale and offset values used to encode the data levels. Color levels can be determined by the display system.

59. Digital Storm Total Difference

The Digital Storm Total Difference product shall be a polar grid of digital accumulation differences. The differences are computed by subtracting the storm total estimates computed by the legacy PPS from the storm total estimates computed by the QPE. Neither estimate will have any bias applied. The product shall be a volume-based product and shall be updated every volume scan. However, this product will only contain a graphical image whenever one of the systems (legacy PPS or dual-polarization QPE) has detected precipitation and is accumulating storm total information. That is, this product will only have a graphical image present when at least one of the systems is generating storm total products

This product will contain 256 data levels on an automatically adjustable linear scale based on the difference with the maximum magnitude (positive or negative). The 0 data level always indicates NO_DATA. The remaining data levels will be scaled at equal steps centered at 0 difference, for the 128th data level. The minimum step will be 0.01 inches. Color levels can be determined by the display system.

TABLE E-1a. PRODUCT GENERATION & ARCHIVE PERFORMANCE LOAD

Product Name	Coverage	Resolution	Products	Archive
1. Reflectivity				
Graphic	0-230	1 x 1 °	lowest 4 + 6.2, 10.0, 14.0	lowest 1
Graphic	0-460	2 x 1 °	lowest 3	lowest 1
Graphic	0-460	4 x 1 °	None	
Digital	0-460	1 x 1 °	All elevations	None
Digital	0-460	0.25 x 0.5 °	Less than/equal 1.5	None
2. Mean Radial Velocity				
Graphic	0-60	0.25 x 1 °	lowest 1	lowest 1
Graphic	0-115	0.5 x 1 °	lowest 4 + 6.2, 10.0	lowest 1
Graphic	0-230	1 x 1 °	14 elevations	lowest 1
Digital	0-230	0.25 x 1 °	All elevations	None
Digital	0-300	0.25 x 0.5 °	Less than/equal 1.5	None
3. Spectrum Width				
	0-60	0.25 x 1 °	lowest 1	lowest 1
	0-115	0.5 x 1 °	None	
	0-230	1 x 1	lowest 4	lowest 1
Digital	0-300	0.25 x 0.5 °	Less than/equal 1.5	None
4. Deleted				
5. Deleted				
6. Composite Reflectivity				
	0-230	1 x 1	1 (16 lvl) 1 (8 lvl)	None None
	0-460	4 x 4	1 (16 lvl) 1 (8 lvl)	yes yes
AP removed	0-230	1 x 1	None	None
AP removed	0-460	4 x 4	None	None
7. Deleted				
8. Echo Tops	0-230	4 x 4	1	yes
9. Deleted				
10. Deleted				
11. Deleted				
12. Velocity Azimuth		1000 ft alt	None	

Product Name	Coverage	Resolution	Products	Archive
Display VAD				
Wind Profile		1000 ft alt	1	None
13. Deleted				
14. Cross Section (230km X 70k')		1 x 0.5	3	None
15. Deleted				
16. Storm Rel Mean Radial Vel Region	50 x 50	0.5 x 1°	1	
Map	0-230	1 x 1°	lowest 4 + 6.2, 10.0, 14.0	lowest 1
17. Vertically Integrated Liquid	0-230	4 x 4	1	yes
18. Storm Tracking	0-460	N/A	1	yes
19. Hail Detection (Enhanced)	0-230	N/A	1	yes
20. Mesocyclone	0-230	N/A	1	yes
Mesocyclone Rapid Update	360 degrees x 124 nm	N/A	All Elevations	None
Mesocyclone Detection	0-230 km	N/A	1	None
Mesocyclone Detection Data Array Product	0-230	N/A	All Elevations	None
Digital Mesocyclone Detection	0-230 km	N/A	All Elevations	None
21. Tornado Vortex Signature	0-230	N/A	1	yes
Tornado Vortex Signature Rapid Update	0-230 km	N/A	1	None.
22. Storm Structure	0-460	N/A	1	None
23. Layer Composite Reflec				
33M-60M	460 x 460	4 x 4	1	None
24M-33M	460 x 460	4 x 4	1	None
Sfc-24M	460 x 460	4 x 4	1	None
AP removed Sfc-24M	460 x 460	4 x 4	1	None
24. Deleted				
25. User Alert Message	N/A	N/A	1	yes
26. Radar Coded Message	N/A	N/A	1	yes
27. Free Text Message	N/A	N/A	1	None
28. Sfc Rainfall Accumulation 1 hr	0-230	2 x 1°	1	yes
3 hr	0-230	2 x 1°	1	None
29. Storm Total Rainfall Accum	0-230	2 x 1°	1	yes
30. Hourly Digital Precip Array	0-230	1/40 LFM	1	yes
31. Supplemental	N/A	N/A	1	yes

Product Name	Coverage	Resolution	Products	Archive
Precipitation Data				
32. User Selectable Precip Accum	0-230	2 x 1°	1	None
33. Hybrid Reflectivity - Digital	0-230	1 x 1°	1	None
- Image	0-230	2 x 1°		
34. Clutter filter control	0-230	1 km x 1.0°	1	Yes
35. ITWS Digital Base Velocity	0-115	1 km x 1 °	None	None
36. Clutter Likelihood Reflectivity	0-230	1 x 1°	Lowest	None
37. Clutter Likelihood Doppler	0-230	1 x 1°	Lowest	None
38. SuperOb	0-230 (adaptable, default=100km)	Default = 5 x 6° (adaptable)	Default = 1 per hour (adaptable between 6 per hour and one every 3 hours)	None
39. High Resolution VIL	0-460	1 x 1°	1	Yes
40. User Selectable Layer Composite Reflectivity	360 degrees x 124 nm	1 degree x 0.54 nm	None	None
41. High Resolution Enhanced Echo Tops	0-345	1 x 1 °	1	None
42 One Hour Snow Accumulation	0-230	1 x 1	1	None
43. Storm Total Snow Accumulation	0-230	1 x 1	1	None
44. User Selectable Snow Accumulation	0-230	1 x 1	1	None
45. Archive III Status Product (ASP)	N/A	N/A	None	Yes
46. Gust Front MIGFA	0-70	N/A	1	No
47. NTDA	2-230km	2 km x 1°	EDR and EDC at all elevations	None
48. Differential Reflectivity ZDR	0-230	1 x 1 °	All Elevations	Lowest 4?
DZD	0-300	0.25 x 1 °	All Elevations	Lowest 2?
49. Correlation Coefficient CC	0-230	1 x 1 °	All Elevations	Lowest 4?
DCC	0-300	0.25 x 1 °	All Elevations	Lowest 2?
50. Specific Differential Phase KDP	0-230	1 x 1 °	All Elevations	Lowest 4?

Product Name	Coverage	Resolution	Products	Archive
DKD	0-300	0.25 x 1 °	All Elevations	Lowest 2?
51. Hydrometeor Classification HC	0-230	1 x 1 °	All Elevations	Lowest 4
DHC	0-300	0.25 x 1 °	All Elevations	Lowest 2?
HHC	0-230	0.25 x 1 °	1	None?
52. Melting Layer	0-230	0.25 x 1 °	All Elevations	TBD
53. One-Hour Accumulation	0-230	2 x 1 °	1	Yes?
54. Digital Accumulation Array	0-230	0.25 x 1 °	1	Yes?
55. Storm Total Accumulation	0-230	2 x 1 °	1	Yes?
56. Digital Storm Total Accumulation	0-230	0.25 x 1 °	1	Yes?
57. Digital User-Selectable Accumulation	0-230	0.25 x 1 °	1	Yes?
58. Digital One-Hour Difference	0-230	0.25 x 1 °	1	None?
59. Digital Storm Total Difference	0-230	0.25 x 1 °	1	None?
Note: Total product generation is depicted (both routine and one time requests).				

TABLE E-1b PRODUCT RESPONSE TIME & THROUGHPUT TIME

Product Name	Throughput (sec)	Response (sec)
Reflectivity	30	90
Mean Radial Velocity	30	90
Spectrum Width	30	90
Combined Shear	no longer available	no longer available
Combined Shear Contour	no longer available	no longer available
Composite Reflectivity	42	90
Composite Reflectivity Contour	no longer available	no longer available
Echo Tops	60	90
Echo Tops Contour	no longer available	no longer available
Severe Weather Analysis (4 products)	no longer available	no longer available
Severe Weather Probability	no longer available	no longer available
Velocity Azimuth Display	60	180
Combined Moment	no longer available	no longer available
Cross Section	N/A	90
Weak Echo Region	no longer available	no longer available
Storm Relative Mean Radial Vel Map/Region	30	90
Vertically Integrated Liquid	30	90
Storm Tracking Information (SCIT)	30	90
Hail Detection (Enhanced)	30	90
Mesocyclone	30	90
Tornado Vortex Signature	30	90
Storm Structure	30	90
Layer Composite Reflectivity Products	30	90
Layer Composite Turbulence Products	no longer available	no longer available
User Alert Message	30	N/A
Radar Coded Message	30	N/A
Free Text Message	30	N/A
Surface Rainfall Accumulation Products	30	90
Storm Total Rainfall Accumulation	30	90
Hourly Digital Precipitation Array	30	90
Supplemental Precipitation Data	30	90
User Selectable Precip Accumulation	30	90
Hybrid Reflectivity	30	90
Clutter Filter Control	30	90
User Selectable Layer Composite Reflectivity Max	42	90
Clutter Likelihood Reflectivity	30	90
Clutter Likelihood Doppler	30	90
SuperOb	90	N/A
High Resolution VIL	30	90
High Resolution Enhanced Echo Tops	30	90
Mesocyclone Rapid Update	30	90
Tornado Vortex Signature Rapid Update (TRU)	30	90
Mesocyclone Detection	30	90
Digital Mesocyclone Detection	30	90
One Hour Snow Accumulation	30	90
Storm Total Snow Accumulation	30	90
User Selectable Snow Accumulation	30	90
Archive III Status Product (ASP)	30	N/A
Gust Front MIGFA	30	N/A

Product Name	Throughput (sec)	Response (sec)
NTDA (EDR and EDC)	TBD	N/A
Differential Reflectivity	30	90
Correlation Coefficient	30	90
Specific Differential Phase	30	90
Hydrometeor Classification	30	90
Melting Layer	30	90
One-Hour Accumulation	30	90
Digital Accumulation Array	30	90
Storm Total Accumulation	30	90
Digital Storm Total Accumulation	30	90
Digital User-Selectable Accumulation	30	90
Digital One-Hour Difference	30	90
Digital Storm Total Difference	30	90

Note: The assumed data rate is 14.4 kbps.

APPENDIX F – DELETED

APPENDIX G – ACRONYMS/ABBREVIATIONS

A _a	Achieved Availability
A _i	Inherent Availability
A _o	Operational Availability
A _s	Service Availability
ADT	Administrative Down Time
AFI	Air Force Instruction
A/D	Analog/Digital
AFGWC	Air Force Global Weather Central
AFWA	Air Force Weather Agency
AFSS	Automated Flight Service Station
ALDT	Administrative and Logistics Down Time
ANSI	American National Standards Institute
ASTM	American Standard Testing Methods
ATCSCC	Air Traffic Control System Command Center
ATE	Automatic Test Equipment
AWIPS	Advanced Weather Interactive Processing System
BITE	Built In Test Equipment
BWS	Base Weather Station
CCB	Configuration Control Board
CI	Configuration Item
CONUS	Continental United States
COTS	Commercial Off The Shelf
CPCI	Computer Program Configuration Item
CWSU	Center Weather Service Unit
DOC	Department of Commerce
DOD	Department of Defense
DOT	Department of Transportation
EIA	Electronic Industry Alliance
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
FAA	Federal Aviation Administration
GDSS	Gage Data Support System
GFP	Government Furnished Property
ICD	Interface Control Document
ITWS	Integrated Terminal Weather System
LAN	Local Area Network
LCC	Life Cycle Cost
LDT	Logistics Down Time
LRU	Line (Lowest) Replaceable Unit
MAX	Maximum
MAXTTR	Maximum Time To Repair
MIN	Minimum
MOC	Meteorological Operational Capability
MSCF	Master System Control Function
MSL	Mean Sea Level
MTBF	Mean Time Between Failures
MTBV	Mean Time Between Visits
MTTR	Mean Time to Repair

NAS	National Airspace System
NCEP	National Center for Environmental Prediction
NEDS	Naval Environmental Display Station
NEXRAD	Next Generation Weather Radar
NIDS	NEXRAD Information Dissemination Service
NIMS	NAS Infrastructure Management System
NOAA	National Oceanic and Atmospheric Administration
NMOC	Naval Meteorology and Oceanography Command
N OCD	Naval Oceanography Command Detachment
NWS	National Weather Service
NWSFO	NEXRAD Weather Service Forecast Office
NWSO	NEXRAD Weather Service Office
OJT	On Job Training
OT	Operating Time
PCPUP	Personal Computer Principal User Processor
PRF	Pulse Repetition Frequency
PRT	Pulse Repetition Time
PUES	Principal User External System
PUP	Principal User Processing
PUPIE	Principal User Processor Interactive Emulator
RDA	Radar Data Acquisition
RF	Radio Frequency
RFC	River Forecast Center
RFI	Radio Frequency Interference
RMS	Remote Monitoring System
RPG	Radar Product Generation
ROC	Radar Operations Center
RPIE	Real Property Installed Equipment
SCC	System Command Center
SPC	Storm Prediction Center
SSM	System Status Monitor
ST	Standby Time
STC	Sensitivity Time Control
TCM	Total Corrective Maintenance
T&E	Test and Evaluation
TPC	Tropical Prediction Center
TPM	Total Preventive Maintenance
TRACON	Terminal Radar Control Facility
TT	Total Time
UTC	Universal Coordinated Time
WAN	Wide Area Network
WARP	Weather and Radar Processor
WINPUP	Windows-based Principal User Processor
WCS	Writeable Control Store
WFO	Weather Forecast Office
WSFO	Weather Service Forecast Office

APPENDIX H – ADAPTATION PARAMETERS

The NEXRAD adaptation parameters that have been identified are listed in Table H-1. This is not a complete list, and no attempt has been made to maintain this list. The list identifies which parameters are applicable to the RDA and RPG functional areas. Following Table H-1, the meaning and use of each adaptation parameter are discussed. The list is intended to be illustrative. The developer will supply a similar list detailing all parameters appropriate to the detailed system design chosen. Nominal values and acceptable ranges for each parameter will be defined by the ROC.

TABLE H-1 NEXRAD ADAPTATION PARAMETERS

<u>ADAPTABLE PARAMETER</u>	RDA	RPG
1. Radar Site Lat/Long Coordinate Position	x	x
2. Product Distribution:		x
Product vs. PUP Functional Area		
Product vs. Principal User's External Systems		x
Product vs. Other User Interface		x
3. Product Selection to Map Background Pairing		x
4. Product to User Display Pairing		x
5. Product to Color Mix Pairing		x
6. Function Key Definition		x
7. Function Key Turn-on Setting (per Display)		x
8. Cursor Home Location (per Display)		x
9. Preview Area Location (per Display)		x
10. Cursor Coordinate Output Location (per Display)		x
11. Data Array(s) Output Location (per Array; per Display)		x
12. Symbol Size; Character Size; Line Width		x
13. Blink Rate		x
14. Inter-character and Inter-line Spacing of Alphanumeric Array Data		x
15. Leader Line to Character of Array Spacing		x
16. Data Type to Line Format Pairing		x
17. Task Overload Shedding Priority (per Functional Area)	x	x
18. Product Overload Shedding Priority (per Functional Area; Interface)	x	x
19. Display Preset Center Location (per-Display)		x
20. Overload Warning Threshold (per Overload Type)	x	x
21. Overload Warning Display (Minimum Time; Maximum Time; Maximum Time Following Overload Recovery)	x	x
22. Processor Storage Capacity (On-line)	x	x
23. Operational Mode (Initial Setting)	x	
24. Deleted		
25. Deleted		
26. Meteorological Phenomena for Alerts		x
27. Communication Timeout Periods		x
28. Alert Areas		x

1. Radar Site Latitude/Longitude Coordinate Position:

A geographic position on the NEXRAD System geographic grid defining the radar site location to within a 1/4 km resolution.

- 2. Product Distribution: Product vs. PUP Functional Area (per RPG interface)
- Product Distribution: Product vs. Principal User External Systems (per RPG interface)
- Product Distribution: Product vs. Other User Interface (per RPG interface)

Tables defining the distribution of NEXRAD products. The Product vs. PUP Functional Area tables will be used by the RPG to control the distribution of products to associated and non-associated PUP functional areas. The Product vs. Principal User External Systems tables will be used by a NEXRAD Unit to control the distribution of NEXRAD products from the NEXRAD Unit to a Principal User External System. The Product vs. Other User Interface tables will be used by the NEXRAD Unit to control product distribution to Other Users.

3. Product Selection to Map Background Pairing

The Product Selection to Map Background Pairing adaptation parameter defines the background map(s) which are to be linked to a product selected for display. The actual display of the map data will be dependent upon the operator selection(s) of map types and the operator selection(s) of map content turn on/turn off.

4. Product to User Display Pairing

The Product to Principal User Display Pairing (per interface) adaptation parameter defines a default turn-on condition for each user display.

5. Product to Color Mix Pairing (per PUP)

The Product to Color Mix Pairing adaptation parameter defines a turn-on color selection for the presentation of each data item of each product.

6. Function Key Definition

The Function Key Definition adaptation parameter defines that set of functions performed automatically when a specific function key is selected (activated). Function key examples are product selection, product content turn on/turn off, map selections, map content turn on/turn off, cursor home, magnification selection, test pattern selection/deselection, off-center, preset center, resolution selection, accept queued output message, alarm acknowledge, and enter.

7. Function Key Turn on Setting (per Display)

The Function Key Turn on setting adaptation parameter defines the initial setting for each Function Key for use during system initialization.

8. Cursor Home Location (per Display)

The Cursor Home Location adaptation parameter defines the display location at which the cursor symbol is displayed.

9. Preview Area Location (per Display)

The Preview Area Location adaptation parameter defines the display location at which the first alphanumeric character will appear on the display when an input message is entered.

10. Cursor Coordinate Output Location (per Display)

The Cursor Coordinate Output Location adaptation parameter defines the display location at which the first character of the cursor coordinate output message will appear on the display when a cursor position control function selection key is activated.

11. Data Array(s) Output Location (per Array; per Display)

The Data Array Output Location adaptation parameters define the size of each alphanumeric data array (number of characters per line, and number of lines) and the display location of the top left character position of each data array.

12. Symbol Size; Character Size; Line Width

The Symbol Size, Character Size, and Line Width adaptation parameters define the alphanumeric and special symbol character font and size in pixels for each character, and the pixels per line for line width.

13. Blink Rate

The Blink Rate adaptation parameter defines the number of times a data item is to be turned on/off in a time period, and the number of display refresh cycles to be used for both the on and the off condition.

14. Inter-character and interline spacing of alphanumeric array data

The Inter-character and interline spacing of array data adaptation parameter defines the number of pixels to be used as spacing between adjacent characters on a line, and the number of pixels to be used as spacing between adjacent lines of characters.

15. Leader Line to Character or Array Spacing

The Leader Line to Character or Array Spacing adaptation parameter defines the number of pixels between the end of a line and the center of a character or the center of a character in one of the four corners of a character array.

16. Data Type to Line Format Pairing

The Data Type to Line Format Pairing adaptation parameter defines the line format (solid, dotted, dashed) to be used for display of each type of line data.

17. Task Overload Shedding Priority (per Functional Area)

The Task Priority for Overload Shedding adaptation parameter defines the order for deleting (not performing) processing tasks when an overload situation occurs. Processing tasks are to be reactivated following elimination of the overload condition. Task reactivation is performed in a manner so as not to cause a cyclic condition between an overload and non-overload condition to occur.

18. Product Overload Shedding Priority (per Functional Area; Interface)

The Data Priority for Overload Shedding/Recovery adaptation parameter defines the order for deleting data when an overload situation occurs on an interface between functional areas, or an interface from a functional area to an external user's system interface or to another user interface. The recovery from an overload condition is performed automatically. Data types are recovered in the reverse of the shedding order. Data types are recovered one data type at a time with the time delay between the recovery of each data type a part of the adaptation parameter.

19. Display Preset Center Location (per Display)

The Display Preset Center Location adaptation parameter defines the geographic coordinate position of the center of the display.

20. Overload Warning Threshold (per Overload Type)

The Overload Warning Threshold adaptation parameter defines the utilization level that, when reached or exceeded, will result in an overload warning being issued. Overload types include

processor, memory, communication interfaces, input data buffers, mass storage, and archival storage utilizations.

21. Overload Warning Display (Minimum Time; Maximum Time Following Overload Recovery)

The Overload Warning Display adaptation parameters define the minimum time an overload message is presented on the display and the maximum time following recovery from an overload condition that an overload message is presented. The minimum time requirement for presentation of an overload takes precedence over the maximum time following recovery from overload requirement.

22. Processor Storage Capacity (Online)

The Processor Storage Capacity adaptation parameter defines the amount of on-line memory available for the operational system.

23. Operational Mode (Initial Setting)

The Operational Mode adaptation parameter defines the radar operational mode to be used following system initialization.

24. Deleted

25. Deleted

26. Meteorological Phenomena for Alerts

The Meteorological Phenomena for Alerts will specify thresholds for all alert categories. Each threshold criterion will contain a set of specific parameter values or phenomenon conditions. This set of adaptable thresholds will be maintained at each RPG.

27. Communication Time-out Periods

The Communication Time-out Periods adaptation parameters define the Time-out periods for the five "timers" identified in the ICD. Specifically, these are Status, Acknowledgment, Busy, Idle, and Interframe.

28. Alert Areas

The Alert Area adaptation parameter allows the PUP operator to define up to two Alert Areas. There is one set of Alert Areas per PUP or RPG Operational Position. Within the defined Alert Area(s) the PUP or RPGOP will be capable of specifying conditions which will cause an alert.

APPENDIX I – OPERATIONAL MODES AND SCANNING STRATEGIES

This Appendix describes Operational Modes and Scanning Strategies and provides a set of criteria for automatic selection/deselection of modes. Additional modes, strategies, and selection/deselection criteria will be defined by the ROC as updates to this Appendix.

1. OPERATIONAL MODES

Operational Modes are designated by capital letters, A, B, C, etc. The following modes are currently defined:

- Mode A - Precipitation/Severe Weather
- Mode B - Clear air observation
- Mode O - Off-line Maintenance

2. SCANNING STRATEGIES

Scanning Strategies are designated by Arabic numerals 1, 2,3, etc. The following strategies are currently defined:

a. Fourteen unique elevation angles from the base elevation angle to +20 degrees in elevation for a volume scan time of 5 minutes. The lowest 6 degrees of elevation will have no gaps between the one-way pattern 3 dB points of adjacent elevation scans. The pulse length is 1.57 msec.

b. Nine unique elevation angles from the base elevation angle to +20 degrees in elevation for a volume scan time of 6 minutes. The lowest 4 degrees of elevation will have no gaps between the one-way pattern 3 dB points of adjacent elevation scans. The pulse length is 1.57 msec.

c. Five unique elevation angles from the base elevation angle to +5 degrees in elevation for a volume scan time of 10 minutes. There will be no gaps between the one-way pattern 3 dB points of adjacent elevation scans. This scanning strategy will be capable of running using either of the two following different pulse lengths:

1. Long pulse $\approx$ 4.71 msec
2. Short pulse $\approx$ 1.57 msec

d. Nine unique elevation angles from the base elevation angle to +20.0 degrees in elevation for a volume scan time of 5 minutes 45 seconds (345.0 seconds). The lowest 5 degrees of elevation will have no gaps between the one-way pattern 3 dB points of adjacent elevation scans. The pulse length is 1.57 μ sec.

e. Fourteen unique elevation angles from the base elevation angle to +20.0 degrees in elevation for a volume scan time of four minutes 42 seconds (282.0 seconds). The elevation angles will maintain constant height uncertainty at all ranges to 460 km and for all heights to 70,000 ft of storm features. This will result in overlapping coverage of one-way 3 dB points of adjacent elevation scans. The pulse length is 1.57 μ sec.

Certain meteorological conditions may limit the effectiveness of standard VCP scanning. Therefore, flexible, dynamic scanning options are designed for integration into the baseline VCP scanning

schemes. If dynamic scanning is implemented, the volume scantime may be different depending on the combination of scanning strategies used. The following dynamic scanning options are currently defined.

a. The ability to determine a volume scan, based on detected return, prior to the completion of the last elevation in the VCP definition. This early termination will cause the system to enter its normal end-of-volume scan transition (RDA antenna retrace, RPG conclude algorithm processing and product generation) to prepare for the start of a new volume scan.

b. Additional elevation scans, not included in standard definition, may be included at the time of execution.

3. SELECTION/DESELECTION CRITERIA

An automatic method for deselection of Mode B, clear air observation will be developed. The criterion used for this will be the Precipitation Category (defined in the Mode Selection Function described in Appendix D). If the Precipitation Category is 1, a Precipitation/Severe Weather operational mode with a scanning strategy determined by adaptation data will be automatically selected. Automatic deselection of Mode B will not occur when the precipitation category is two (2).

APPENDIX J – METEOROLOGICAL PHENOMENA FOR ALERTS

<u>ALERT CATEGORIES</u>	<u>RANGE OF COVERAGE (km)</u>	<u>THRESHOLD CRITERIA</u>
a. Grid Group		
Velocity (lowest elev)	0-115	Knots: 6 values
Composite Reflectivity	0-345	dBZ _e : 6 values
Echo Tops	0-230	K-feet: 4 values
VIL	0-230	Kg/m ² : 6 values
b. Volume Group ²² (non-grid, algorithm outputs)		
VAD (lowest elev.)	NA	Knots: 6 values
Maximum Hail Size	0-230	Inches: 6 values
TVS	0-115	Detected
Max Storm Refl	0-460	dBZ _e : 6 values
Prob. of Hail	0-230	%: 6 values
Prob. of Severe Hail	0-230	%: 6 values
Storm Top	0-460	K-feet: 6 values
Max 1-hr rainfall accumulation	0-230 ²³	inches: 4 values
MDA Strength Rank		
c. Forecast Group ²⁴ (Storm oriented)		
Maximum Hail Size	0-230	Inches: 6 values
TVS	0-115	Detected
Max Storm Refl.	0-460	dBZ _e : 6 values
Prob. of Hail	0-230	%: 6 values
Prob. of Severe Hail	0-230	%: 6 values
Storm Top	0-460	K-feet: 6 values
Max Strength Rank		

Expansion provisions will be made to accommodate additional alert categories as follows:

Group 11 additional

Group 29 additional

Group 310 additional

²²See the NEXRAD Algorithm Report (December 1985) for definitions of outputs listed under Volume and Forecast Groups.

²³Location of value not required.

²⁴See the NEXRAD Algorithm Report (December 1985) for definitions of outputs listed under Volume and Forecast Groups.

APPENDIX K – RADAR CODED MESSAGE CODE

The Radar Coded Message, as described in Item 26, Appendix E, NEXRAD Products, is composed of three parts, preceded by a communications header. Part A, Reflectivity, and Part B, VAD Winds, will be encoded automatically. Part C, Remarks, will be encoded automatically or at the PUP as specified in that part. In the groups below, capital letters represent the fixed part of the group, and small letters represent variables. The message is encoded as follows:

Header. The header is encoded as follows:

cccc The communications node (PUP site identifier).

ROBUU The product category for unedited radar coded message.

sidd (C/R) Four letter RDA site identifier.
(Example: ccccROBUU sidd (C/R))

Part A: Reflectivity

Part A of the Radar Coded message (RCM) contains an intermediate graphic product and a tabular listing of alphanumerics. Data in the Radar Coded Message are located with respect to a polar stereographic grid. The local grid at each antenna site is designed to be a subject of the National Radar Grid so that data may be readily composited.

The National Radar Grid has a resolution of 1/4 LFM (Limited Fine Mesh Model) which is 47.625 km at 60 degrees north latitude. The vertical axis of the grid is parallel to the 105 degree west longitude meridian.

At each site, a local grid is chosen having 25 rows and 25 columns, with the antenna site located within the central box. The 25 rows and columns of the grid are assigned letters A through Y, so that the box containing the antenna site is always box MM. Box AA is at the upper left. As shown in Figure K-1, each box is further subdivided to form an overall 1/16 LFM grid.

The graphic product contains reflectivity data for the 1/16 LFM grid over the radar area of coverage out to 460 km. The reflectivity intensity value for each grid box is determined by assigning the maximum value of all Composite Reflectivity boxes whose centers are contained within the 1/16 LFM grid square.

The RCM is based on the 16 level, 4 km x 4 km Composite Reflectivity product but contains only 9 data level categories; 6 for data and 1 for missing or below threshold data within 230 km and 2 for data beyond 230 km. For data beyond 230 km, a separate threshold is provided for which: a) all data above that threshold are labeled as level 8, and b) all data below that threshold are labeled as level 9.

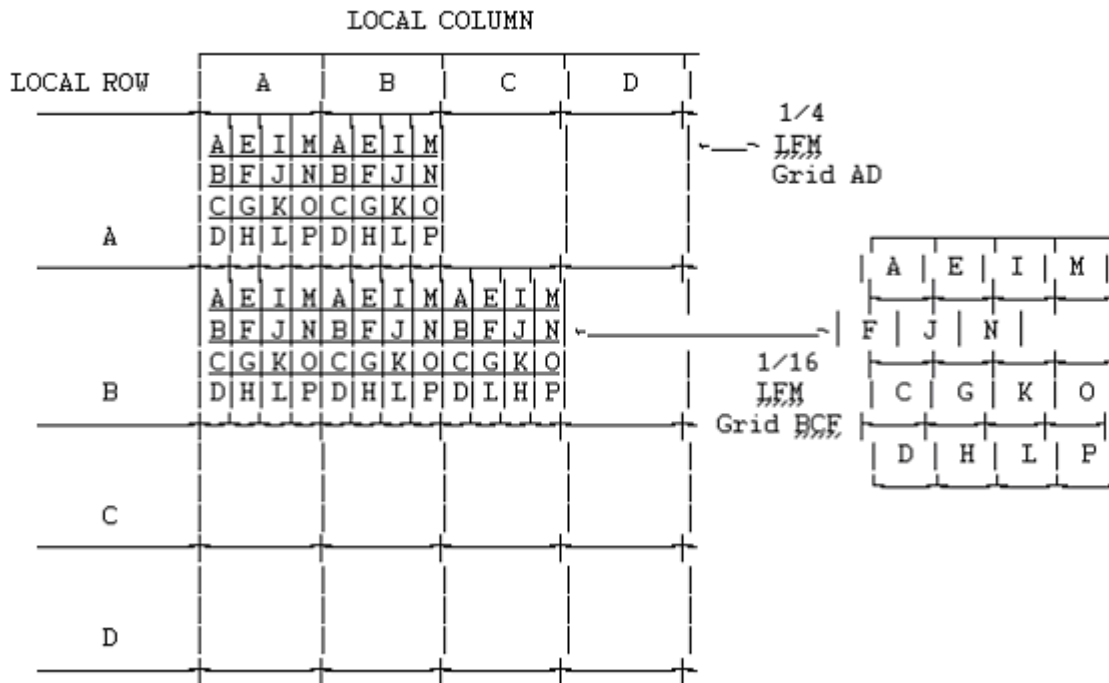


Figure K-1 Limited Fine Mesh Grid

Within the tabular listing, data are provided for the maximum echo top. The height, and the position where provided, is derived from the Echo Tops product. The listing also shows the locations of the largest centroids within 230 km of the radar using the 1/16 LFM grid and provides the forecast centroid speed and direction, as available from the Storm Position Forecast algorithm.

Part A of the message is encoded as follows:

/NEXRAA	Part A indicator.
sidd	Four letter RDA site identifier
ddmmyyddd	The day (dd) of the month (mm), the year (yy), and the time (ddd), to the nearest minute, in Greenwich Mean Time (GMT).
unedited	Status of message
RADNE	A group to encode no reportable reflectivity intensity values is provided, i.e., field NInnnn is zero.
RADOM	A group to encode radar down for maintenance is provided.
/MDnnnn	A group of six characters to encode operational mode is provided. See Appendix I. Choices are PCPN and CLAR. (Example: /MDPCPN)
/SCnnnn	A group of six characters to encode scan strategy is provided. See Appendix I. Choices are 1405 (14 scans in 5 minutes), 0906, 0510, etc. (Example: /SC1405)
/NInnnn:	The total number (nnnn) of intensities (NI) reported in the following field (gggi) will be encoded. (Example: /NIO144)

gggi Reflectivity intensity is mapped onto the 1/16 LFM grid (ggg). Encode locations and intensities by a series of groups made up of three letters (1/16 LFM) followed by the maximum intensity of the designated grid box. The three letters (in order) are row, column, and sub-grid. The numbers following, represent intensities in succeeding sub-grid boxes in that row. That is, encode each 1/16 LFM grid box from west to east, starting with the northern-most row with data, followed by the next southern row, etc. In the interest of compacting the message, successive intensities of different or similar values may be listed after a single location as long as the intensities are continuous. When succeeding grid boxes contain the same intensity value, the number of succeeding boxes with the same value may be designated by a letter of the alphabet. That is, if 4 succeeding 1/16 LFM grid boxes (a total of five boxes) are at level 2, they could be coded as GGG2D. The "2D" may also be followed by different intensity values. Location/intensity groups are separated by a comma. (Example: ABF112D331,BCA1211)

/MThhh:ggg The location and height (MSL) of the maximum echo top (MT) within 230km radius of the radar are encoded using the three letter grid designator (ggg) and assigning the height coinciding with echo top product in hundreds of feet (hhh). (Example: /MT320:NLB)

/NCENnn: The total number (nn) of centroids (NCEN) reported in this portion of the message is encoded. This number corresponds to the corrected centroids below. (Example: /NCEN04:)

Cnnggg dddfff The centroid (C) number (nn), location (grid box) (ggg), direction from which it is moving (in one degree increments) (ddd), and its speed (fff) in knots, are encoded. Successive groups are separated by commas. If, during editing, data are deleted in a grid box that contains a centroid, this group will be corrected by deleting this centroid. (Example: C03QMB240012)

/ENDAA(C/R) A group to indicate the end of Part A.
The following is a summary example of the components of Part A.
/NEXRAA sidd 2812881330 Unedited (C/R)

/MDnnnn /SCnnnn /NInnnn:

gggi ... i,gggi...i

/MThhh:ggg

/NCENnn: Cnnggg dddfff,Cnnggg dddfff

/ENDAA (C/R)

Part B: VAD Winds

Part B of the RCM contains a single profile of the horizontal wind information derived from the output of the VAD algorithm.

Part B of the message is encoded as follows:

/NEXRBB Part B indicator.
sidd Four letter RDA site identifier
ddmmyytttt The day (dd) of the month (mm), the year (yy), and the time (tttt), to the nearest minute, in Greenwich Mean Time (GMT).
VADNA The optional entry VADNA encoded for instances when no VAD wind data are available for the last 15 minutes, if appropriate.
hhhcddffff Coded heights (hhh) in hundreds of feet MSL; confidence (c) level, using RMS for the coded height; wind direction (ddd) and wind speed (fff), in knots, coincide with those derived from the VAD Winds product. The confidence level is encoded as a single letter in accordance with the following:
 A = RMS of 1 m s; B = RMS of 2 m s;
 C = RMS of 3 m s; D = RMS of 4 m s;
 E = RMS of 5 m s; F = RMS of 6 m s;
 G = RMS of greater than or equal to 7 m s.

Wind direction and speed, as output from the VAD algorithm, are reported at up to 19 heights, in feet MSL. Default heights are:

1,0006,00012,00025,000
2,0007,00014,00030,000
3,0008,00016,00035,000
4,0009,00018,00050,000
5,00010,00020,000

(Example: 080C240060)

/ENDBB End of Part B indicator.

(C/R)

The following is a summary example of the components of Part B.

/NEXRBB sidd 2812881330 (C/R)

hhhcddffff,hhhcddffff,hhhcddffff

/ENDBB (C/R)

Part C: Remarks

Part C of the Radar Coded Message contains remarks in an alphanumeric format. Automatically generated remarks provide information on the locations of tornadic vortex signatures, mesocyclones, centroids, storm tops, and hail indices.

The automated portion of Part C is encoded as follows:

/NEXRCC	Part C indicator.
sidd	Four letter RDA site identifier
ddmmyytttt	The day (dd) of the month (mm), the year (yy), and the time (tttt), to the nearest minute, in Greenwich Mean Time (GMT).
/NTVSnn:	The total number (nn) of tornadic vortex signatures (NTVS) detected by the TVS algorithm and reported in Part C is encoded. (Example: /NTVS03:).
TVSnnnggg	The location (ggg) and number identifier (nn) of each tornadic vortex signature (TVS) is encoded using the three-letter grid box designator. (Example: TVS02NLB).
/NMESnn:	The total number (nn) of mesocyclones detected by the Mesocyclone Detection algorithm that meet or exceed the Minimum Display Filter Strength Rank threshold (default = strength rank 5) and reported in Part C is encoded. (Example: /NMES05:).
Mnnnggg	The location (ggg) and strength rank (rr) of each mesocyclone that meets or exceeds the Minimum Display Filter Strength Rank threshold (M) are encoded using the three-letter grid box designator. (Example: M05JLC).
/NCENnn:	The total number (nn) of centroids (NCEN) reported in Part C is encoded. (Example: /NCEN08:)
Cnnnggg ShhhHi	The height (hhh) in hundreds of feet (MSL), of the storm top (S), as derived from the Storm Structure algorithm, for each centroid (C) identified in Part A to include location (ggg) is encoded. The centroid identifier number (nn) is the same as given in Part A. The hail (H) index (i), as provided by the Hail algorithm, is also given as one of the four following data levels:

N - no hail; P - possible or probable hail; H - hail;

U - unknown.

(Example: C04QQD S440HP).

End of Message

At the end of the message, the following group is sent:

/ENDALL (C/R)	A group to indicate end of message shall be provided.
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APPENDIX L – DELETED

APPENDIX M – DELETED

ANNEX A – EVOLUTIONARY WSR-88D REQUIREMENTS *VERSION 2.0 – DELETED*

The Annex A - Evolutionary WSR-88D Requirements section was moved to a separate document called "Evolutionary Requirements".